Geotechnical and Foundation Design Considerations

ANSI/API RECOMMENDED PRACTICE 2GEO
FIRST EDITION, APRIL 2011

ADDENDUM 1, OCTOBER 2014

ISO 19901-4:2003 (Modified), Petroleum and natural gas industries—Specific requirements for offshore structures, Part 4—Geotechnical and foundation design considerations

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Upstream Segment

ANSI/API RECOMMENDED PRACTICE 2GEO
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ADDENDUM 1, JULY 2014

ISO 19901-4:2003 (Modified), Petroleum and natural gas industries—Specific requirements for offshore structures, Part 4—Geotechnical and foundation design considerations

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API Foreword

The API Subcommittee on Offshore Structures (SC 2) voted to adopt a modified version of ISO 19901-4:2003 as American National Standard ANSI/API Recommended Practice 2GEO.These modifications from the ISO standard have been incorporated directly into the text and marked with a change bar in the margin.

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Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

International Standards are drafted in accordance with the rules given in the ISO/IEC Directives, Part 2.

The main task of technical committees is to prepare International Standards. Draft International Standards adopted by the technical committees are circulated to the member bodies for voting. Publication as an International Standard requires approval by at least 75 % of the member bodies casting a vote.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. ISO shall not be held responsible for identifying any or all such patent rights.

ISO 19901-4 was prepared by Technical Committee ISO/TC 67, *Materials, equipment and offshore structures for petroleum, petrochemical and natural gas industries*, Subcommittee SC 7, *Offshore structures*.

ISO 19901 consists of the following parts, under the general title *Petroleum and natural gas industries — Specific requirements for offshore structures*:

— Part 1: Metocean design and operating considerations

— Part 2: Seismic design procedures and criteria

— Part 3: Topsides structure

— Part 4: Geotechnical and foundation design considerations

— Part 5: Weight control during engineering and construction

— Part 6: Marine operations

— Part 7: Stationkeeping systems for floating offshore structures and mobile offshore units

ISO 19901 is one of a series of standards for offshore structures. The full series consists of the following International Standards.

— ISO 19900, *Petroleum and natural gas industries — General requirements for offshore structures*

— ISO 19901 (all parts), *Petroleum and natural gas industries — Specific requirements for offshore structures*

— ISO 19902, *Petroleum and natural gas industries — Fixed steel offshore structures*

— ISO 19903, *Petroleum and natural gas industries — Fixed concrete offshore structures*

— ISO 19904, *Petroleum and natural gas industries — Floating offshore structures*

— ISO 19905-1, *Petroleum and natural gas industries — Site-specific assessment of mobile offshore units — Part 1: Jack-ups*

— ISO/TR 19905-2, *Petroleum and natural gas industries — Site-specific assessment of mobile offshore units — Part 2: Jack-ups commentary*

— ISO 19906, *Petroleum and natural gas industries — Arctic offshore structures*

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Introduction

The API offshore structures standards constitute a common basis covering those aspects that address design requirements and assessments of all offshore structures used by the petroleum and natural gas industries worldwide. Through their application the intention is to achieve reliability levels appropriate for manned and unmanned offshore structures, whatever the type of structure and the nature of the materials used.

It is important to recognize that structural integrity is an overall concept comprising models for describing actions, structural analyses, design rules, safety elements, workmanship, quality control procedures and national requirements, all of which are mutually dependent. The modification of one aspect of design in isolation can disturb the balance of reliability inherent in the overall concept or structural system. The implications involved in modifications, therefore, need to be considered in relation to the overall reliability of all offshore structural systems.

The offshore structures International Standards are intended to provide a wide latitude in the choice of structural configurations, materials and techniques without hindering innovation. Sound engineering judgment is therefore necessary in the use of these International Standards.

The overall concept of structural integrity is described above. For foundations, some additional considerations apply. These include the time, frequency and rate at which actions are applied, the method of foundation installation, the properties of the surrounding soil, the overall behavior of the seabed, effects from adjacent structures and the results of drilling into the seabed. All of these, and any other relevant information, need to be considered in relation to the overall reliability of the foundation.

The design practice for the foundations of offshore structures has proved to be an innovative and evolving process over the years since the 1950s. This evolution is expected to continue and is encouraged. Therefore, circumstances can arise when the procedures described herein (or elsewhere) are insufficient on their own to ensure that a safe and economical foundation design is achieved.

Seabed soils vary. Experience gained at one location is not necessarily applicable at another. The scope of the site investigation for one structure is not necessarily adequate for another. Extra caution is necessary when dealing with unfamiliar soils or foundation concepts. This document is intended to provide wide latitude in the choice of site investigation techniques and foundation solutions, without hindering innovation. Sound engineering judgment is therefore necessary in the use of this document.

For an offshore structure and its foundations, the action effects at the interface between the structure's subsystem and the foundation's subsystem(s) are internal forces, moments and deformations. When addressing the foundation's subsystem(s) in isolation, these internal forces, moments and deformations may be considered as actions on the foundation's subsystem(s) and this approach is followed in this document.

Some background to and guidance on the use of this document is provided for information in Annex A. Guidance on foundations in carbonate soils is provided for information in Annex B. There is, as yet, insufficient knowledge and understanding of such soils to produce normative requirements.

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Petroleum and natural gas industries — Specific requirements for offshore structures —

Part 4:
Geotechnical and foundation design considerations

1 Scope

This document contains requirements and recommendations for those aspects of geoscience and foundation engineering that are applicable to a broad range of offshore structures, rather than to a particular structure type. Such aspects are

— site characterization,

— soil and rock characterization,

— design and installation of foundations supported by the seabed (shallow foundations),

— identification of hazards,

— design of pile foundations, and

— soil-structure interaction for risers, flowlines, and auxiliary subsea structures.

Aspects of soil mechanics and foundation engineering that apply equally to offshore and onshore structures are not addressed. The user of this document is expected to be familiar with such aspects.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

API RP 2A-WSD, 21st Edition, *Recommended Practice for Planning, Designing and Constructing Fixed Offshore Platforms — Working Stress Design*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

3.1
drained condition
condition whereby the applied stresses and stress changes are supported by the soil skeleton and do not cause a change in pore pressure

3.2
effective foundation area
reduced foundation area having its geometric center at the point where the resultant action vector intersects the foundation base level

3.3
seafloor
interface between the sea and the seabed

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3.4
seabed
materials below the sea in which a structure is founded, whether of soils such as sand, silt or clay, cemented materials or of rock

NOTE 1 The seabed can be considered as the half-space below the seafloor.

NOTE 2 Offshore foundations are most commonly installed in soils, and the terminology in this part of ISO 19901 reflects this. However, the requirements equally apply to cemented seabed materials and rocks. Thus, the term “soil” does not exclude any other material at or below the seafloor.

NOTE 3 As yet there are no universally accepted definitions of the various types of soil and rock, see A.6.4.3.

3.5
settlement
permanent downward movement of a structure as a result of its own weight and other actions

3.6
undrained condition
condition whereby the applied stresses and stress changes are supported by both the soil skeleton and the pore fluid and do not cause a change in volume

3.7
undrained shear strength
maximum shear stress or shear stress at a specified shear strain, in an undrained condition

4 Symbols

4.1 General

Commonly used symbols are listed as follows; other symbols are defined in the text following the applicable formula. It should be noted that symbols can have different meanings between formulae.

4.2 Symbols for stability of shallow foundation

A actual foundation area

A' effective foundation area of the foundation depending on the load eccentricity

A_o actual foundation area per unit length, of an infinitely long footing

A_h embedded vertical cross-sectional area of foundation

A_p end area of skirt tip

A_s side surface area of foundation skirt embedded at a particular penetration depth (including both sides)

$A_{idealized}$ idealized rectangular foundation area, for irregular foundation shapes

b_c, b_q, b_γ individual correction factors related to foundation base inclination

B minimum lateral foundation dimension (also foundation width)

B' minimum effective lateral foundation dimension (also foundation effective width)

C compression index of the soil over the load range considered

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d_c, d_q, d_γ	individual correction factors related to foundation embedment depth
D_b	depth below seabed to base level
e	load eccentricity
e_o	initial void ratio of the soil
e_1	load eccentricity along foundation length
e_2	load eccentricity along foundation width
f	unit shaft friction resistance along foundation skirts, during installation
F	correction factor
g_c, g_q, g_γ	individual correction factors related to foundation seafloor surface inclination
G	elastic shear modulus of the soil
h	soil layer thickness
H	acting horizontal loads, for relevant loading condition
H_d	*maximum total horizontal load applied to the base of the foundatio*n at failure under undrained *conditions*
H_d'	*maximum total horizontal load applied to the base of the foundation at failure under drained conditions*
H_{ult}	ultimate horizontal capacity in yield surface design method
ΔH	horizontal soil resistance due to active and passive earth pressures on foundation skirts
i_c, i_q, i_γ	individual correction factors related to foundation load inclination
K_c, K_q, K_γ	correction factors, which account for load inclination, footing shape, depth of embedment, inclination of base, and inclination of the seafloor
K_p	*coefficient of passive earth pressure*
K_{rd}	drained horizontal soil reaction coefficient
K_{ru}	undrained horizontal soil reaction coefficient
L	maximum lateral foundation dimension (also foundation length)
L'	maximum effective lateral foundation dimension (also foundation effective length)
M	overturning moment
M_{ult}	ultimate moment capacity in yield surface design method.
N_c	a dimensionless constant, 5.14

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N_q , N_γ	dimensionless functions of ϕ'
p'_o	vertical effective overburden stress at base level (skirt tip level when skirts are used)
q_o	initial effective vertical stress at level of a given soil layer
Δq	added effective vertical stress, at level of a given soil layer
q	unit end bearing pressure on the foundation skirt tip, during penetration
q_c	CPT cone tip resistance, in stress units, for calculation of skirt penetration resistance
Q	acting vertical load, for relevant loading condition
Q_d	maximum total vertical load applied to the base of the footing at failure (*excluding* weight of soil plug inside skirts) under undrained conditions
Q_d'	maximum total vertical load applied to the base of the footing at failure (*excluding* weight of soil plug inside skirts) under drained conditions
Q_f	shaft friction resistance along foundation skirts during skirt penetration
Q_o	maximum vertical load, per unit length, of an infinitely long footing
Q_p	end bearing resistance at foundation skirt tip during skirt penetration
Q_r	total soil resistance during skirt penetration
Q_{ult}	ultimate vertical capacity in yield surface design method.
R	radius of the base of a circular foundation
s_u, c_u	undrained shear strength of soil
s_{uo}, c_{uo}	undrained shear strength of soil at foundation base level
$s_{u\ ave}$	average soil strength from seabed to base level
s_{u2}	equivalent shear strength below base level
s_c, s_q, s_γ	individual correction factors related to foundation shape
T	torsional moment
u_v, u_n	vertical and horizontal short-term elastic displacements at foundation base
u_v	vertical long-term settlement of a soil layer
β	ground inclination angle in radians, in calculation of inclination factors
δ	interface friction angle at soil-footing interface (in Clause 7.15)
ϕ'	angle of internal friction of sand for drained triaxial conditions

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γ' effective unit weight of soil

κ rate of increase of undrained shear strength with depth, for linear shear strength profiles

υ Poisson's ratio of the soil

ν base inclination angle in radians, in calculation of inclination factors

θ_r, θ_t overturning and torsional short term undrained rotations of foundation base

4.3 Symbols for pile foundation design

A_p gross end area of the pile, $A_p = \frac{\pi \times D^2}{4}$

A_r pile displacement ratio, $A_r = \frac{A_w}{A_p} = 1 - \left(\frac{D_i}{D}\right)^2$

A_w cross sectional steel area of pile at pile tip, $A_w = \frac{\pi}{4} \cdot \left(D^2 - D_i^2\right)$

A_s side surface area of pile

C_1, C_2, C_3 Coefficients determined as a function of ϕ'

D pile outside diameter

D_{50} mean effective soil particle diameter

D_i pile inside diameter, $D_i = D - 2\,WT$

D_{CPT} diameter of the CPT tool, D_{CPT} = 36 mm for a standard cone with a base area of 10 cm^2

D_r relative density of a sand; For CPT-based Method 1, $0 \le D_r \le 1.0$; for CPT-based Method 4, $D_r > 1.0$ should be accepted and used

E_S initial modulus of subgrade reaction

$f(z)$ unit shaft friction, in stress units

$f_c(z)$ unit shaft friction in compression, in stress units

$f_p(z)$ unit shaft friction between the sand soil plug and the inner pile wall, for CPT-based Method 4

$f_t(z)$ unit shaft friction in tension, in stress units

h distance above pile tip = $L - z$

k rate of increase with depth of initial modulus of subgrade reaction, for lateral soil resistance-displacement p-y curves for sand

K_o coefficient of lateral earth pressure at rest

L embedded length of the pile below the original seafloor

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L_s	length of the soil plug in the sand layers
N_q	dimensionless bearing capacity factor
p_a	atmospheric pressure, in stress units, (e.g. p_a = 100 kPa)
$p'_o(z)$	effective vertical stress at depth z
$p'_{o,tip}$	effective vertical stress at the pile tip
$p'_m(z)$	effective mean stress at depth z
$p'_{oh}(z)$	effective horizontal stress at depth z
P_o	pile outer perimeter = πD
q	unit end bearing at the pile tip, in stress units
$q_c(z)$	CPT cone-tip resistance at depth, z, in stress units
$q_{c,f}(z)$	Reduced CPT cone-tip resistance at depth, z, in stress units , to account for general scour
$q_{c,av,1,5D}$	average value of $q_c(z)$ between 1.5D above the pile tip and 1.5D below the pile tip
$q_{c,tip}$	cone tip resistance at the pile tip
Q	mobilized end bearing capacity in Q-z curves, in force unit
Q_c	axial pile ultimate capacity in compression, in force units
$Q_{f,c}$	shaft friction capacity in compression, in force units
$Q_{f,t}$	shaft friction capacity in tension, in force units
$Q_{f,i,clay}$	cumulative shaft friction capacity of the clay layers within the soil plug, for CPT method 3
Q_p	end bearing capacity, in force units
Q_t	axial pile ultimate capacity in tension, in force units
s_u	undrained shear strength of the soil at the point in question, in stress units
WT	pile wall thickness
t	mobilized soil-pile adhesion, for axial shear transfer t-z curves
t_{max}	f(z) = maximum soil-pile adhesion or unit shaft friction at depth z, for axial shear transfer t-z curves
t_{res}	residual soil-pile adhesion or unit shaft friction
z	depth below the original seafloor
z	local pile axial deflection, for axial shear transfer t-z curves
z	axial pile tip displacement, for Q-z curves

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z'	final depth below seafloor, after general scour
z_{res}	axial pile displacement at which the residual soil-pile adhesion, t_{res}, is reached
α	dimensionless shaft friction factor, for cohesive soils
β	dimensionless shaft friction factor, for cohesionless soils
γ'	submerged, or effective, soil unit weight
ε_c	strain at one-half the maximum deviator stress, for lateral soil resistance-displacement p-y curves for soft clay
δ_{cv}	sand constant volume friction angle at the interface between the sand and the pile wall
ϕ'	angle of internal friction of sand, for drained *triaxial* conditions
Δz_{GS}	general scour depth
Δz_{LS}	local scour depth
e	base natural logarithms ≈ 2.718
ln	natural logarithm (base e)

5 General requirements

5.1 General

The foundation shall be designed to carry static and dynamic (repetitive as well as transient) loads without causing excessive deformation of or vibrations in the structure. Special attention shall be given to the effects of repetitive and transient actions on the structural response, as well as on the strength of the supporting soils. The possibility of movement of the seabed shall be considered. Any actions resulting from such movements on foundation members shall be considered in the design. The potential for disturbance to foundation soils by conductor installation or shallow well drilling shall be assessed (see 5.3). Furthermore, the guidance herein does not necessarily apply to so-called “problem” soils such as carbonate material or volcanic sands or highly sensitive clays (see Annex B).

5.2 Testing and instrumentation

Where there is uncertainty regarding the behavior of foundations, testing or instrumentation should be undertaken. Possible methods include the following.

Load testing.

Load testing or large-scale field testing should be performed where there is particular uncertainty in the foundation capacity and where safety and/or economy are of particular importance.

Model tests.

Model tests should be performed where

1) the foundation configuration differs significantly from earlier configurations where operational experience exists,

2) the soil conditions differ significantly from those where operational experience exists,

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3) new methods of installation or removal are envisaged, or

4) a high degree of uncertainty exists as to how the structure or its foundation will behave.

Temporary instrumentation.

Structures should be fitted with temporary instrumentation where

5) the installation method presupposes the existence of measured data for control of the operation, or

6) an installation method is to be applied with which little or no experience has been gained.

Permanent instrumentation.

Structures should be fitted with permanent instrumentation where

7) the safety or behavior of the foundation is dependent on active operation,

8) the foundation configuration, the soil conditions, or the actions differ substantially from those with which experience has been gained,

9) there is a need for monitoring of the whole foundation with regard to penetration, settlement, tilt, or other behavior, or

10) the method of removal presupposes the existence of measured data for control of the operation.

5.3 Conductor installation and shallow well drilling

The planning for conductor installation and shallow well drilling shall take into account the potential for disturbance to foundation soils and the consequent risk of a reduction in stability of the structure or of adjacent conductors.

Soil disturbances during drilling operations can result from hydraulic fracture, washout (uncontrolled enlargement of the drilled hole), or shallow gas pockets. Hydraulic fracture occurs where drilling fluid pressure is too high and fluid is lost into the formation, possibly softening the surrounding soil. Washout generally occurs in granular soils and can, in part, be induced by high drilling fluid circulation rates or drilling without mud. Washout can produce large voids in the soil structure and lead to stress relief in the surrounding soils. These incidents can be accompanied by loss of circulation of drilling fluids, return of these fluids to the seafloor other than through the conductor, or the creation of seafloor craters. Thereby the stability of foundations can be reduced and displacements increased. These detrimental effects can occur whether the drilling takes place after installation of the structure or before, e.g. through a pre-installed template or for an exploration well.

Records of conductor installation and shallow well drilling shall be available to the designer of the structure. The implications for foundation soils of any incidents of inadequate grouting, excessive loss of circulation, return of drilling fluids to the seafloor other than through the conductor, or creation of seafloor craters should be assessed. The cuttings from the well drilling operation, if allowed to accumulate on the seafloor, should be taken into account in the foundation design, installation procedure and structure removal.

6 Geotechnical data acquisition and integrated geoscience studies

6.1 Geotechnical assessment

The determination of geotechnical parameters and the assessment of geological hazards and constraints result from an integrated study of the area using geophysics, geology and geotechnical engineering. Geophysical data are acquired to develop a geological model so as to better understand depositional and other processes and features of an area. The geophysical data are also used to help interpret the stratigraphy from geotechnical boreholes, to define lateral variability across a site, and to provide guidance on optimizing the location of the proposed facilities. Incorporation of geotechnical data into the geological model gives insight into the potential impact of geological conditions on man-made facilities, such as structures, pipelines, anchors and wellheads.

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6.2 Shallow geophysical investigation

Shallow geophysical investigation can provide information about soil stratigraphy and evidence of geological features, such as slumps, scarps, irregular or rough topography, mud volcanoes, mud lumps, collapse features, sand waves, slides, faults, diapirs, erosional surfaces, gas bubbles in the sediments, gas seeps, buried channels, and lateral variations in stratum thicknesses. The areal extent of shallow soil layers can sometimes be mapped if good correspondence is established between the soil boring and *in situ* test information and the results from the seabed surveys.

The types of equipment for performing shallow geophysical investigation that should be considered are discussed below.

a) Echo sounders or swathe bathymetric systems (in which a series of sweeps of the bathymetric equipment are used) define water depths and seafloor morphology. On complex seafloors, swathe systems have the advantage of providing higher data density and better definition of variable topography. Seismic three-dimensional data acquired for exploration purposes also provide useful data for developing water-bottom (bathymetry) maps.

   These data should only be used for preliminary evaluations because the resolution could be of the order of a few meters depending on the variability of the topography.

Sub-bottom profilers (tuned transducers) define structural features within the near-surface sediments.

NOTE These systems can also provide data to develop water-bottom maps.

Side-scan sonar defines seafloor features and seafloor reflectivity.

NOTE Backscatter measurements from some swathe systems can also provide morphological information.

Seismic sources, such as boomers or minisparkers, can define the structure to deeper depths up to approximately 100 m below the seafloor and either single or tuned arrays of sparkers, air guns, water guns or sleeve-exploders can define structure to deeper depths and can tie in with deep seismic data from reservoir studies. Seismic source signals are received either with single channel analogue or multi-channel hydrophones. Digital processing of the recorded signals enhances the quality of the images recorded and removes extraneous noise and multiples from the recorded signals.

Seabed refraction equipment provides information on the stratification of the top few meters of the seabed.

Shallow sampling of near surface sediments using drop, piston or grab sampling, or vibrocoring together with cone penetrometer tests along geophysical tracklines can be useful for calibration of results and improved definition of the shallow geology.

Direct observation of the seafloor using a remotely operated vehicle (ROV) or manned submersible can also provide important confirmation or characterization of geological conditions.

6.3 Geological modelling and identification of hazards

6.3.1 General

The nature, magnitude, and return intervals of potential active geological processes should be evaluated by site investigation techniques. Judicious use of analytical modelling can provide input for determination of the effects of active geological processes on structures and foundations. Due to uncertainties associated with definition of these processes, a parametric approach to studies is also helpful in the development of design criteria.

A geological model is constructed by hypothesizing a depositional process. The geophysical data should be mapped within the context of the hypotheses made. Features within the same geological period should be mapped together. Features not associated with a particular process should be mapped separately. If necessary, the mapping strategy should be adjusted to fit the model until agreement exists between the data and the model. The results of the geological modelling phase should ideally allow the interpreter to discuss in a report how

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features have developed over time, in order to allow assessment of how the features can affect future man-made developments.

Some of the more familiar geological processes, events and conditions are discussed in 6.3.2 to 6.3.7.

6.3.2 Earthquakes

Seismic actions shall be considered in design of structures for areas that are determined to be seismically active. Areas are considered seismically active on the basis of the historical record of earthquake activity, both in frequency of occurrence and in magnitude, or on the basis of a tectonic review of the region (for more details, see API 2EQ [66]).

Seismic considerations for such areas shall include investigation of the subsurface soils for instability due to liquefaction, submarine slides triggered by earthquake activity, proximity of the site to seismogenic faults, the characteristics of the ground motions expected during the life of the structure, and the acceptable seismic risk for the type of operation intended. Structures in shallow water that can be subjected to tsunamis shall be investigated for the effects of the resulting actions.

6.3.3 Fault planes

In some offshore areas, fault planes can extend to the seafloor with the potential for vertical and horizontal movement. Fault movement can occur as a result of tectonic activity, removal of fluids from deep reservoirs or long-term creep related to large-scale sedimentation or erosion. Siting of facilities in close proximity to fault planes intersecting the seafloor should be avoided, if possible.

If circumstances dictate siting structures near potentially active faults, the effect of future fault movement on the foundation shall be assessed. If these effects are shown to be detrimental, the magnitude and time scale of expected movement shall be estimated on the basis of a geological study for use in the design of structures.

See Annex A for more guidance.

6.3.4 Seafloor instability

Movements of the seafloor can be caused by ocean wave pressures, earthquakes, soil self-weight, hydrates, shallow gas, faults, or other geological processes. Weak, underconsolidated sediments occurring in areas where wave pressures are significant at the seafloor are susceptible to wave-induced movement and can be unstable under very small slope angles. Earthquakes can induce failure of seafloor slopes that are otherwise stable under the existing soil self-weight and wave loads.

Rapid sedimentation (such as actively growing deltas), low soil strength, soil self-weight, and wave-induced pressures are generally controlling factors for the geological processes that continually move sediment downslope. Important design considerations under these conditions include the effects of large-scale movement of sediment (i.e. mud slides and slumps) in areas subjected to strong wave pressures, downslope creep movements in areas not directly affected by wave/seafloor interaction and the effects of sediment erosion and/or deposition on structure performance.

The scope of site investigations in areas of potential instability shall focus on identification of metastable geological features surrounding the site and definition of the soil engineering properties required for modelling and estimating seafloor movements.

Estimates of soil movement as a function of depth below the seafloor based on geotechnical analyses can be used to predict actions on structural members. Geological studies employing historical bathymetric data can be useful for estimating deposition rates during the design life of the facility.

6.3.5 Scour and sediment mobility

Scour is the removal of seabed soils by currents and waves. Such erosion can be due to a natural geological process or can be caused by structural components interrupting the natural flow regime above the seafloor.

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From observations, seafloor variations can usually be characterized as some combination of the following.

a) Local scour.

Steep-sided scour pits around foundation components such as piles and pile groups, as seen in flume models.

Global scour.

Shallow scoured basins of large extent around a structure, possibly due to overall structure effects, multiple structure interaction, or wave-soil-structure interaction.

Overall seabed movement of sand waves, ridges, and shoals that would also occur in the absence of a structure.

Such movements can result in seafloor lowering or rising, or repeated cycles of these. The addition of man-made structures often changes the local sediment transport regime that can aggravate erosion, cause accumulation, or have no net effect.

Scour can result in removal of vertical and lateral support for foundations, causing undesirable settlements of shallow foundations and overstressing of foundation components. Where scour is a possibility, it shall be taken into account in design and/or its mitigation shall be considered.

6.3.6 Shallow gas

The presence of either biogenic or petrogenic gas in the pore water of shallow soils is an important consideration to the engineering of the foundation. *In situ* natural gas can be both gaseous or bound with water to form a solid (known as hydrate). In addition to being a potential drilling hazard during both site investigation soil borings and oil well drilling, the effects of shallow gas can be important to foundation engineering. The effect of dissolution and expansion of gas in recovered soil samples shall be taken into account in developing geotechnical parameters for design.

6.3.7 Seabed subsidence

The nature of the soil conditions and the reservoir and extraction processes should be investigated to establish whether subsidence of the seabed is likely to occur during the field life. Subsidence, where this is a possibility, shall be taken into account in design.

The potential for seabed subsidence due to reservoir compaction shall be considered. The magnitude of surface settlements will depend on the reservoir dimensions, compressibility of reservoir rocks and the anticipated pressure drop. Where the magnitude of subsidence is uncertain, an increase in air gap should be considered.

6.4 Geotechnical investigation

6.4.1 General

Knowledge of the soil conditions existing at the construction site of any structure is necessary to develop a safe and economical design. Site investigations shall be performed to determine the various soil strata and their corresponding physical and engineering properties. Results of previous integrated geoscience studies and experience at the site can enable the design and installation of additional structures with minimal or no additional geotechnical investigation.

The initial step for a geotechnical investigation is a review of available geological, geophysical, and previous soil investigation data. The purpose of this review is to identify potential constraints and aid in planning the subsequent data acquisition phases of the site investigation.

Geophysical surveys should be performed before the geotechnical investigation. Geotechnical surveys and geotechnical investigation data should be combined in an engineering geological model of the region so as to

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develop the required design parameters. The on-site studies shall extend throughout the depth and areal extent of soils that will affect or be affected by installation of the foundations.

6.4.2 Soil investigation and testing

The soil investigation program should be defined after review of the geophysical results and the geology at the site. In general, an on-site geotechnical investigation should include

a) sampling for soil classification and engineering property testing, and

b) in situ soil profiling and strength testing.

Details of the field investigation and subsequent laboratory testing program depend on soil variability, environmental design conditions (e.g. earthquake actions) to be considered in the foundation design, the structure type and geometry, conductor, riser and pipeline requirements, and the presence of geohazards (e.g. slope instability), which can affect the entire construction site.

Depending upon the required investigation depth and sample quality, suitable soil samples can be obtained from geotechnical boreholes, freefall piston corers or vibrocorers. Variations in the vertical soil profile can also be assessed in great detail from continuous piezocone penetration tests (PCPT) and geophysical borehole logging. The PCPT can be performed in a geotechnical borehole or from the seafloor, depending upon the required investigation depth.

In situ strength testing should particularly be considered for investigations where sampling disturbance and/or poor recovery are expected. Sample quality largely depends on the control of the drilling and sampling process, soil type and occurrence of gas in the pore fluid. Sample disturbance and recovery problems are notable in silica sands, carbonate materials and soft soils. *In situ* vane shear test results are used in conjunction with shear strengths measured on retrieved samples to assess soil strength in soft to firm clays. PCPT results combined with the former data can be used to establish a continuous shear strength profile in clays. PCPT data in sands are used to estimate *in situ* relative density, pile and skirt friction and end bearing values.

The objective of laboratory testing is to determine the strength-deformation-consolidation properties of a given soil deposit. Samples should be tested as soon as possible after sampling. A combination of shipboard and land-based laboratory testing is recommended to assess possible transportation, storage, and/or ageing effects.

The required sophistication of the soil sampling and preservation techniques, *in situ* testing, and laboratory testing programs are a function of the structure's design requirements and the need to assess geohazards that can affect the structure. For novel structural concepts, deepwater applications, and structures in areas with potential slope instability, the geotechnical programs should be increased and designed to provide the data necessary for detailed geotechnical analyses. In addition, special geological testing (e.g. X-ray radiography or age dating) can be required to assess geological processes. X-ray radiography can also aid in assessing sample quality prior to extruding samples for laboratory testing.

Geotechnical investigations in seismically active areas should include tests to determine dynamic soil properties and liquefaction potential.

For small structures, the influence depth under the foundation is shallow. The near seafloor soil conditions in soft clay or loose sands and silts are difficult to investigate and are often neglected or misinterpreted. Site investigation tools to be considered for very shallow investigations include, but are not limited to, *in situ* vane tests, cone penetration tests, plate load tests, box core samplers and seabed seismic refraction equipment.

If practical, the soil sampling and testing program should be defined after review of the geophysical results. On-site soil investigation should include one or more soil borings and/or corings to provide samples suitable for engineering property testing and a means to perform in-situ testing, if required. The number and depth of borings and/or corings will depend on the soil variability in the vicinity of the site and the platform configuration. Likewise, the degree of sophistication of soil sampling and preservation techniques, required laboratory testing, and the need for *in situ* property testing are a function of the platform design requirements and the adopted design philosophy.

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As a minimum requirement, the foundation investigation for pile-supported structures should provide the soil engineering property data needed to determine the following parameters: axial capacity of piles in tension and compression, load-deflection characteristics of axially and laterally loaded piles, pile drivability characteristics. and mudmat bearing capacity.

The required scope of the soil sampling, in-situ testing, and laboratory testing programs is a function of the platform design requirements and the need to characterize active geologic processes that may affect the facility. For novel platform concepts, deepwater applications. platforms in areas of potential slope instability. and gravity-base structures, the geotechnical program should be tailored to provide the data necessary for pertinent soil-structure interaction and pile capacity analyses.

6.4.3 Identification and classification of soils and rocks

Soils and rocks shall be identified and classified in accordance with a recognized published standard (Annex A).

6.4.4 Carbonate soils

When performing site investigations in frontier areas or areas known or suspected to contain carbonate material, the investigation should include diagnostic methods to determine the existence of carbonate soils. Carbonate deposits are variably cemented and range from lightly cemented with sometimes significant void spaces to extremely well cemented. Therefore, in planning a site investigation program, there should be enough flexibility in the program to switch between soil sampling, rotary coring, and *in situ* testing as appropriate. Tests shall be performed to establish the carbonate content. Particularly in sands and silts that contain in excess of 15 % to 20 % carbonate material, foundation behavior can be adversely affected. In these soils, a carefully developed field and laboratory testing program can be warranted (Annex B).

7 Stability of shallow foundations

7.1 General

Recommendations pertaining to the aspects of shallow foundation design described in API 2A-WSD are given in 7.2 to 7.16.

It is primarily intended that these recommendations be applied to the design of temporary foundations (such as jacket mudmat foundations with shallow skirts, used for temporary support during installation) under idealized conditions. The methods are intended for combinations of dead load, variable live load, environmental (i.e. wind, wave and current) load and thermal load (such as arising from pipelines); where these arise from transient or cyclic actions they are considered as quasi-static loads. Specialist advice shall be sought where more complex dynamic analysis is required, such as where the inertia of the structure or foundation soils is important. This includes seismic loading.

These recommendations may also be considered for foundations that are to be used for permanent support. This may be applicable only where permanent foundations are subject to simple loading and soil conditions, for which the recommendations remain valid; or for situations where foundation failure would only result in displacements and rotations that could be shown to have minimal environmental, safety and economic consequences. Advice shall be sought from specialist geotechnical engineers where the recommendations in this document are to be considered for permanent foundations.

Alternative design approaches may be more applicable for design of shallow foundations used in more complicated and critical situations, such as for foundations in complex seabed conditions or foundations subject to complex multi-directional cyclic loading conditions (including those where cyclic uplift loads may largely offset the foundation dead weight). This may include use of a partial factor design approach, such as outlined in ISO 19901-4. In particular, advanced analysis techniques such as finite element modeling may be used to confirm that design requirements are met for more complex situations. Additional comments are provided in 7.3.

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7.2 Principles

In the following discussion bearing failure constitutes any failure mode that could result in excessive combinations of vertical displacement, lateral displacement, or overturning rotation of the foundation; while pure sliding or torsional failure corresponds to a failure mode where the foundation translates or twists only in a horizontal plane.

The following general principles are to be considered in assessing stability of shallow foundations.

a) Ultimate foundation stability should be analyzed by limit equilibrium methods ensuring equilibrium between applied loads and foundation resistance. Appropriate factors of safety may then be adopted as outlined in 7.3. Due consideration shall be given to potential consequences of excessive displacement and deformation of the foundation soil.

b) For temporary foundations, design may alternatively be based on displacement (rather than stability) criteria, allowing additional movement under environmental actions. Differential settlement must be considered. The appropriateness of adopting this approach will depend on the type of structure and its installation, and should be the subject of a risk assessment.

c) If calculation methods other than limit equilibrium are used, an explanation of any differences between calculated foundation capacities and those obtained using limit equilibrium methods should be provided.

d) For embedded foundations (including surface foundations with skirts) loads acting at base level (typically skirt tip level for skirted foundations) shall be used. Transformation of loads from seafloor to base level shall take account of additional load applied to embedded members, such as that arising from water pressure acting at seafloor level adjacent to the embedded member.

e) Undrained calculations are typically adopted where no drainage, and hence no dissipation of excess pore pressure, occurs during loading. This may occur as a result of the rate of loading or the impermeable nature of the soil. In contrast, drained calculations are typically adopted where no excess pore pressures arise during loading.

The methods outlined in 7.4 to 7.6 are based on use of the effective area concept. The methods may not be appropriate for use with highly compressible soils or layered soils, and for skirted foundations on soft soils or subject to high overturning moment. Specialist advice shall be sought in these conditions and additional guidance is provided below.

7.3 Acceptance criteria

7.3.1 General

Due consideration of displacement and deformation during the life of the foundation shall be incorporated in the design. Where displacement and deformation are critical, more complex analysis approaches may be warranted.

Foundations designed using the approaches in 7.4, 7.5, 7.6, 7.15, and 7.16 shall have an adequate margin of safety against failure under the design loading conditions. Design loads shall be determined with due consideration of the design life of the foundation.

API 2A-WSD outlines the minimum factors of safety that are recommended for the specific failure modes.

7.3.2 Variations in safety factor

Where geotechnical data are sparse or site conditions are particularly uncertain, or where high uncertainty exists in relation to potential failure mechanisms or methods of analysis, increases in the above safety factors may be warranted.

The difference in required minimum safety factor between bearing failure and sliding or torsional failure is intended to account for the relative consequences of each failure mode, and higher factors may be applicable in some situations.

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7.3.3 Use in design

7.3.3.1 General

The minimum recommended factors of safety are intended to provide acceptable levels of reliability for shallow foundation design of typical offshore platforms. Safety margins are achieved by reducing the capacity calculated using equations in the relevant section by the relevant safety factor. Note that in some cases (for drained loading) a reduction in vertical load may actually result in a reduction in reliability by making sliding more critical, and care should be taken to ensure an appropriate range in loading (including lowest possible vertical load) is considered in selection of design loads.

It is recommended that the equations are used to derive envelopes that capture the ultimate capacity of the foundation under a range of loading conditions, and that the above safety factors are subsequently applied to these envelopes to derive envelopes of allowable load. Examples are provided in Figure 1 for both undrained and drained conditions, which also illustrate how the safety factors should apply. Note that, in the case of undrained loading, the ultimate capacity envelope is factored by 1.5 against both the vertical load, Q, and the horizontal load, H, axes (not just the H-axis). This is to ensure a minimum 1.5 factor for all load applications and results in a marginal reduction in the allowable load envelope for combinations of high Q and H.

NOTE Since the WSD method is adopted, the maximum storm and operating environmental conditions with no additional load factors should be used to calculate the foundation reaction loading.

Once an envelope of allowable load is derived, the design loads may be transposed onto this envelope to ensure the minimum safety factors are met. As noted in 7.4, 7.5 and 7.6, in the case of skirted foundations the method recommended to calculate the maximum total vertical load at failure assumes there is no difference in the depth of soil inside and outside the skirts. However

— in some cases the soil height above skirt tip may be higher inside the skirt than outside the skirt, such as where significant scour has occurred, or

— in some cases the soil height above skirt tip may be lower inside the skirt than outside the skirt, such as where the foundation has penetrated deeper than the depth of the skirts.

In cases where a difference does exist, this may be accounted for by adjusting the design vertical load. This may be achieved by modifying the design vertical load by the following:

$$\Delta Q = (p'_{\text{in}} - p'_{\text{out}})\, A \tag{1}$$

where

ΔQ is the change in design vertical load to account for differences in vertical effective stress at skirt tip level;

p'_{in} is the *in situ* vertical effective stress at skirt tip level inside the skirts;

p'_{out} is the *in situ* vertical effective stress at skirt tip level outside the skirts;

A is the actual foundation area.

A similar method may be used for embedded shallow foundations that are designed without skirts. Care should be taken to ensure the design loads are adjusted appropriately in these cases.

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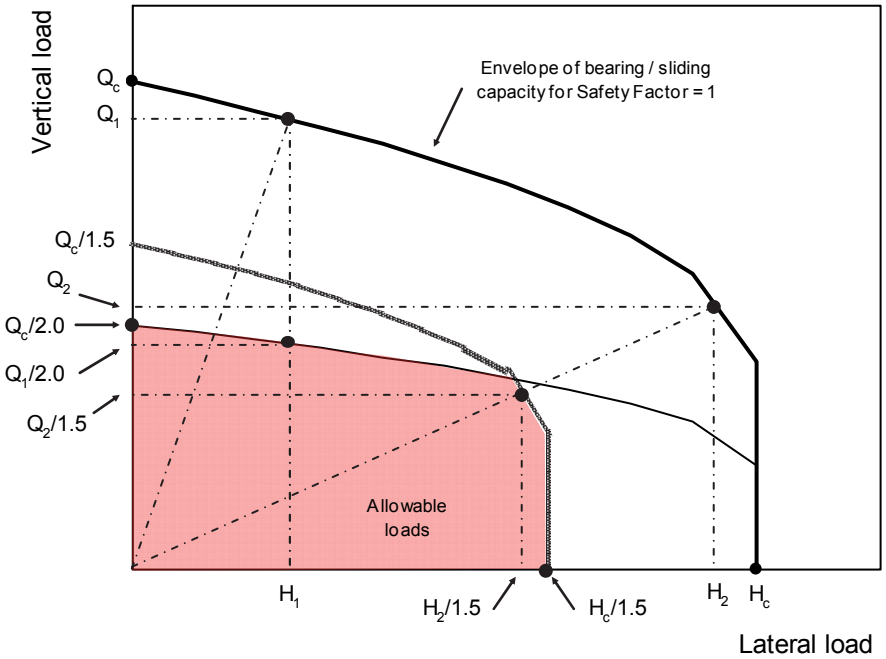
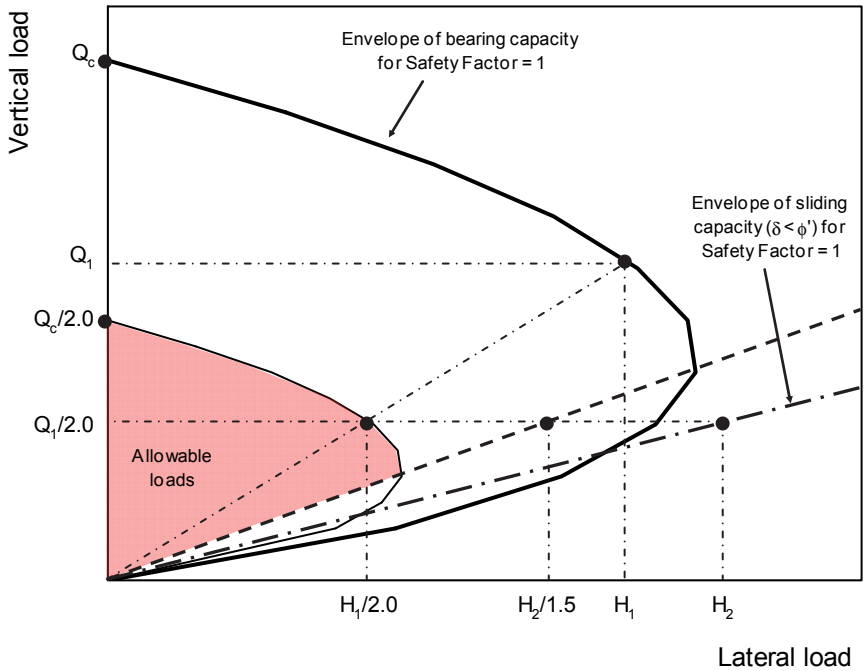


Undrained conditions



Drained conditions

Figure 1—Typical envelopes for undrained and drained conditions

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These safety factors should be used after the effects of cyclic loading have been taken into account. Discussion of derivation of cyclic soil strength is provided in 7.7.

7.3.3.2 Additional design considerations

The conditions described in 7.3.3.2.1 through 7.3.3.2.6 may require further consideration

7.3.3.2.1 Tensile stresses beneath foundation

Generally, tensile stress (relative to ambient water pressure) beneath skirtless foundations is to be avoided because of the potential disturbance to the underlying seabed due to pumping scour, an erosional mechanism whereby rapid movement of water may lead to undermining of the foundation. A global stability check should be performed to confirm no tensile stress occurs, with the following conditions:

— a minimum safety factor of 1.5 against average soil tension is recommended for individual foundations;

— a minimum factor of safety of 1.0 against localized soil tension anywhere under the foundation is recommended, accounting for flexibility of the foundation.

Tensile stresses within the soil may be acceptable in some cases, such as for skirted foundations that can be shown to resist tension through generation of negative excess pore pressures. Specialist advice shall be sought in relation to design of these foundations. Apart from consideration of the permeability of the soil, drainage paths, duration of action and geometry of the foundation, uplift capacity should be analyzed as a reverse bearing capacity problem with a minimum recommended factor of safety factor equal to 2.0.

7.3.3.2.2 Non-standard soils or soil profiles

The analysis methods outlined above have been developed primarily for use in seabed conditions comprising uniform all sand or all clay profiles. The methods may require more careful consideration when used in other soils, such as silts and carbonate sediments. Specialist *in situ* and laboratory testing may be warranted to determine the characteristics of these materials for use in design. Due care is also required for foundation design in more complex (e.g. layered) soil profiles, such as where ‘punch through’ of foundations into underlying soils may occur.

7.3.3.2.3 Interaction with other structures

Potential influence of adjacent structures, such as jack-up spudcans or conductors, should be considered.

7.3.3.2.4 Multiple foundations

For foundations comprising several (connected) foundations, redistribution of loading between individual foundations may be considered, provided the structure can accommodate the resulting distribution, and the soil response is sufficiently ductile.

7.3.3.2.5 Consideration of surrounding seabed conditions

Foundation design shall make allowance for ground conditions outside the base area where this may impact the derivation of foundation capacity. Although relevant for all foundations, this may be particularly relevant for foundations where failure is associated with deep seated failure surfaces.

7.3.3.2.6 Carbonate soils

Carbonate soils may require particular consideration. Carbonate soils comprise a large percentage of the ocean floor, and for the most part are biogenic (i.e. comprised of the accumulated skeletal remains of plant and animal life). Carbonate soils are susceptible to crushing and are more compressible than (for example) terrigenous silica deposits. This difference from siliceous soils is an important characteristic of carbonate soils which profoundly affects mechanical behavior.

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Generally, shallow foundations are suitable for use on carbonate sediments, although any evaluation of such foundations should account for the important difference that exists between such material and silica sands or normal clays. Carbonate sands and silts generally have higher friction angles than silica sands and silts, but are more compressible, and these two factors influence bearing capacity in opposite ways. Carbonate sands and silts are also generally less permeable than equivalent silica material, leading to longer drainage times for a given size of foundation. The tendency for volume reduction on shearing, particularly under cyclic actions, combined with longer drainage times, leads to potential for bearing failure induced by pore pressure generation due to cyclic loading. It should also be noted that the undrained cyclic strength of carbonate sands is often lower than that of silica sands. The high compressibility of most carbonate sediments results in relatively large consolidation settlements, and can give rise to large settlements induced by cyclic actions. Shallow foundations are attractive for carbonate sediments that exhibit a significant degree of cementation, since they give high bearing capacities, good resistance to cyclic actions and low potential for settlements. However, layered profiles of variably cemented and uncemented sediments should be treated cautiously, with due account taken of the risks and consequences of a punch-through type of failure.

7.3.4 Alternative method of design based on yield surfaces

It is generally accepted that the effective area method is conservative in some instances, especially where large lateral load and overturning moment act in a consistent direction. An alternative method of design involves use of explicit yield functions to derive encompassing yield surfaces directly in vertical, horizontal and overturning moment space. Additional discussion is provided in Annex A.

7.4 Undrained bearing capacity — constant shear strength with depth

When approximately constant undrained shear strength exists to a depth below the base of the foundation equal to at least two-thirds (2/3) of the foundation width, a simplification of constant soil strength may be adopted. However, the impact of lower soil strength below this depth on foundation capacity shall also be considered.

The maximum total vertical load (Q_d) which a footing can support at its base under undrained conditions is

$$Q_d = (s_u N_c K_c) A' \tag{2}$$

where

Q_d is the maximum total vertical load applied to the base of the footing at failure (excluding weight of soil plug inside skirts) under undrained conditions;

s_u is the undrained shear strength of soil (see 7.7);

N_c is a dimensionless constant equal to 5.14;

A' is the effective area of the foundation depending on the load eccentricity;

K_c is the correction factor which accounts for load inclination, footing shape, depth of is the embedment, inclination of base, and inclination of the seafloor.

This method is based on assessment of bearing capacity for surface foundations, or skirted foundations where the weight of the soil plug is balanced by the weight of soil overburden outside the skirts. Where this is not the case, the design load should be appropriately adjusted as outlined in 7.3.3.

A method for determining the correction factor and the effective area is given in Annex A. Two special cases of Equation 2 are frequently encountered. For a vertical central load applied to a foundation at seafloor level where both the foundation base and seafloor are horizontal, Equation 2 is reduced as shown in Equation 3 and Equation 4 for two foundation shapes.

a) Infinitely long strip footing:

$$Q_o = 5.14\, s_u A_o \tag{3}$$

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where

Q_o is the maximum vertical load per unit length of footing;

A_o is the actual foundation area per unit length.

b) Circular or square footing:

$$Q_d = 6.05\, s_u\, A \tag{4}$$

where

A is the actual foundation area.

7.5 Undrained bearing capacity — linearly increasing shear strength

Seabed sediments can naturally display undrained strength that increases with depth. This clause provides a method to analyze the bearing capacity where the strength can be assumed to increase linearly with depth, and it would be inappropriate to assume a uniform strength.

The maximum total vertical load, Q_d, which a footing can support at its base under undrained conditions is

$$Q_d = F\left(s_{uo} N_c + \frac{\kappa B'}{4}\right) K_c A' \tag{5}$$

where

Q_d is the maximum total vertical load applied to the base of the footing at failure (excluding weight of soil plug inside skirts) under undrained conditions;

F is the correction factor given as a function of $\kappa B'/s_{uo}$;

κ is the rate of increase of undrained shear strength with depth;

s_{uo} is the undrained shear strength of soil at foundation base level (see 7.7);

N_c is a dimensionless constant equal to 5.14;

B' is the minimum effective lateral foundation dimension;

A' is the effective area of the foundation depending on the load eccentricity;

K_c is the correction factor which accounts for load inclination, footing shape, depth of embedment, inclination of base, and inclination of the seafloor surface.

This method is based on assessment of bearing capacity for surface foundations, or skirted foundations where the weight of the soil plug is balanced by the weight of soil overburden outside the skirts. Where this is not the case, the design load should be appropriately adjusted as outlined in 7.3.3.

A method for determining the correction factors and the effective area is given in Annex A.

7.6 Drained bearing capacity

The maximum total vertical load, Q_d', which a footing can support at its base under drained conditions is

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$$Q_d' = \{p'_o (N_q - 1) K_q + 0.5\gamma' B' N_\gamma K_\gamma\} A' \quad (6)$$

where

Q_d' is the maximum total vertical load applied to the base of the footing at failure (excluding weight of soil plug inside skirts) under drained conditions;

N_q is (exp [π tan ϕ']) (tan^2(45° + ϕ'/2)), a dimensionless function of ϕ';

N_γ is an empirical dimensionless function of ϕ' that can be approximated by 1.5(N_q – 1) tan ϕ';

ϕ' is the effective friction angle of Mohr envelope (see 7.7);

γ' is the effective unit weight of soil;

p'_o is the vertical effective stress at base level (skirt tip level when skirts are used);

B' is the minimum effective lateral foundation dimension;

A' is the effective area of the foundation depending on the load eccentricity;

K_q, K_γ are the correction factors which account for load inclination, footing shape, depth of embedment, inclination of base, and inclination of the seafloor. The subscripts q and γ refer to the particular term in the equation.

This method is based on assessment of bearing capacity for surface foundations, or skirted foundations where the weight of the soil plug is balanced by the weight of soil overburden outside the skirts. Where this is not the case, the design load should be appropriately adjusted as outlined in 7.3.3.

Equation 6 for drained bearing capacity has deliberately omitted any component due to an effective cohesion, c', and accompanying bearing capacity factor, N_c. This is mainly because the occasions when it might be appropriate to include a component for bearing capacity due to a presumed effective cohesion are extremely rare, and specialist geotechnical advice shall be sought before any such inclusion. More advice is given in Annex A.

A complete description of the K factors, as well as curves showing the numerical values of N_q and N_γ as a function of ϕ' are given in Annex A.

Two special cases of Equation 6 are frequently encountered. For a vertical central load applied to a foundation at seafloor level where both the foundation base and seafloor are horizontal, Equation 6 is reduced as follows for two foundation shapes.

a) Infinitely long strip footing:

$$Q_o = 0.5\, \gamma' B N_\gamma A_o \quad (7)$$

where

Q_o is the maximum vertical load, per unit length;

B is the minimum lateral foundation dimension.

b) Circular or square footing:

$$Q_d' = 0.3\, \gamma' B N_\gamma A \quad (8)$$

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7.7 Shear strength used in bearing capacity calculations

The design of shallow foundations is strongly dependent on the quality of the site investigation performed, and the methods used for determining characteristic strength and deformation characteristics, both in situ and in the laboratory. In particular, the uncertainty in determining the characteristic shear strength can be large and either prevent optimization or result in lower than intended safety margins.

Such sources of uncertainty are:

— imprecise definition of the soil strength parameters used in the analysis methods;

— variability in strength measurements (which depends on the scope and content of the soil investigation, sampling disturbance, methods of testing, etc.);

— strategies, methods, and procedures used for assessment of the characteristic strength, which vary from one designer or design environment to another;

— amount of testing forming the basis for proposed characteristic shear strengths, statistical variability, and bias.

These uncertainties are relevant for both drained and undrained shear strengths but special care should be taken when the undrained shear strength of sand is applied in the design calculations.

The following general practices are recommended.

a) For strongly dilatant soils, like dense sand or hard clays, high undrained shear strengths may be used in design only if the potential loss of dilatancy on shear surfaces has been explicitly considered.

b) For very soft and soft clays, results from unconsolidated undrained triaxial tests should be used with care. Unconfined compression tests should generally not be used without supplementary strength test data. In very soft and soft silty clays, neither unconsolidated undrained triaxial tests nor unconfined compression tests are reliable and should not generally be used. Consolidated undrained triaxial tests with pore pressure measurement, simple shear tests, *in situ* vane, and penetrometer tests (where the correlation between penetration resistance and soil strength is known for a particular soil) are more reliable techniques for determining undrained shear strength of soft clays and silty clays.

c) Soils display undrained shear strength anisotropy, thus triaxial compression, triaxial extension or simple shear strengths can be different. Care should be taken to adopt an appropriate strength in assessing foundation capacity, and any assumptions made in this regard should be clearly documented.

d) When undertaking drained bearing capacity calculations for sands using the method outlined in 7.6, the angle of friction (ϕ') should be based on triaxial test conditions. The friction angle used should be derived based on tests performed at appropriate stress levels.

e) Foundation stability under cyclic loading conditions can be assessed using pseudo-static analysis provided appropriately derived cyclic shear strengths are used.

f) In many instances cyclic performance of non-cohesive soils may be assessed using cyclic undrained soil strength derived in a manner similar to that used for cohesive soils. This should take into account the potential for dissipation of excess pore pressure to occur over the duration of cyclic loading.

g) Rate effects should be considered when assessing foundation response to rapid loading events.

h) Where applicable the effect of soil consolidation on strength may be considered in design. This will typically increase overall foundation capacity.

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7.8 Response of shallow foundations to static and pseudo-static loading

The maximum foundation displacement under static or pseudo-static loading may affect the structural integrity of the platform, its serviceability, and its components. Equations for evaluating the static short-term and long-term displacement of shallow foundations are given below. These equations are applicable to idealized conditions, and a discussion of the limitations is given in Annex A.

7.8.1 Short-term displacement (undrained loading)

For foundation materials which can be assumed to be isotropic and homogeneous; for the condition where the structure base is circular, rigid, and rests on the soil surface; and where the anticipated displacements are elastic, the deformations of the base under various loads may be estimated as follows:

Vertical: $$u_{\mathrm{v}} = \left(\frac{1-v}{4GR}\right)Q \tag{9}$$

Horizontal: $$u_{\mathrm{h}} = \left(\frac{7-8v}{32(1-v)GR}\right)H \tag{10}$$

Rocking: $$\theta_{\mathrm{r}} = \left(\frac{3(1-v)}{8GR^3}\right)M \tag{11}$$

Torsion: $$\theta_{\mathrm{t}} = \left(\frac{3}{16GR^3}\right)T \tag{12}$$

where

u_{v}, is the vertical displacement;

u_{h} is the horizontal displacement;

θ_{r} is the overturning rotation;

θ_{t} is the torsional rotation;

Q is the vertical load;

H is the horizontal load;

M is the overturning moment;

T is the torsional moment;

G is the elastic shear modulus of the soil;

v is Poisson's ratio of the soil;

R is the radius of the base.

These solutions can also be used for approximating the response of a square base of equal area.

Equations for rigid, embedded circular foundations are provided in Reference [67] and [68].

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Solutions also exist that account for non-uniform soil profiles (e.g. linearly increasing soil strength), general embedded foundations, flexible foundations, and non-uniform base geometries. In addition, consideration should be given to use of numerical analysis methods for more complex situations.

Note that G is not a unique soil parameter and depends on the level of strain applied to each soil element. Care should be taken to select an appropriate (representative) value in design using Equation 9 through Equation 12. Any assumptions made in this regard shall be clearly documented.

7.8.2 Long-term displacement (primary settlement)

An estimate of the vertical settlement of a soil layer under an imposed vertical load can be determined by the following equation:

$$u_{\mathrm{v}} = \left(\frac{hC}{1+e_{\mathrm{o}}} \right) \log_{10} \frac{q_{\mathrm{o}} + \Delta q}{q_{\mathrm{o}}} \tag{13}$$

where

u_{v} is the vertical settlement;

h is the layer thickness;

e_{o} is the initial void ratio of the soil;

C is the compression index of the soil over the load range considered;

q_{o} is the initial effective vertical stress;

Δq is the added effective vertical stress.

Where the vertical stress varies within a thin layer, as in the case of a diminishing stress with depth, estimates may be determined by using the stress at the midpoint of the layer. Thick homogeneous layers should be subdivided for analysis. Where more than one layer is involved, the estimate is simply the sum of the settlement of the layers. Compression characteristics of the soil are determined from appropriate consolidation tests.

7.8.3 Long-term displacement (secondary settlement)

Depending on the duration of loading, additional displacement due to secondary compression (creep) may also need to be assessed.

7.8.4 Long-term displacement (regional)

Additional settlement may occur in the long-term due to a global lowering of the seafloor due to oil and gas extraction (see 6.3.7). This settlement component is relative to sea (not seafloor) level, and is not significant for temporary foundations.

7.9 Response of shallow foundations to environmental loading

Offshore foundations are subject to environmental loading from current, waves, ice, wind, and earthquakes. In addition, loads may be applied by thermal loading. Both the influence of the foundation on the structural response and the integrity of the foundation itself shall be considered.

In some instances environmental loads may be considered as pseudo-static loads, and assessed using the calculations outlined in 7.8. Examples of where this may be appropriate include current loading.

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However, other loading types are cyclic in nature and require more detailed methods of analysis. For example, cyclic wave loading of offshore structures may result in large average and cyclic foundation displacements, as well as volumetric strain after completion of cyclic loading. Specialist advice shall be sought when such analysis is required for foundation design.

7.10 Hydraulic stability

7.10.1 Scour

Positive measures should be taken to avoid erosion and scour of the soil beneath or near the structure base. Examples of such measures are

— scour skirts penetrating through erodible layers into scour resistant materials or deep enough to eliminate the scour hazard, or

— scour protection placed around the edges of the foundation.

Sediment transport studies may be carried out to assess the scour potential of the seabed material around the foundation.

7.10.2 Piping

The foundation should be designed to prevent excessive hydraulic gradients (piping conditions) in the soil due to environmental loadings or operations performed during or subsequent to structure installation.

7.11 Installation and removal

Penetration of foundation skirts into the seabed may be facilitated by providing under-pressure (relative to the ambient hydrostatic pressure) inside the skirt compartments under the foundation. In these cases, installation procedures shall be planned to avoid damage to the soil, including plug uplift, erosion, and piping. If removal is anticipated, an analysis should be made of the actions generated during removal to ensure they can be accomplished with the means available.

7.12 Shallow foundations equipped with seabed penetrating skirts

For many situations, skirts around the periphery improve the performance of shallow foundations. These skirts penetrate the seabed. If the foundation area is large, interior skirts should also be provided to form skirt compartments under the plate foundation. The presence of the skirts will in most cases

a) increase foundation resistance to overturning moment, bearing load, sliding load and torsion load,

b) decrease the vertical and horizontal displacements and rotations, and

c) improve erosion resistance under the foundation.

In addition to assessing foundation stability, installation studies should be performed to ensure that the skirts can fully penetrate to their intended depth below seafloor. For assessment of bearing capacity, the foundation level and depth of embedment may be considered to be the depth to which the skirts penetrate, provided that it can be shown that governing failure modes do not exist within the skirt compartments.

For assessment of sliding capacity, additional capacity under pure sliding conditions may come from the soil strength at skirt tip level (s_{uo}) being greater than that at seafloor level. Lateral resistance can also be provided above skirt tip level by embedded members, such as through lateral (active and passive) resistance and side shear. This is discussed further in Annex A. Care must be taken to ensure internal mechanisms (within the soil plug for skirted foundations) do not lead to lower overall capacity; while care should also be taken in conditions where installation disturbance or soil conditions may lead to lower resistance above skirt tip level (e.g. in conditions where soil tension cracking on the active side of the skirt may occur).

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In addition, skirts may significantly increase the ability of the shallow foundation to sustain short term tensile total stresses (relative to ambient water pressure) beneath the foundation, sustained by negative excess pore pressures within the soil. Cyclic tensile stresses (again, relative to ambient water pressure) from waves with a few seconds duration can often be demonstrated as acceptable in design; while longer duration tensile stresses may potentially be carried by skirted foundations on clays with low permeability. Confirmation that tensile stresses can be sustained for a given design should be performed on a case by case basis by specialist geotechnical engineers. The ability of the soil to withstand tensile stresses has been discussed further in 7.3.

Where foundation loads are not provided at the level of the skirt tip(s), but stability analysis is to be performed at this level, care should be taken to appropriately transfer loads to skirt tip. Additional guidance is provided in Annex A.

7.13 Shallow foundations without seabed penetrating skirts

Foundations without skirts, which remain at the seafloor, can require special consideration. In particular, the consequences of loss of contact between the foundation and the seafloor shall be assessed.

7.14 Installation effects

The shallow foundations considered herein are often placed on the seafloor from an installation vessel. The vertical heave motions induced by the vessel's motion characteristics and created by the environmental boundary conditions cause the touchdown to be an impact between the seafloor and the structure. This impact is normally controlled by limiting weather criteria and carefully planned and well controlled installation operations. However, it is still the experience that many small structures suffer a foundation failure during installation, mainly in soft soil conditions. Many such incidents can be avoided by observing the following recommendations.

a) If the installation is performed under controlled conditions by use of a heave compensator and low rate of descent towards the seafloor (< 0.2 m/s), no extra safety margin is needed.

b) If the installation is performed without any control on the rate of descent (no heave compensation), penetration in excess of the failure displacement can occur. The consequences of such additional penetration should be investigated. Further guidance on this problem is given in Annex A.

7.15 Sliding stability

7.15.1 General

In assessment of sliding stability due consideration shall be given during site investigation and subsequent interpretation to the possible occurrence of discrete layers of low strength soil, which may provide a preferential failure surface.

7.15.2 Surface foundations

When the vertical load capacity has been established by equations in 7.4 to 7.6, the maximum horizontal load should be limited to the capacity determined for the extreme condition of pure sliding as defined by the following equations.

a) Undrained case:

$$H_d = s_{uo} A \tag{14}$$

where

H_d is the maximum total horizontal load applied to the base of the foundation at failure under undrained conditions;

s_{uo} is the shear strength at base level.

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b) Drained case

$$H_d' = Q \tan \phi' \quad (15)$$

where

H_d' is the maximum total horizontal load applied to the base of the foundation at failure under drained conditions;

Q is the actual vertical load acting during the relevant loading condition.

In Equations 14 and 15, it is assumed that full soil resistance may be mobilized along the interface between the foundation and the soil (i.e. full soil-soil contact is assumed). This should be assessed on a case by case basis.

For drained conditions it may be more appropriate to consider use of a different interface friction angle (δ) between the foundation soil and the structure. In addition, sliding along a thin weak layer of low permeability at shallow depth below the foundation base within the sand mass may need to be assessed.

For undrained cases it may be appropriate to use a soil adhesion coefficient to reduce the undrained strength at foundation level. Alternatively, in cases where an undrained response is anticipated in the bulk of the soil mass it may nevertheless be appropriate to account for a drained interface between the foundation base plate and soil. In addition, the possibility of sliding along a sand seam within a competent clay layer should be considered.

7.16 Torsional stability

Torsional stability requires consideration, and additional analysis shall be performed if torsion loads are significant.

Torsional loads decrease the overall bearing capacity of shallow foundations. Torsional loads may also act to reduce sliding capacity of shallow foundations. Correction factors that account for torsional loads are not available for use with the bearing capacity methods in 7.4 to 7.6, or the assessment of pure sliding in 7.15, and specialist advice shall be sought where this requires consideration.

In assessment of torsional stability due consideration should be given during site investigation and interpretation to the possible occurrence of discrete layers of low strength along which displacements could preferentially occur.

8 Pile foundation design

Recommended design criteria for pile foundations can be found in API 2A-WSD.

8.1 Pile capacity for axial compression

8.1.1 General

Pile capacity for axial compression, as discussed in 8.1.2 through 8.1.5, relates to the axial resistance of a pile when the pile head is subjected to compressive loads along the pile axis. Pile capacity for axial tension is addressed in 8.2.

Pile capacities are commonly determined using the simplified calculation model described in 8.1.2; the parameters that are used in this model are determined in accordance with 8.1.3 through 8.1.5. This simplified model has been developed and applied in many years of offshore practice and represents the current industry standard. However, the model does not provide any information about axial pile displacements which are important for serviceability requirements, especially in non-extreme conditions for loads due to permanent, variable and operating environmental loads that are generally well below the design loads. Axial pile behavior aimed at meeting service requirements is referred to as axial pile performance and is discussed in 8.3. Methods for determining pile performance are described in C.8.6.2 and C.8.6.3.

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The simplified model for pile capacity presented in 8.1.2 is based on a (quasi-)static and monotonic application of the axial loads and has no ability to reflect the complex occurrences that take place in the interaction between pile and soil during field conditions. To enhance understanding of the limitations of the model and to assist in applying engineering judgment to its results, it is useful to gain a better insight in the actual occurrences through the investigation of pile performance (see 8.3).

The relationships between mobilized axial shear transfer between pile and soil and the local pile displacement, and between mobilized end bearing resistance and the pile tip displacement can be determined using 8.4.

8.1.2 Ultimate axial pile capacity

The ultimate axial capacity of piles in compression, including belled piles, Q_c, should be determined by the equation:

$$Q_c = Q_{f,c} + Q_p = f(z)\, A_s + q\, A_p \tag{16}$$

where

$Q_{f,c}$ is the shaft friction capacity in compression, in force units;

Q_p is the end bearing capacity, in force units;

$f(z)$ is the unit shaft friction, in stress units;

A_s is the side surface area of pile;

q is the unit end bearing at the pile tip, in stress units;

A_p is the gross end area of the pile;

z is the depth below the original seafloor.

For open-ended pipe piles, the total end bearing capacity, Q_p, shall not exceed the sum of the end bearing capacity of the internal plug and the end bearing on the pile tip wall annulus. In computing the design loads in compression on the pile, the weight of the pile shall be considered.

In determining the capacity of a pile, consideration shall be given to the relative deformations between the soil and the pile as well as to the compressibility of the soil-pile system. In some circumstances, a more explicit consideration of axial pile performance effects on pile capacity is warranted. Additional discussion of these effects is given in 8.3 and C.8.3.

The foundation configurations should be based on those that experience has shown can be installed consistently, practically and economically under similar conditions with the pile size and installation equipment being used. Alternatives for possible remedial action in the event that design objectives cannot be obtained during installation should be investigated and defined prior to construction.

In the case of belled piles, the shaft friction values on the pile section should be those given in 8.1 and 8.2. Shaft friction on the upper bell surface and, possibly, on the pile for some distance above the bell should be discounted in computing the shaft friction resistance, $Q_{f,c}$ The end bearing area of a pilot hole, if drilled, should also be discounted in computing the total bearing area of the bell.

8.1.3 Shaft friction and end bearing in cohesive soils

There are a number of methods for calculating the shaft friction and end bearing in cohesive soils. The method described below has been developed and applied over many years and is the current industry standard. However, caution should be exercised in its application as there are many more variables which affect pile capacity than

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those included in the design equations. This matter is discussed below and in C.8.1.3. For pipe piles in cohesive soils, the unit shaft friction, $f(z)$, in stress units, at depth, z, can be calculated using Equation 17.

$$f(z) = \alpha\, s_u \tag{17}$$

where

α is the dimensionless shaft friction factor, for clays;

s_u is the undrained shear strength of the soil at the point in question, in stress units.

The factor α can be computed by Equation 18:

$$\alpha = 0.5\,\psi^{-0.5} \text{ for } \psi \le 1.0 \tag{18}$$

$$\alpha = 0.5\,\psi^{-0.25} \text{ for } \psi > 1.0$$

with the constraint that $\alpha \le 1{,}0$

where

$$\psi = \frac{s_u}{p'_o(z)} \text{ at depth, } z; \tag{19}$$

$p'_o(z)$ = effective vertical stress at depth z.

A discussion of appropriate methods for determining the undrained shear strength, s_u, and effective overburden stress, $p'_o(z)$, including the effects of various sampling and testing procedures is included in C.8.1.3. For underconsolidated clays (clays with excess pore pressures undergoing active consolidation), α can usually be taken as 1.0.

Due to the shortage of pile load tests in soils having $s_u/p'_o(z)$ ratios greater than three, Equation 18 should be applied with considerable care for high $s_u/p'_o(z)$ values. Similar judgment should be applied for deep penetrating piles in soils with high undrained shear strength. Low plasticity clays should be treated with particular caution, see C.8.1.3.

For very long piles some reduction in capacity may be warranted, particularly where the shaft friction degrades on continued displacement. This effect is discussed in more detail in C.8.1.3.

Where the pile tip is in cohesive soils, the unit end bearing, q, in stress units, may be computed using Equation 20.

$$q = 9\, s_u \tag{20}$$

The shaft friction, $f(z)$, acts on both the inside and the outside of the pile. The total axial resistance for pile compression is the sum of the external shaft friction, the end bearing on the pile wall annulus, and the total internal shaft friction or the end bearing of the plug, whichever is the lesser. For piles considered to be plugged, the bearing pressure may be assumed to act over the entire cross-section of the pile. For unplugged piles, the bearing pressure acts on the pile wall annulus only. That a pile is considered plugged or unplugged shall be based on static calculations. A pile can be driven in an unplugged condition but behave as plugged under static loads

Shaft friction resistance and end bearing capacity computed on the basis of the requirements above represent long-term capacities. Axial capacity immediately after installation is usually lower, especially in underconsolidated to slightly overconsolidated clays. This is dependent on the development of excess pore pressure in the soil during installation and its subsequent dissipation with time. When the design loads are applied to a pile foundation

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shortly after installation, the capacity of a pile immediately after installation and the increase in capacity with time are important design considerations. More discussion on the soil–pile set-up behavior is provided in C.8.1.3.

For piles driven in undersized drilled holes, piles jetted in place or piles drilled and grouted in place, the selection of shaft friction values shall take into account the soil disturbance resulting from installation. In general, $f(z)$ shall not exceed values for driven piles; however, in some cases, for drilled and grouted piles in overconsolidated clay, $f(z)$ can exceed these values. In determining $f(z)$ for drilled and grouted piles, the strength of the soil-grout interface, including potential effects of drilling mud, shall be considered. A further check shall be made of the allowable bond stress between the pile steel and the grout as recommended in API 2A-WSD, Section 10.4.4.

In layered soils, shaft friction values, $f(z)$, in the cohesive layers shall be as given by Equation 17 through Equation 19. End bearing values for piles tipped in cohesive layers with adjacent weaker layers may be taken as given in Equation 20 provided that

a) the pile achieves a penetration of two to three diameters or more into the layer in question, and

c) the tip is approximately three diameters or more above the bottom of the layer to preclude punch through.

Where these distances are not achieved, some modification of the end bearing is usually necessary. Where adjacent layers are of comparable strength to the layer of interest, the proximity of the pile tip to the interface is not a concern.

8.1.4 Shaft friction and end bearing in cohesionless soils

This clause provides a simple method for assessing pile capacity in cohesionless soils. Clause C.8.1.4 presents other, recent and more reliable methods. These are based on direct correlations of pile unit friction and end bearing data with cone penetration test (CPT) results. In comparison to the Main Text method described below, these CPT-based methods are considered fundamentally better, have shown statistically closer predictions of pile load test results and, although not required, are in principle the preferred methods. These methods also cover a wider range of cohesionless soils than the Main Text method. However, offshore experience with these CPT methods is limited and hence more experience is needed before they are recommended for routine design, instead of the main text method. CPT-based methods should be applied only by qualified engineers who are experienced in the interpretation of CPT data and understand the limitations and reliability of these methods. Following installation, pile driving (instrumentation) data may be used to give more confidence in predicted capacities.

For pipe piles in cohesionless soils, the unit shaft friction at a given depth, $f(z)$, in stress units, may be calculated by Equation 21.

$$f(z) = \beta\, p'_o(z) \tag{21}$$

where:

β is the dimensionless shaft friction factor, for sands;

$p'_o(z)$ is the effective vertical stress at depth z.

In the absence of specific data, β values for open-ended pipe piles that are driven unplugged may be taken from Table 1. For full displacement piles (i.e. closed-ended or fully plugged open-ended piles) values of β may be assumed to be 25 % higher than those given in Table 1. For long piles, $f(z)$, does not necessarily increase linearly with the overburden stress as implied by Equation 21. In such cases, it may be appropriate to limit f to the values given in Table 1.

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Table 1—Design parameters for cohesionless siliceous soil

Relative Density[a]	Soil Description	Shaft Friction Factor[b] β (-)	Limiting Shaft Friction Values kPa (kips/ft^2)	End Bearing Factor N_q (-)	Limiting Unit End Bearing Values MPa (kips/ft^2)
Very loose Loose Loose Medium dense Dense	Sand Sand Sand-silt[c] Silt Silt	Not applicable[d]	Not applicable[d]	Not applicable[d]	Not applicable[d]
Medium dense	Sand-silt[c]	0.29	67 (1.4)	12	3 (60)
Medium dense Dense	Sand Sand-silt[c]	0.37	81 (1.7)	20	5 (100)
Dense Very dense	Sand Sand-silt[c]	0.46	96 (2.0)	40	10 (200)
Very dense	Sand	0.56	115 (2.4)	50	12 (250)

NOTE The parameters listed in this table are intended as guidelines only. Where detailed information, such as CPT records, strength tests on high quality samples, model tests, or pile driving performance, is available, other values may be justified.

[a] The definitions for the relative density percentage description are as follows:

- Very loose, 0 – 15;
- Loose, 15 – 35;
- Medium dense, 35 – 65;
- Dense, 65 – 85;
- Very dense, 85 – 100.

[b] The shaft friction factor β (equivalent to the "K tan δ" term used in previous editions of API 2A-WSD) is introduced in this document to avoid confusion with the δ parameter used in the Annex.

[c] Sand-silt includes those soils with significant fractions of both sand and silt. Strength values generally increase with increasing sand fractions and decrease with increasing silt fractions.

[d] Design parameters given in previous editions of API 2A-WSD for these soil/relative density combinations may be unconservative. Hence, it is recommended to use CPT-based methods from the annex for these soils.

For end bearing of piles in cohesionless soils, the unit end bearing, q, in stress units, may be computed using Equation 22.

$$q = N_q \, p'_{o,\,tip} \tag{22}$$

where

$p'_{o,\,tip}$ is the effective vertical stress at the pile tip;

N_q is the dimensionless bearing capacity factor.

Recommended N_q values are presented in Table 1. For long piles, q does not necessarily increase linearly with the overburden stress as implied by Equation 22. In such cases, it is appropriate to limit q to the values given in Table 1. For plugged piles, the bearing pressure may be assumed to act over the entire cross-section of the pile. For unplugged piles, the bearing pressure acts on the pile annulus only. In this case, additional resistance is offered by friction between the soil plug and the inner pile wall. Whether a pile is considered to be plugged or unplugged shall be based on static calculations using a unit shaft friction on the soil plug equal to the outer shaft friction. Note that a pile can be driven in an unplugged condition, but can behave as plugged under static loads.

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Load test data for piles in cohesionless soils (see Reference [69] and [70]) indicate that variability in capacity predictions using the simple method described herein can exceed those for piles in clay. These data also indicate that the above method is conservative for short offshore piles (< 45 m, [150 ft]) in dense to very dense sands loaded in compression and can be unconservative in all other conditions. Therefore, in unfamiliar situations, the designer should account for this uncertainty through the selection of conservative design parameters and/or higher safety factors. This is especially important where force redistributes after the development of maximum resistance occurs, which can lead to an abrupt (brittle) failure — such as is the case for short piles in tension.

For soils that do not fall within the ranges of relative density and soil description given in Table 1, or for materials with unusually weak grains or compressible structure, Table 1 is not necessarily appropriate for selection of design parameters. For example, very loose soils or soils containing large amounts of mica or volcanic grains may require special laboratory or field tests for selection of design parameters. Of particular importance are sands containing calcium carbonate (see A.6.4.4), which are found extensively in many areas of the oceans.

For piles driven in undersized drilled or jetted holes in cohesionless soils, the values of $f(z)$ and q should be determined by some reliable method that accounts for the amount of soil disturbance due to installation, but they shall not exceed the values for driven piles.

In layered soils, shaft friction values, $f(z)$, in cohesionless layers should be computed according to Table 1. End bearing values for piles tipped in cohesionless layers with adjacent soft layers may also be taken from Table 1, provided that

a) the pile achieves penetration of two to three diameters or more into the cohesionless layer, and

b) the tip is at least three diameters above the bottom of the layer to preclude punch-through.

Where these pile tip penetrations are not achieved, some modification in the tabulated values may be necessary. Where adjacent layers are of comparable strength to the layer of interest, the proximity of the pile tip to the layer interface is not a concern.

8.1.5 Shaft friction and end bearing of grouted piles in rock

The unit shaft friction of grouted piles in jetted or drilled holes in rock shall not exceed half the uniaxial compressive strength of the rock or grout, but in general should be much less than this value. The reduction depends on pile construction factors (such as roughness on the side of the hole) and on rock mass factors (such as the presence of discontinuities). The sidewall of the hole can develop a layer of slaked mud or clay, which will never gain the strength of the rock. The bond stress between the pile and the grout shall be checked in accordance with API 2A-WSD.

The end bearing capacity of the rock shall not exceed the uniaxial compressive strength of the rock or grout multiplied by a bearing capacity factor appropriate for the type of rock. In general, the end bearing capacity should be much less or ignored in the design, depending on pile construction factors, such as degree of removal of drill cuttings from the base of the hole, and on rock mass factors, such as the presence of discontinuities within the rock mass. The limiting end bearing capacity for this type of pile can be governed by stresses in the grout or in the pile steel.

Design values for (static) unit shaft friction and end bearing can be found in various publications (see References [139], [146], and [147]). It is noted that most publications on this subject refer to relatively “stubby” stiff piles as used in onshore practice (bored piles). Owing to the brittle response applicable to unit shaft friction, design values given in these publications can be unconservative for long flexible piles as used in offshore practice. In addition, consideration shall be given to the fact that cyclic loads will adversely affect the axial capacity of such piles.

8.2 Pile capacity for axial pullout loads

The pile axial capacity in tension, Q_t, is less than or equal to, but shall not exceed $Q_{f,c}$, the total shaft friction capacity in compression. In computing the tensile design load on the pile, the weight of the pile shall be considered; the weight of the soil plug may also be considered, if this can be justified. For cohesive soils, $f(z)$ shall

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be the same as stated in 8.1.3. For cohesionless soils, $f(z)$ shall be computed according to 8.1.4. For rock, $f(z)$ shall be the same as stated in 8.1.5.

8.3 Axial pile performance

8.3.1 Static axial behavior of piles

Pile response is affected by load directions, load types, load rates, loading sequence, installation technique, soil type, axial pile stiffness, and other parameters.

Some of these effects for cohesive soils have been observed in both laboratory and field tests.

In some circumstances, e. g. for soils that exhibit strain-softening behavior and/or where the piles are axially flexible, the actual capacity of the pile may be less than that given by Equation 16. In these cases an explicit consideration of these effects on ultimate axial capacity may be warranted. Note that other factors such as increased axial capacity under loading rates associated with storm waves may counteract the above effects. For more information see the commentary in API 2A-WSD as well as C.8.3.2 and Reference [204].

8.3.2 Cyclic axial behavior of piles

Cyclic loads (including inertial loads) due to environmental conditions such as storm waves and earthquakes can have two potentially counteractive effects on the static axial capacity. Repetitive loads can cause a temporary or permanent decrease in resistance and/or an accumulation of deformation. Rapidly applied loads can cause an increase in resistance and/or stiffness of the pile. Very slowly applied loads can cause a decrease in resistance and/or stiffness of the pile. The resultant influence of cyclic loads will be a function of the combined effects of the magnitudes, cycles, and rates of change of applied loads, the structural characteristics of the pile and the types of soils, see C.8.3.2.

The pile–soil, resistance–displacement, t-z and Q-z, characterizations are discussed in 8.4.

8.4 Soil reaction for piles under axial compression

8.4.1 General

The relationship between mobilized soil-pile shear transfer and local pile displacement at any depth is described using a t-z curve. Similarly, the relationship between mobilized end bearing resistance and axial pile tip displacement is described using a Q-z curve.

8.4.2 Axial shear transfer t-z curves

Various empirical and theoretical methods are available for developing curves for axial shear transfer and pile displacement, t–z curves.

Theoretical curves described in Reference [71] may be constructed. Empirical t-z curves based on the results of model and full-scale pile load tests may follow the procedures described in Reference [72] in clay soils or described in Reference [73] for granular soils. Additional curves for clays and sands are provided in Reference [74].

Load deflection relationships for grouted piles are discussed in Reference [75].

Curves developed from pile load tests in representative soil profiles or based on laboratory soil tests that model pile installation can also be justified. In the absence of more definitive criteria, the t–z curves in Figure 2 are recommended for non-carbonate soils. The definitions for these recommended curves are shown in Table 2.

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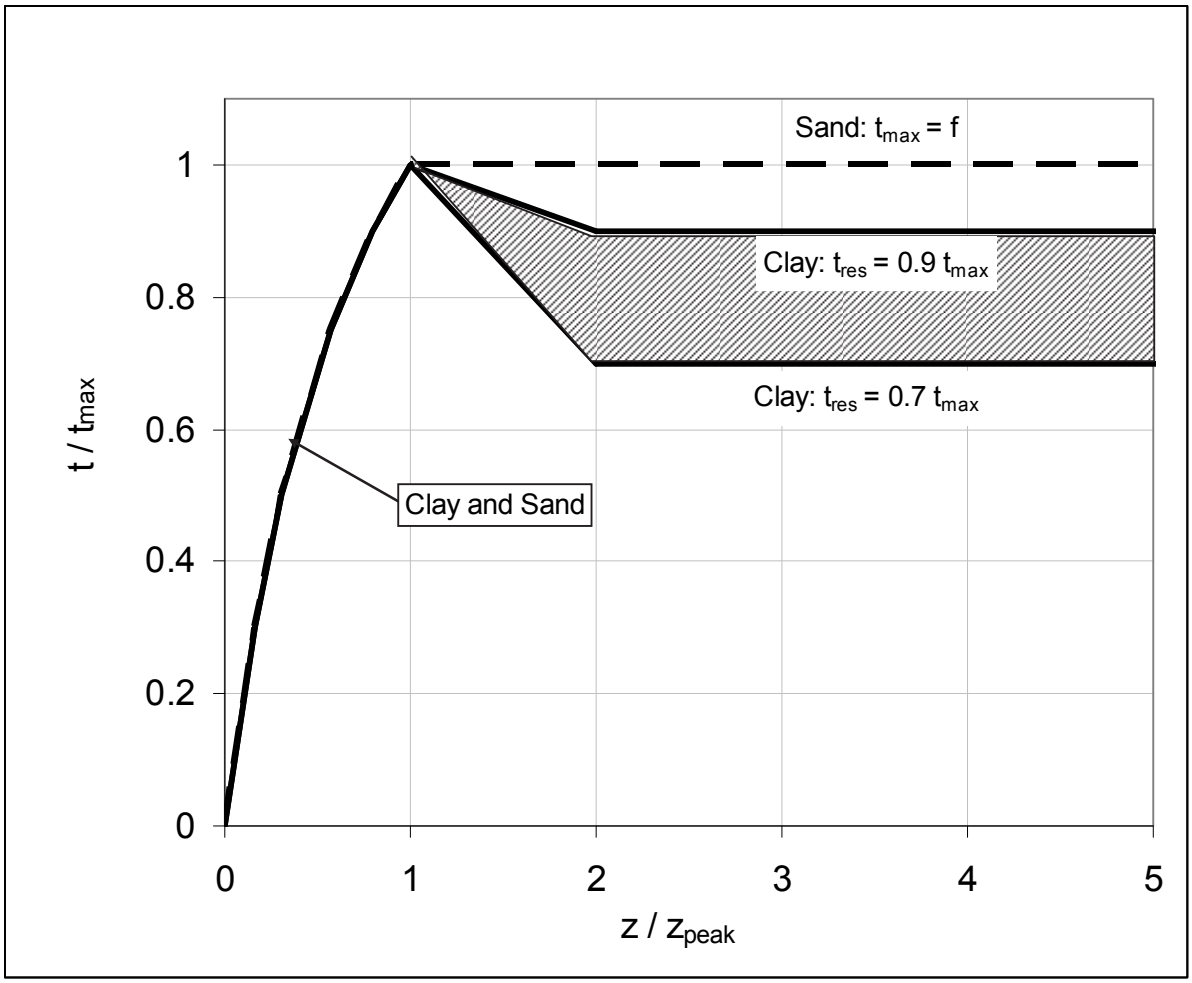

Key

z is the local pile axial deflection;

D is the pile outside diameter;

t is the mobilized soil-pile adhesion;

t_{max} is the maximum soil pile adhesion or unit shaft friction computed in accordance with 8.1;

t_{res} is the residual soil-pile adhesion or unit shaft friction;

z_{res} is the axial pile displacement at which the residual soil-pile adhesion, t_{res}, is reached.

Figure 2—Typical axial pile load transfer—Displacement (t-z) curves

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Table 2—Definition of *t-z* curves

z/z_{peak}	t/t_{max}	
	Clays	Sands
0.16	0.30	0.30
0.31	0.50	0.50
0.57	0.75	0.75
0.80	0.90	0.90
1.0	1.00	1.00
2.0	0.70 to 0.90	1.00
∞	0.70 to 0.90	1.00

Key

z is the local pile axial deflection;

z_{peak} is the displacement to maximum soil pile adhesion or unit skin friction;

D is the outside diameter;

t is the mobilized soil-pile adhesion;

t_{max} = $f(z)$ = maximum soil-pile adhesion or unit shaft friction, computed according to 8.1.

A typical value for z_{peak} of one per cent of the pile outer diameter (i.e. z_{peak}/D = 0.01) is recommended for routine design purposes. However there is significant uncertainty on this value and values ranging from 0.25 % to 2.0 % of the pile diameter may be considered in cases where axial pile stiffness is critical for design.

The shape of the *t-z* curve at displacements greater than z_{peak} as shown in Figure 2 should be carefully considered. Values of the residual adhesion ratio t_{res} / t_{max} and the axial pile displacement at which it occurs, z_{res}, are a function of soil stress-strain behavior, stress history, pile installation method, pile load sequence and other factors. The value of t_{res} / t_{max} for clays can range from 0.70 to 0.90. Laboratory, *in situ*, or model pile tests can provide valuable information for determining values of t_{res} / t_{max} and z_{res} for various soils.

8.4.3 End bearing resistance-displacement, *Q–z*, curve

The end bearing capacity should be determined as described in 8.1. However, relatively large pile tip movements are required to mobilize the full end bearing resistance. A pile tip displacement up to 10 percent of the pile diameter may be required for full mobilization in both sand and clay soils. In the absence of more definitive criteria the curve shown in Figure 3 is recommended for both sands and clays.

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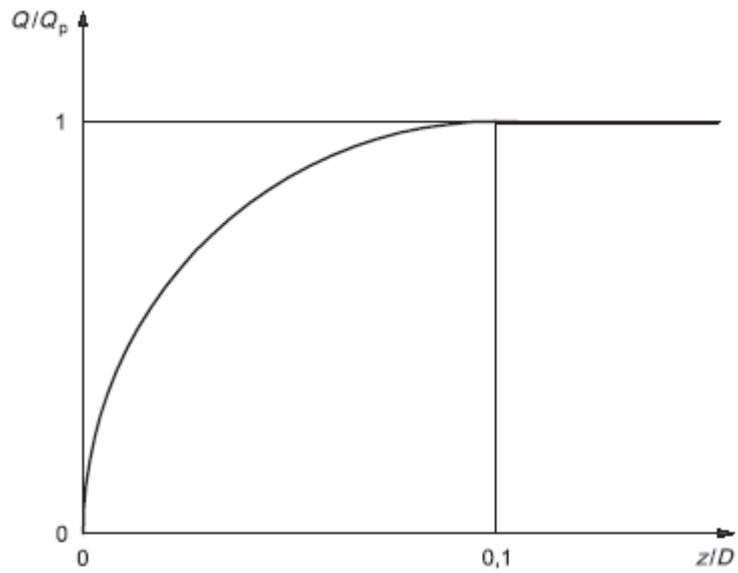

Key

z is the axial pile tip displacement;

z_{peak} is the displacement to maximum soil pile adhesion or unit skin friction;

Q is the mobilized end bearing capacity, in force units;

Q_p is the end bearing capacity, in force units, in accordance with 8.1.

z / D	Q / Q_p
0	0
0.002	0.25
0.013	0.50
0.042	0.75
0.073	0.90
0.100	1.00
∞	1.00

Figure 3—Pile end bearing capacity-displacement curve

8.5 Soil reaction for piles under lateral loads

8.5.1 General

The lateral resistance of the soil near the surface is significant to pile design, and the possible effects of scour on this resistance shall be considered. Estimates of these effects are discussed in C.8.5. In the absence of more definitive criteria, the procedures given in 8.5.2 to 8.5.7 may be used for constructing the relationships between lateral soil resistance and lateral displacement, p–y curves.

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8.5.2 Lateral capacity for soft clay

For static lateral loads, the ultimate unit lateral bearing capacity, p_uD, (in units of force per unit length) for soft clay (s_u ≤100 kPa or 2000 lb/ft^2) has been found to vary between 8 s_uD and 12 s_uD except at shallow depths where failure occurs in a different mode due to low overburden stress. Cyclic loads cause deterioration of lateral bearing capacity below that for static loads. In the absence of more definitive criteria, the following value of the lateral capacity shall be used:

p_uD increases from 3 s_uD to 9 s_uD as z increases from 0 to z_R according to Equation 23:

$$p_uD = 3\, s_uD + \gamma'\, z\, D + J\, s_u z \tag{23}$$

but p_uD is limited by Equation 24:

$$p_uD = 9\, s_uD \text{ for } z \geq z_R \tag{24}$$

where

p_uD is the ultimate resistance, units of force per unit length;

p_u is the ultimate resistance, units of pressure;

s_u is the undrained shear strength of the soil at the point in question, in stress units;

D is the pile outside diameter.

γ' is the submerged soil unit weight;

J is the dimensionless empirical constant with values ranging from 0.25 to 0.5 having been determined by field testing;

NOTE A value of 0.5 is appropriate for Gulf of Mexico clays.

z is the depth below the original seafloor;

z_R is the depth below soil surface to bottom of reduced resistance zone.

For a condition of constant strength with depth, Equation 23 and Equation 24 are solved simultaneously to give:

$$z_R = \frac{6D}{\dfrac{\gamma' D}{s_u} + J} \tag{25}$$

For non-uniform soils, Equation 23 and Equation 24 may be solved by plotting the two equations for p_uD vs. depth. The point of first intersection of the two equations is taken to be z_R. In general, z_R is in excess of 2.5 pile diameters. These empirical relationships do not necessarily apply where strength variations are erratic. These equations also do not apply in case of scour. Guidance for scour conditions is given in C.8.5.

8.5.3 Lateral soil resistance–displacement *p-y* curves for soft clay

Lateral soil resistance-displacement relationships for piles in soft clay are generally non-linear. The p-y curves for short-term static loads may be generated from Table 3. For the case where equilibrium has been reached under cyclic loads, the p–y curves may be generated from Table 4.

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Table 3—Mobilized lateral resistance – Displacement data for short-term static loads

p / p_u	y / y_c
0.00	0.0
0.23	0.1
0.33	0.3
0.50	1.0
0.72	3.0
1.00	8.0
1.00	∞

Key

p_u is the ultimate lateral capacity, unit of pressure;

p is the mobilized lateral resistance, unit of pressure;

y is the local pile lateral displacement;

y_C equals 2.5 × ε_c × D;

D is the pile diameter;

ε_c is the strain at one-half the maximum deviator stress in laboratory undrained compression tests of undisturbed soil samples.

Table 4—Mobilized lateral resistance–Displacement data for equilibrium conditions of cyclic loads

$z > z_R$		$z < z_R$	
p / p_u	y / y_c	p / p_u	y / y_c
0	0	0.00	0.0
0.23	0.1	0.23	0.1
0.33	0.3	0.33	0.3
0.50	1.0	0.50	1.0
0.72	3.0	0.72	3.0
0.72	∞	$0.72\, z / z_R$	15.0
		$0.72\, z / z_R$	∞

Key

Z is the depth below seafloor;

z_R is the depth below seafloor to bottom of reduced capacity zone for uniform soils (see Equation 25);

p_u is the ultimate lateral capacity, unit of pressure;

p is the mobilized lateral resistance, unit of pressure;

y is the local pile lateral displacement;

y_c equals 2.5 × εc ×D;

D is the pile diameter;

ε_c is the strain at one-half the maximum deviator stress in laboratory undrained compression tests of undisturbed soil samples.

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8.5.4 Lateral capacity for stiff clay

For static lateral loads, the ultimate bearing capacity (force per unit length, $p_u D$) of stiff clay (s_u > 100 kPa or 2000 lb/ft^2) is similar to that for soft clay. However, due to rapid deterioration under cyclic loading, the lateral resistance shall be reduced for cyclic design considerations in accordance with acceptable best practice or available data.

8.5.5 Lateral soil resistance–Displacement (*p-y*) curves for stiff clay

While stiff clays also have non-linear stress-strain relationships, they are generally more brittle than soft clays. In developing stress-strain curves and subsequent *p-y* curves for cyclic loads, good judgment should reflect the rapid deterioration of load capacity at large deflections for stiff clays in accordance with acceptable best practice or documented data.

8.5.6 Lateral capacity for sand

The ultimate lateral bearing capacity for sand has been found to vary from a value at shallow depths determined by Equation 26 to a value at deep depths determined by Equation 27. At a given depth the equation giving the smallest value of p_u should be used as the ultimate bearing capacity. These equations can be unconservative for layered soil conditions when the sand is overlain by soft clay.

$$p_{us} = (C_1 z + C_2 D)\,\gamma' z \tag{26}$$

$$p_{ud} = C_3 D\,\gamma' z \tag{27}$$

where

p_u is the ultimate resistance (force/unit length), (s = shallow, d = deep);

γ' is the submerged soil unit weight;

z is the depth below the original seafloor;

ϕ' is the angle of internal friction of sand;

D is the pile outside diameter;

C_1, C_2, C_3 are the coefficients determined as follows as function of ϕ';

$$C_1 = \frac{(\tan\beta)^2 \tan\alpha}{\tan(\beta-\phi')} + K_o \times \left[\frac{\tan\phi' \times \sin\beta}{\cos\alpha \times \tan(\beta-\phi')} + \tan\beta \times \left(\tan\phi' \times \sin\beta - \tan\alpha\right)\right];$$

$$C_2 = \frac{\tan\beta}{\tan(\beta-\phi')} - K_a;$$

$$C_3 = K_a \times \left[(\tan\beta)^8 - 1\right] + K_o \times \tan\phi' \times (\tan\beta)^4.$$

where

$$\alpha = \frac{\phi'}{2};$$

$$\beta = 45 + \frac{\phi'}{2};$$

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$K_o = 0.4$;

$$K_a = \frac{1 - \sin\phi'}{1 + \sin\phi'} .$$

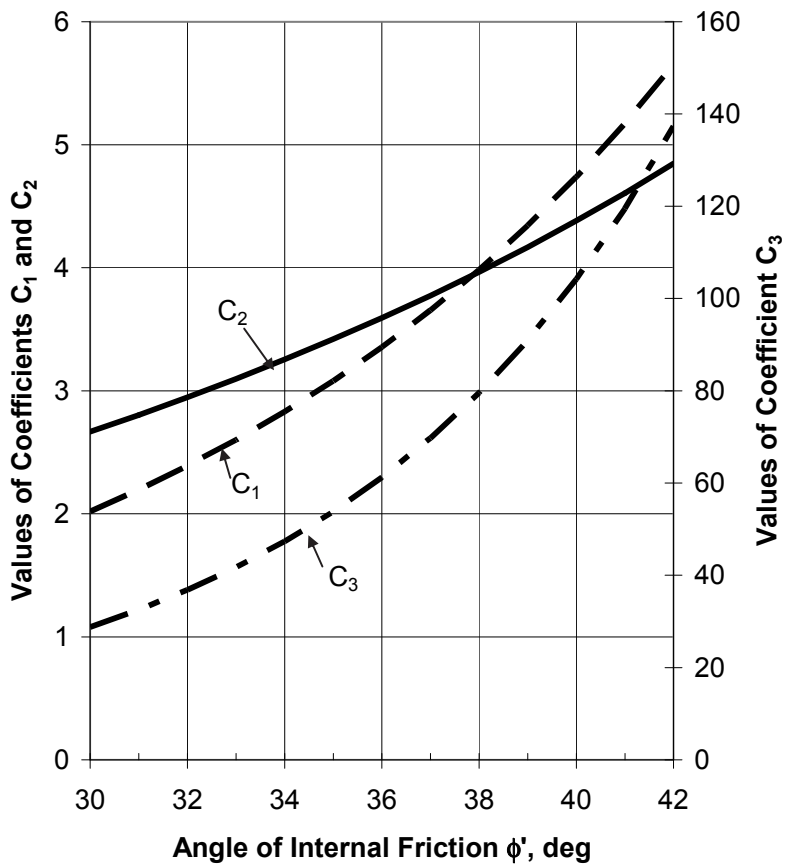

Figure 4—Coefficients C_1, C_2, C_3, as function of ϕ′

8.5.7 Lateral soil resistance–Displacement (*p-y*) curves for sand

The lateral soil resistance-deflection (*p-y*) relationship for sand is also non-linear and in the absence of more definitive information may be approximated at any specific depth, z, by Equation 28.

$$p = A \times p_u \tanh\left[\frac{k \times z}{A \times p_u} y\right] \tag{28}$$

where

A is the factor to account for cyclic or static loading condition, evaluated by

A = 0.9 for cyclic loading;

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$$A = \left(3.0 - 0.8\frac{z}{D}\right) \geq 0.9 \text{ for static loading;}$$

p_u is the ultimate lateral resistance at depth z;

k is the rate of increase with depth of initial modulus of subgrade reaction, see Table 5;

y is the lateral deflection at depth z;

z is the depth below the original seafloor.

The database for lateral soil-pile behavior in sands consists of free-head tests on piles in clean sands, with angles of internal friction ranging from 34° to 42°, as determined by shear box tests, drained triaxial tests or correlations with in situ tests.

Extrapolation of these data to soils outside the limits of experience, particularly to those sands with angles of internal friction less than 30°, should be done with caution. In particular, laboratory test results on such soils should be critically reviewed for evidence of anomalous behavior and for the presence of considerable fractions of cohesive soils, either of which could require a different formulation for the p–y relationships.

In the absence of more definitive information, the values of the initial modulus of subgrade reaction, k, given in Table 5 are recommended.

Table 5—Initial modulus of subgrade reaction values

ϕ'	k	
	MN/m^3	(lb/in^3)
25°	5.4	(20)
30°	11	(40)
35°	22	(80)
40°	45	(165)

8.6 Pile group behavior

8.6.1 General

Consideration should be given to the effects of closely spaced adjacent piles on the load and deflection characteristics of pile groups. Generally, for pile spacing less than eight diameters, group effects have to be evaluated.

8.6.2 Axial behavior

For piles embedded in clays, the group capacity may be less than a single isolated pile capacity multiplied by the number of piles in the group; conversely, for piles embedded in sands the group capacity may be higher than the sum of the capacities in the isolated piles. The group settlement in either clay or sand would normally be larger than that of a single pile subjected to the average pile load of the group.

8.6.3 Lateral behavior

For piles with the same pile head fixity conditions and embedded in either cohesive or cohesionless soils, the pile group would normally experience greater lateral deflection than that of a single pile under the average pile load of the corresponding group. The major factors influencing the group deflections and load distribution among the piles are the pile spacing, the ratio of pile penetration to the diameter, the pile flexibility relative to the soil, the dimensions of the group, and the variations in the shear strength and stiffness modulus of the soil with depth.

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An API-sponsored project [76] found, of the four group analysis methods examined in this study, the following methods to be the most appropriate for use in designing group pile foundations for the given loading conditions:

a) advanced methods, such as PILGP2R, for defining initial group stiffness (see Reference [76]);

b) the Focht-Koch (1973) method [77] as modified by Reese et al. (1984) [78] for defining group deflections and average maximum pile moments for design event loads. Deflections are probably underpredicted at loads giving deflections of 20 percent or more of the diameter of the individual piles in the group;

c) largest value obtained from the Focht-Koch and (b) methods for evaluating maximum pile load at a given group deflection.

9 Soil-structure interaction for risers, flowlines and auxiliary subsea structures

9.1 Site characterization

9.1.1 General considerations

Characterization of site conditions is necessary to develop a safe and economical layout and design of all facilities for a deepwater development including flowlines and risers. The site characterization required specifically for flowline design and riser touchdown point analyses should be included within the overall site characterization scope of work for the development. The scope of work may be optimized for a specific preliminary field layout and facilities design, but some final changes should be expected to accommodate final changes in field layout. The scope of work should anticipate the requirements for optimal design and code compliance.

9.1.2 Desktop assessment of site conditions

The initial step in site characterization is a desktop assessment of site conditions based on a review of available 2D and 3D exploration seismic data, regional and site specific geology and geotechnical data from engineering files, literature, or government files. The purpose of this review is to identify potential development constraints and to aid in planning and developing the scope of work for the subsequent shallow high resolution geophysical survey and geotechnical investigation.

9.1.3 Shallow high resolution geophysical survey

Upon completion of the desktop assessment, a high-quality, shallow high-resolution 2D geophysical survey should be performed over the entire areal extent of the development site prior to performing the geotechnical survey. The primary purpose of a shallow high resolution geophysical survey is to provide high resolution seismic data for a refinement of the shallow geology, geohazards, and seafloor features as expected from the desktop assessment, together with defining the uniformity and continuity of soil stratigraphy over the development site.

Shallow high resolution geophysical surveys in deepwater are typically performed using an autonomous underwater vehicle (AUV) equipped with the following:

a) multi-beam echo sounder for definition of water depth;

b) side scan sonar to define seafloor features;

c) sub-bottom profiler for definition of geologic structure and features.

Surveys can also be conducted using deep-towed equipment.

The scope of the geophysical survey field work typically also includes shallow soil sampling to help ground truth the side-scan sonar and sub-bottom profiler data with respect to soil type and stratigraphy at and near the seafloor and obtain soil data for preliminary design of flowlines and risers. The soil sampling equipment typically includes a triggered gravity drop core sampler with a core barrel length ranging from 3 to 6 m and a box core sampler with a typical maximum sampling depth of 0.5 m. Laboratory miniature vane and torvane plus possibly fall

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cone tests are performed on the ends of gravity drop core liner segments to measure soil shear strength. In-situ laboratory miniature vane and hand operated miniature T-bar penetrometer tests are recommended to be performed on the soil sample inside the box core sampler, followed by the recovery and preservation of sub-samples from the box core sampler.

9.1.4 Geotechnical investigation

9.1.4.1 General

The purpose of the geotechnical investigation is to explore and define the soil conditions across the development site, including soil stratigraphy, soil type, and the pertinent soil properties for design of flowlines and risers. The results of the geophysical survey should be integrated with the results of the desktop assessment and any existing geotechnical data to guide the final location of flowlines and riser touchdown points and to develop a scope of work for the final geotechnical investigation.

9.1.4.2 Soil sampling and in-situ testing

The total scope of geotechnical work for a development should include some combination of the following soil sampling and in-situ testing techniques:

a) drilled soil borings with downhole soil sampling and in-situ testing;

b) continuous seabed cone penetration, T-bar or ball tests to penetrations as shallow as 3 m for flowlines and steel catenary risers and as deep as 40 m, for riser towers and top tension risers;

c) large-diameter piston drop cores (core barrel length up to 20 to 30 m);

d) large-diameter push samples (core barrel length up to 20 m);

e) gravity drop cores and box cores.

9.1.4.3 Laboratory soil testing

Accurate determination of an appropriate undisturbed undrained shear strength profile is fundamental to the geotechnical design of flowlines and risers. The only laboratory strength tests that can be performed on near-seafloor, very soft clay soil samples are motorized laboratory miniature vane, hand-operated vane shear device such as the torvane, and fall cone. In addition, the remolded shear strength should also be measured using the miniature vane and fall cone to evaluate soil sensitivity.

In addition to the above described strength testing, total (bulk) density, Atterberg limits, and moisture content should also be performed. Optional tests may include grain size distribution, specific gravity, carbonate content, and pH. Depending on the intended application, thermal conductivity and electrical resistivity tests may also be performed to assess the insulating and corrosive properties of the soil.

Laboratory soil testing should be performed in accordance with recognized industry standards such as ASTM Book of Standards, Volume 04.08: Soil and Rock (1) or BS 1377:1990, ‘British Standard Methods of Test for Soils for Civil Engineering Purposes’.

9.1.4.4 Interpretation of soil design parameters

The undisturbed shear strengths measured in the field and laboratory should be plotted together with the interpreted shear strength based on in-situ penetrometer tests to develop a design shear strength profile. A design remolded shear strength profile can be based on data from a cyclic penetrometer test carried out in situ or in box cores, an interpretation of actual remolded strength test results on recovered soil, or by dividing the undisturbed design shear strength by an interpreted average soil sensitivity considered appropriate for the site. Upper and lower bound design and remolded shear strength profiles may be developed for use in parametric studies. Interpretation of design shear strength should be performed by an experienced geotechnical engineer/consultant qualified to consider all the factors influencing the measurement of shear strength.

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If information about the regain of strength of remolded soil due to reconsolidation or thixotropy is required, thixotropy tests can be performed on site-specific samples after various storing times. Thixotropy strength testing can be done using the miniature vane and fall cone.

A design submerged unit weight profile should be interpreted from a plot of measured unit soil weights versus penetration. If required, the total unit weight of the soil can be computed using the measured moisture content and specific gravity for the soil, and assuming the soil is 100 percent saturated.

The over-consolidation ratio (OCR) of the soil cannot typically be measured in near-seafloor sediments. It can be estimated through extrapolation from deeper depths or from the measured water content. An understanding of in-situ OCR and expected dilatant or contractant behavior of the soil when sheared may prove useful for the design of flowlines. An understanding of the remolded strength conditions of the soil near the mudline can also be useful for better understanding the soil response.

9.1.5 Integrated study

Finalization of the site characterization may require the integration of the geotechnical data, geological study and the shallow high resolution data, depending on the uniformity of geologic and soil conditions. Such an integrated study can develop maps showing the areal extent of different soil or geologic units and isopach maps showing the depth below the seafloor for different soil or seismic horizons and the thickness of different soil or geologic units. The results of the integrated study can be used to assess restraints imposed on flowline and riser design by seafloor features, geohazards, and soil conditions. This integrated study is an extension and update to the desktop assessment discussed in Clause 9.1.2.

9.2 Steel catenary risers

9.2.1 Introduction

The geotechnical properties of the seabed can influence the design conditions for steel catenary risers (SCRs) in two of the following aspects:

— an ultimate limit state associated with excessive bending and tensile stresses in the riser wall;

— a fatigue limit state associated with cumulative damage to the riser from motion-induced changes in bending stress in the region of the touchdown point.

In an SCR, the maximum curvature occurs within the suspended part of the catenary and the seabed stiffness has a negligible effect on the maximum curvature. Thus the seabed properties have essentially no influence on the maximum in-plane bending stresses within the riser. However, the seabed properties have a significant influence on the shear force in the riser, and hence changes in bending moment due to environmentally-induced motions of the riser. The properties thus affect fatigue calculations. In addition, the seabed properties will affect local out-of-plane curvature of the riser during extreme environmental events or large transverse or out-of-plane motions, particularly where the riser has become partially embedded within the touchdown zone. They may also affect transient bending moments induced during any position changes of the floating facility from which they are suspended.

9.2.2 Design for ultimate limit state

An ultimate limit state can arise under extreme environmental events that cause out-of-plane motion, particularly where the riser has become embedded, or lies within a trench, thus giving rise to high lateral soil resistance and locally high curvature of the riser.

Specialist geotechnical advice should be sought in order to quantify the lateral soil resistance, which will usually exceed normal frictional resistance for pipelines lying on the seabed surface (see Clause 9.5). During out-of-plane motion the riser will encounter resistance from the sides of any trench that has formed, or soil berms lying to either side of the pipe.

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9.2.3 Design for fatigue

The stress ranges used in the fatigue analysis of SCRs are calculated from the changes in riser stress caused by first and second order motions. Within the touchdown zone (TDZ) these motions can be simplified to moving the touchdown point (TDP) in-line with the riser and assessing the resulting changes in bending moment. A sketch of the change in maximum pipeline stresses arising from bending moments (Figure 5 shows stresses, not moments) in the TDZ due to example riser motions with both high and low values of soil stiffness is shown in Figure 5.

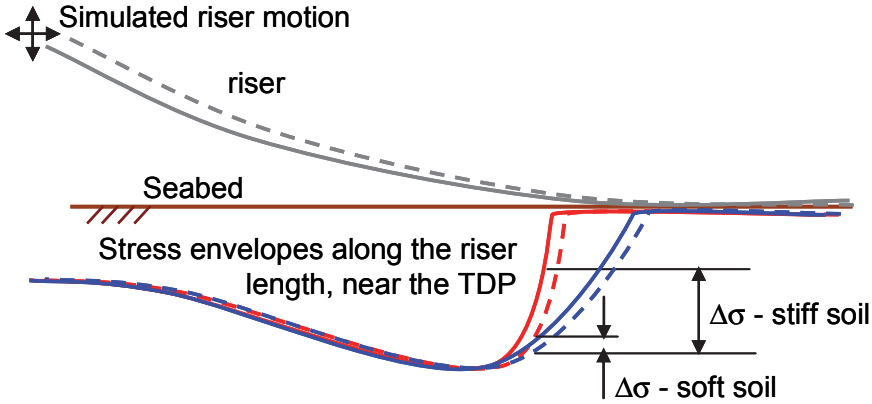

Figure 5 – Example stress changes for fatigue calculations [212]

The cyclic stress range in the TDZ depends on the rate of change of the bending moment and thus the shear force. Analysis shows that the maximum shear force varies approximately linearly with the logarithm of the soil stiffness. Fatigue laws follow a power law relationship, with damage proportional to a high power (typically about 5) of the cyclic stress amplitude [229]. Even relatively minor differences in the shear force can therefore have a significant effect on the estimated fatigue life, and hence the non-linear response of the soil needs to be considered.

Either small or large waves can dominate the fatigue damage in the touchdown zone. The majority of fatigue damage can occur from either large waves (not necessarily the most extreme) with low probability of occurrence or continuous motions from small day-to-day waves.

9.2.4 Seabed-riser response in vertical plane

9.2.4.1 Background

Riser interaction with the seabed involves complex non-linear processes including plastic penetration during initial touchdown, softening during cycles of upward and downward motion and potential suction-induced tensile resistance prior to breakaway. In most cases, design is undertaken using simplified models where the riser-soil interaction is idealized by a series of linear springs, with zero tension capacity, distributed along the riser throughout the touchdown zone. Ideally, the choice of spring stiffness should consider the amplitude of vertical displacement and other effects such as the cyclic motion of the riser. While the soil response will also be affected by out-of-plane motion of the riser, the discussion here is restricted to vertical stiffness of the seabed.

Bridge et al. (2004) gave the conceptual description of the seabed resistance shown in Figure 6 for a robust load cycle involving soil-riser separation. Following initial riser penetration into the seabed, unloading occurs as the pipe is uplifted. The soil response in the early stages of uplift is much stiffer than that under conditions of virgin penetration as shown in the 'unloading' curve in Figure 6. With continued uplift the net resistance force goes into tension ('pipe-soil suction' in Figure 6) until maximum uplift resistance of the soil is reached and the pipe begins to detach from the soil. Uplift resistance decreases until the pipe completely detaches from the soil. Upon re-penetration the pipe comes back into contact with the soil, with the re-loading stiffness typically being less than the unloading stiffness. Upon completion of a full load cycle, the load path does not return to the initial point of departure from the backbone curve; rather the pipe penetrates a small additional depth into the soil.

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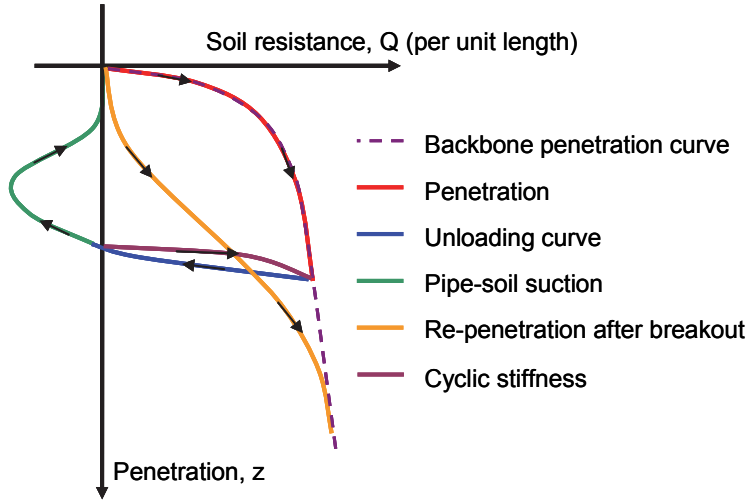

Figure 6 – Conceptual diagram of seabed stiffness [213]

NOTE The uplift resistance is referred to here as 'suction', although, strictly speaking, under submerged conditions pore pressures will normally remain positive. For consistency with much of the published literature, the term 'suction' is retained, understanding that it refers to a net upward force acting on the seabed.

9.2.4.2 Plastic penetration resistance

The backbone penetration curve in Figure 6 may be estimated by considering the seabed strength profile and an appropriate bearing capacity factor for a given penetration. For conditions where the soil strength profile increases approximately linearly with depth, the limiting penetration resistance per unit length may be expressed as shown by Equation 29 [210]:

$$Q_{\mathrm{u}} = N_{\mathrm{c}}\, s_{\mathrm{u}}\, D \approx a\left(\frac{z}{D}\right)^{b} s_{\mathrm{u}}\, D \tag{29}$$

where

s_{u} is the shear strength at the pipe invert;

D is the pipe diameter;

z is the depth to the pipe invert;

N_{c} is a bearing capacity factor;

a is a parameter fitted to results of finite element analyses, with an average value of ~6;

b is a parameter fitted to results of finite element analyses, with an average value of ~0,25.

NOTE 1 See Reference [239] regarding parameters a and b.

Allowance for buoyancy effects should also be included.

NOTE 2 s_{u} refers to an average shear strength (between that measured in triaxial compression, extension and simple shear), or that deduced from a field penetrometer test such as the T-bar.

In certain regions of the world, a crust of higher strength soil exists in the upper 0.5 to 1 m, before the strength profile reverts to a linear trend. The potential for the SCR to punch through the crust, and the consequences for fatigue studies, deserves careful consideration.

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9.2.4.3 Secant stiffness

The soil resistance behavior depicted in Figure 6 may be characterized in terms of equivalent springs having secant stiffness k_v supporting the riser pipe; the secant stiffness k_v in the vertical plane is defined by Equation 30.

$$k_v = \frac{\Delta Q}{\Delta z} \tag{30}$$

where

ΔQ is the change in vertical force per unit length of pipe;

Δz is the change in vertical displacement.

The non-linearity of the riser-soil interaction will lead to a variation in seabed stiffness along the length of the touchdown zone, which may be estimated based on the soil strength profile $s_u(z)$ and the predicted trench geometry, i.e. trench depth as a function of distance within the touchdown zone. The spatial variation in seabed stiffness is also affected by the temporal variability of the actual point of touchdown.

A hyperbolic model has been proposed for estimating the soil stiffness up to the point at which maximum penetration or maximum suction (uplift) is mobilized [211] [213]. The model may be expressed as shown in Equation 31.

$$K = \frac{k_v}{N_c\, s_u} = \frac{f\, K_{max}}{f + K_{max}\, \Delta z / D} \tag{31}$$

where

N_c is a bearing factor defining the backbone curve during virgin penetration;

K_{max} is the maximum value of the normalized secant stiffness on initial unloading or reloading;

D is the pipe diameter;

f is the asymptotic value of $\Delta Q/N_c s_u D$ at large displacements (i.e. $f \sim I\,(Q_{initial} - Q_{limit})\, I/N_c s_u D$, where Q_{limit} is Q_{max} for penetration or suction),

NOTE For soft clays, Reference [213] suggested a value of K_{max} of about 250, which is consistent with the first load cycle of small amplitude laboratory model tests in kaolin [224].

9.2.4.4 Uplift and breakaway

When the riser pipe is continuously uplifted, a maximum soil uplift resistance is reached, after which uplift resistance declines and breakaway of the pipe from the seabed occurs. The resistance of the soil to uplift can lead to bending stresses in uplift exceeding those in lay-down. In contrast, separation of the pipe from the soil tends to relieve bending stresses in the pipe. Accordingly, realistic estimates of the magnitude of maximum suction resistance and the displacement levels associated with suction mobilization and when breakaway occurs are important to characterize soil-riser interaction accurately.

Soil resistance to uplift of the riser pipe can arise from two mechanisms. The first is the ability of the soil to resist suction. The second is the resistance mobilized by backfill of soil above the pipe created by processes such as deposition or collapse of the side walls of the trench.

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The maximum suction force per unit length of pipe, Q_{smax}, depends on a number of factors such as the effects of cyclic movement, the pipe velocity and the time over which the uplift resistance is sustained. These may be expressed through factors applied to the (compression) bearing capacity as shown in Equation 32 [213].

$$Q_{smax} = f_c\, f_V\, f_t\, N_c\, s_u\, D \tag{32}$$

where

f_c is a dimensionless cyclic factor;

f_V is a dimensionless velocity factor;

f_t is a dimensionless time factor.

NOTE For recommendations for f_c, f_V, and f_t refer to Reference [213].

For conditions of cyclic and fatigue loading, Reference [213] recommended the use of a remolded strength s_{ur} rather than intact strength s_u, in addition to a cyclic reduction factor f_c, although it may be more consistent to use the original intact shear strength as the benchmark, relying on the various factors to quantify adjustments in estimating Q_{smax}.

Video surveys have shown that risers often cut a trench of significant depth in the seabed, as is considered further in Clause 9.5. Additional uplift resistance will occur where partial backfilling of the trench occurs, leaving the pipe embedded. The trench backfill is likely to be a product of mixing with water as well as remolding; therefore, its strength is likely to be less than the remolded strength of the seabed soil.

For conditions of no trench backfill with uplift resistance being mobilized purely from suction, Reference [213] proposed the following relationship shown by Equation 33 for breakout displacement Δz_b, i.e. the uplift displacement at which the pipe completely detaches from the soil, measured from the point at which the net force, Q, becomes negative.

$$\Delta z_b = \zeta_V\, \zeta_t\, D \tag{33}$$

where

ζ_V is a dimensionless velocity factor;

ζ_t is a dimensionless time factor.

NOTE Reference [213] provides recommendations for the factors ζ_V and ζ_t based on STRIDE and CARISIMA data.

For the case of a backfilled trench, general relationships analogous to Equation 33 have not appeared in the published literature. However, relationships for uplift-displacement behavior have been developed on an ad hoc basis for specific sites (e.g. Reference [242]).

9.2.4.5 Stiffness adjustment for cyclic loading

Cyclic loading is recognized to degrade the soil stiffness. Based on cyclic model pipe-soil tests in kaolin, Reference [224] reported values of normalized cyclic stiffness, K, of less than 5, in contrast to monotonic K values ranging from approximately 250 (see 9.4.3) down to 40 at $\Delta z/D$ about 0.025. Model tests also suggested that the stiffness reduces by a factor of 10 to 20 where soil-riser separation and re-contact occurs [225].

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The magnitude of the distance along the pipe over which soil-riser separation occurs will vary with water depth, riser properties, type of floating facility and environmental conditions. Table 6 summarizes typical motions of the riser in the touchdown region for different storm conditions for spar platforms in the Gulf of Mexico. For typical riser diameters of 0.3 m to 0.4 m, these distances correspond to about ±15 m for day-to-day wave loading, and ±25 m for extreme storm conditions. Transverse or out-of-plane vessel motions can lead to trebling of the range of separation.

Table 6 – Summary of distance and occurrence of SCR TDP motions [242]

Motion	Probability of occurrence	Limit of in-plane TDP motions	Limit of transverse TDP motions
Day-to-day	95 %	±43D	±0.5D
Extreme storm	2.5 %	±70D	±1D
Second order vessel motions	2.5 %	±200D to ±260D	±7D

On the catenary side of the mean TDP, the relevant soil stiffness may be taken as that associated with large displacements, and hence an order of magnitude lower than the maximum unload-reload stiffness. On the flowline side of the TDP, the gradual decay in magnitude and frequency of the motions suggests that, in the zone between 'day-to-day' and 'extreme storm' conditions in Table 6, the 'average' operative value of stiffness should be increased in steps towards the maximum value. Histograms of the vertical SCR motion can assist in assessing the magnitude and spatial variation of stiffness decay.

In practice, software used for assessing fatigue damage may be limited in the extent to which a spatially varying seabed stiffness can be implemented. An equivalent effect can be achieved with a suitable non-linear model for riser-seabed interaction, incorporating a relatively soft backbone curve and higher unload-reload stiffness, as indicated in Figure 6 Such non-linear models are now starting to become available in commercial software packages for riser design.

9.2.5 Trenching

9.2.5.1 Process

Riser movements at the touchdown zone produce seafloor trenching that affects stress analysis and fatigue performance of the riser. Trenching is a process of soil scour and plowing due to riser responses to global motions of the floating system. The process generally takes durations of months to years of cumulative soil displacements to form trenches, but may be accelerated with greater incidence of environmental events, heave-prone vessels or vortex-induced vibration.

9.2.5.2 Features

Observations of riser trenches reported by Reference [214] reveal a typical, ladle-shaped profile that tends to be deepest near the nominal touchdown point and gradually slopes up to the natural seafloor towards the flowline. The riser is often buried inside the trench in the region of the mean touchdown point. In plan, trenches are generally shaped as a bell mouth, with the widest portion on the vessel side of the touchdown and progressively narrowing towards the flowline. A generalized trench shape is provided in Figure 7.

Trench dimensions for a specific riser depends on several factors, including the following:

a) soil conditions;

b) riser dimensions and configuration (e.g. pipe-in-pipe);

c) environmental conditions;

d) time after installation;

e) vessel type.

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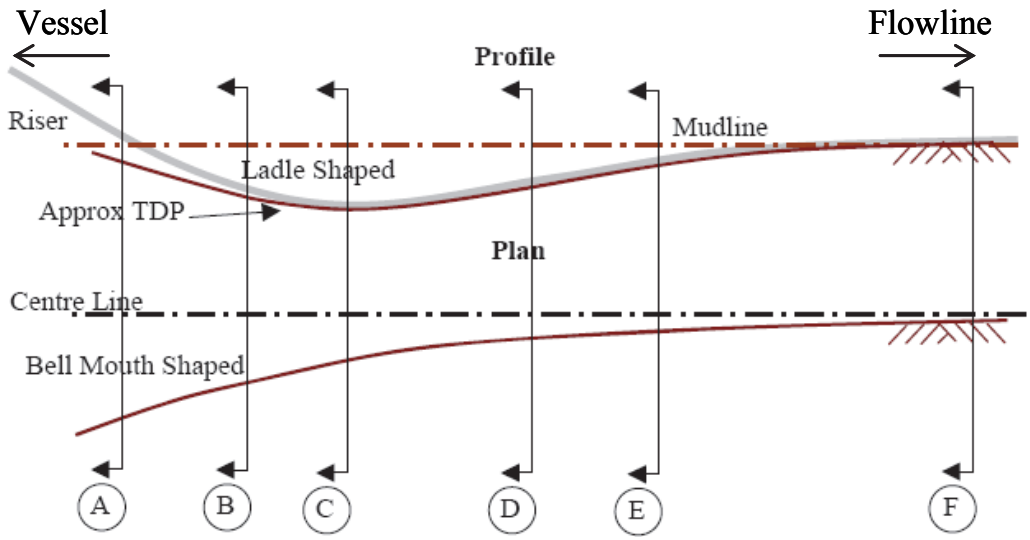

Figure 7 – Overview of Generalized Trench [214]

9.2.5.3 Influence on analysis

The impact of a trenched seafloor on SCR touchdown region fatigue performance is not well defined due to competing factors related to trench geometry and soil conditions. In contrast, trenching may cause large bending stresses in the riser for transverse motions at or near the touchdown. The riser may be overstressed in cases of abrupt interaction with a trench during extreme environmental events or large transverse or out-of-plane motions. Also, persistent transverse motions inside a trench by prevailing environmental conditions, second-order motions and vortex-induced vibration can adversely impact the fatigue life at or near the riser touchdown.

9.2.6 Three-dimensional motion

Under extreme conditions, such as for broken mooring lines, transverse or out-of-plane motions of the vessel or platform can lead to 3-dimensional motion of the riser sufficient to cause it to 'break out' from the current trench. Consideration should therefore be given to wide sweeps of the nominal SCR touchdown point as the surface vessel translates through large horizontal distances, which may be in the range 5 % to 10 % of the water depth. This distance is likely to be larger than the largest trench width considered in previous sections. The surface vessel may swing back and drag the riser large distances past the original SCR trench, before reaching equilibrium at a position that is not quantifiable in a failure scenario like this.

Under normal (i.e. with intact moorings) operations in major storms and large current events, it is also possible for the floating support structure to lift an SCR out of its trench and set it down in virgin soil many trench widths away as the surface and subsea environmental conditions build up, possibly change direction, and then ease off.

In either of the above situations a new trench may form at the new location, which will affect the fatigue life of the riser. A more critical issue, however, is that the break-out resistance from the original trench may result in high localized bending moments and a potential ultimate limit state. The effect of the high break-out resistance during three-dimensional motion of a trenched riser should be quantified.

9.3 Top tension riser

9.3.1 Introduction

The design of top tension risers near or below the blowout preventer (BOP) stack is dependent on the riser (conductor)-soil interaction. The following two design issues should be considered:

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a) the reaction of the riser at the ultimate limit state when the host facility has moved a considerable distance from the mean position;

b) the fatigue that occurs within the riser as a result of repeated cyclic motions with a range of amplitudes and frequencies.

The first problem occurs during a move-off condition or during environmental loading. The move-off case can be an intentional static type of condition that occurs when the host facility is moved to facilitate drilling operations or when the host temporarily loses station keeping capabilities.

The fatigue problem is governed by cyclic loads that occur throughout the life cycle of the riser-conductor system. These loads can occur from:

— environmental wave, wind and current loads on the host facility;

— vortex induced vibrations (VIV) on the host facility or riser;

— environmental loads (wave and current) mostly in the top portion of the riser.

Analyses have shown that peak loads are not the major contributor to the fatigue damage. Rather, smaller more frequent loads are responsible for most fatigue damage. Therefore, characterization of the soil response at small amplitude displacements is particularly important for the fatigue problem. Previous work indicates that the criteria in API 2A-WSD, 21st Edition, may underestimate the lateral stiffness of the soil, especially at low amplitude displacements. It is often difficult to assess whether soft or stiffer soil stiffness estimates are the more conservative. Stiffer estimates suggest that the critical cyclic bending moments will occur above the mudline while softer estimates would suggest the opposite.

Finally, temperature effects will also have an impact on the conductor. Increased temperatures can result in:

— upheaval forces on the conductor,

— changes in the shear strength of the soil around the conductor, or

— dissociation of seafloor hydrates.

The impact of temperature changes on the soil properties along the conductor is likely to be more severe for axial loads. However, the axial capacity is quite robust due to the length of the conductor and associated casing strings. The impact on lateral loads is mitigated due to the constant temperature condition imposed by seawater at the mudline.

9.3.2 Soil response

The response of the near seafloor portion of top tension riser-conductors to fatigue loading is highly dependent on soils to about 15 m to 18 m. below the mudline. The overall challenge of assessing this lateral response is analogous to that of a laterally loaded pile. Accordingly, lateral soil springs provided for offshore piles have often been used to assess the lateral response of a conductor. Despite the apparent similarities, the following differences exist.

a) The soil springs for piles were originally developed for steel jackets subjected to large storm loads. As such, the primary intent was to characterize the soil near yield while less attention was paid to the soil response at smaller displacements.

b) The maximum moments developed in a pile are relatively insensitive to the lateral soil response. Differences in the soil springs will tend to change the location rather than the magnitude of the maximum moment. Since the wall thickness for offshore piles is usually constant, inaccuracies in the *p-y* springs will have a lesser impact. Top tension risers-conductors, however, are assembled in sections with connectors. The connectors are critical locations for fatigue.

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c) Recommendations from API 2A-WSD, 21st Edition, and the supplement/errata published in 2007 were based on tests with loads applied over a few days while the loads that cause fatigue are applied over a much shorter time period.

d) The relationship between cyclic bending moment and fatigue life is highly non-linear so that with many cycles fatigue life is highly sensitive to soil stiffness.

9.3.3 Development of *p-y* springs via finite element (FE) analyses

Instead of relying exclusively on API 2A-WSD, 21st Edition, recommendations for lateral load analyses of riser-conductor systems, the *p-y* springs can alternatively be developed using the finite element method. The following provides guidelines for this approach.

An important aspect of developing *p-y* springs with the finite element method is developing a representative soil model for riser-conductor problems. The small strain or initial shear modulus is an important part of this soil model. The initial shear modulus, G_{max}, can be determined from resonant column testing on samples taken during the site investigation. Samples should be taken from depths where the maximum fatigue damage might occur, for example the upper 15 m. of soil. The resonant column test results should be adjusted for actual field conditions, for example using the correlations proposed in Reference [231].

Values of G_{max} obtained from resonant column tests should also be modified to account for:

— the increase in shear modulus that occurs after primary consolidation;

— the reduction in shear modulus due to the lower rate of loading expected for riser-conductor types of loading, compared to the higher rate in the resonant column tests; and

— cyclic degradation, with the influence on steel stress and fatigue behavior being bracketed between pre- and post-cyclic behavior of the soil mass.

The increase per log cycle in time due to secondary consolidation has been estimated as proportional to the square root of the soil plasticity index [231]. This increase will be offset in part by the reduction due to the lower rate of loading which, in the absence of site specific data, may be approximated as a 10 % reduction in shear modulus per log difference in frequency relative to the resonant column tests.

Alternately, Reference [207] presents the following for a normalized shear modulus (G_{max}/s_{uDSS}) for normally consolidated clays:

$$\frac{G_{max}}{s_{uDSS}} \approx \frac{300}{PI/100} \tag{32}$$

where

s_{uDSS} is the undrained shear strength from direct simple shear tests;

PI is the plasticity Index of the soil.

Generally, values of normalized shear modulus for over-consolidated clays are lower than those for the normally consolidated data [207].

Reference [230] shows example *p-y* curves developed with the finite element approach compared with API 2A-WSD, 21st Edition, recommendations (Figure 8 [230]). Also shown on the figure are centrifuge results for tests where the load was applied in less than one minute. The green and black curves show the direct comparison between the centrifuge results and finite element model. The blue curve represents API 2A-WSD, 21st Edition, recommendations.

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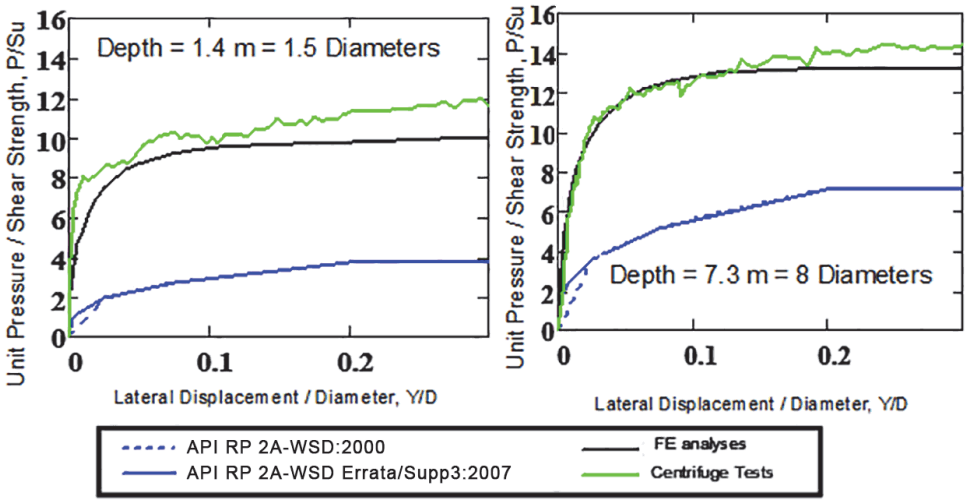

Figure 8 – Comparison between *p-y* springs measured in centrifuge tests on kaolinite and computed using the FEA technique with the API 2A-WSD, 21st Edition, recommendations [230]

Additional details for developing *p-y* curves through numerical modelling procedures are described in Reference [241].

9.3.4 Additional considerations

In addition to soil stiffness at small strains the soil should also include effects of work hardening. This work hardening results from either elasto-plastic isotropic hardening of the soil at larger strains or kinematic hardening from cyclic loading.

Figure 9 [230] presents results that demonstrate this potential impact. This figure shows that cyclic loading can degrade the soil stiffness. If the static *p-y* backbone curve is used in the analysis at the mid-point for the cyclic load, the actual unload-reload secant stiffness (as is used in several riser fatigue software products) is greater. Therefore, using the tangent stiffness from the backbone curve can produce conservative results (more fatigue damage) when the critical fatigue point is below the mudline. However, if the critical fatigue point is above the mudline, softer soil springs can result in a non-conservative solution.

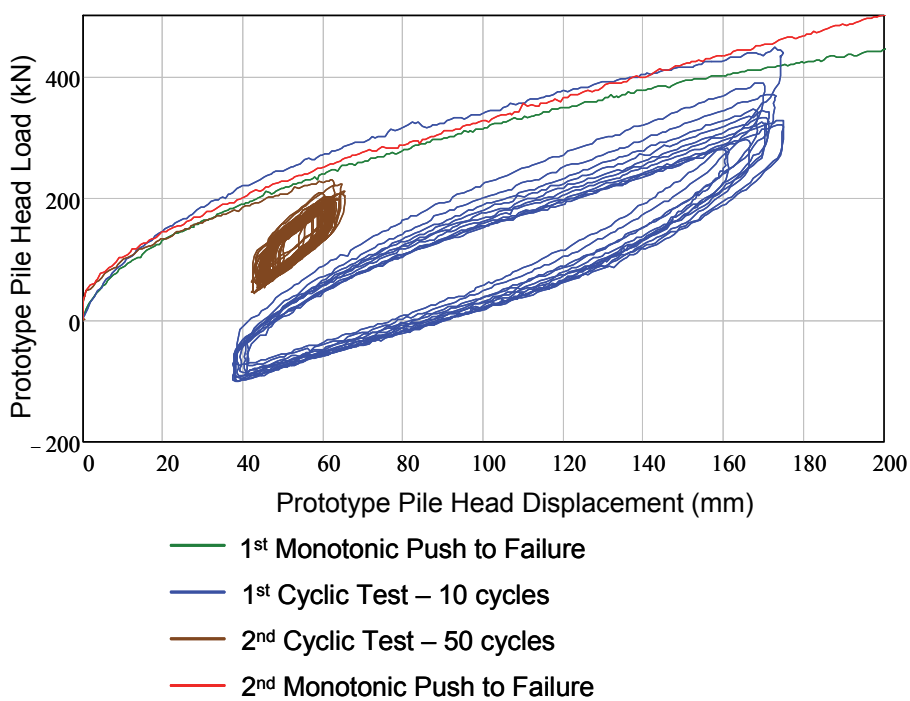

Figure 9 – Expanded view of pile head displacement versus pile head load [230]

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9.3.5 Summary and recommendations for top tension risers

The following lists a summary and recommendations for top tension risers.

a) Top tension riser-conductor design should consider both ultimate and fatigue limit state. Because of variable wall thicknesses and irregular geometries, the critical bending moment can occur either above or below the mudline.

b) The *p-y* curves specified by API 2A-WSD, 21st Edition, criteria for piles provide a significantly softer response than *p-y* curves developed via finite element analyses. Advanced soil models should be used based on site-specific soils data that captures both small displacement soil behavior and work hardening from cyclic loading.

c) Although finite element modelling produces a stiffer soil response, physical model tests demonstrate cyclic softening with repeated load cycles.

d) Whenever the critical fatigue point is below the seafloor, the tangent stiffness from adjusted static *p-y* curves based on finite element analyses can be conservative, because the tangent stiffness at the mid-point of cycling can be less than the load-unload stiffness. If the critical fatigue point is above the mudline the *p-y* curves given by API 2A-WSD, 21st Edition, can be non-conservative. For this case the amount of cyclic degradation should be conservatively estimated, i.e. less degradation.

e) Developing *p-y* curves with finite element analysis requires expert skills and is time consuming. If the critical bending moments are below the mudline, the curves given by API 2A-WSD, 21st Edition, can be initially used. If this initial attempt leads to unacceptable failure levels, then the site-specific data can be used to determine the benefit from increasing soil stiffness.

f) For a drilling riser with heavier lower stacks (LMRP, BOP), the soft soil (low soil stiffness) may cause the system natural frequency to move into wave energy zone. So using stiffer soil (or fixed at mudline) does not guarantee the results to be in the conservative side even if the critical fatigue point is above the mudline.

9.4 Riser tower foundations

9.4.1 Introduction

The riser tower concept consists of a free-standing riser assembly, incorporating several risers in a bundle configuration, tensioned from the top by a buoyancy tank, and anchored to the sea-bed. The connection to the surface vessel or platform is generally ensured by flexible jumpers.

A riser tower supports axial tension generated by buoyancy and by cyclic wave action, and is secured into the sea-bed. A significant part of the tension acts permanently during the life of the development.

9.4.2 Foundation options

There are a number of possible foundation options: gravity base, suction caissons and driven piles.

The concept selection for the foundation should be based on technical and economic criteria, taking into account the soil data, the installation aspects, as well as the in-place performance. A preliminary conceptual study should be performed to select the most appropriate foundation type.

9.4.3 Loads and safety factor

9.4.3.1 Loads

Design loads should be evaluated for the following conditions:

— foundation installation and retrieval;

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— normal operating conditions;

— extreme conditions.

Load combinations should be selected so as to anticipate the most unfavorable result, for each of the stability mechanisms and deformation analyses performed.

9.4.3.2 Recommended safety factors

Safety factors should be as recommended by API 2T for driven piles and gravity base anchors and by API 2SK for suction caissons.

NOTE 1 The API 2SK safety factors were developed with NO consideration of permanent uplift loads on suction caissons. Reference [223] provides information on the potential uplift response under sustained loading.

NOTE 2 It should be noted that API 2T does not address the case of gravity loads explicitly. Additionally, Section 10.3.3 of API 2T states that:

> For axial pile design where the weight of the foundation system is less than approximately 10 % of the ultimate axial capacity, the underwater weight of the foundation system may be subtracted from the applied loads in determining the safety factor of the foundations. For other weight-dominated systems, the foundation system weight should be added to the resistance side of the equation.

9.4.4 Soil design parameters

High quality in-situ measurements and/or soil borings should be obtained to select the soil design parameters.

The depth of the geotechnical borings should exceed the foundation depth by at least 3 anchor diameters unless a more regional site characterization has shown that there are no major changes in stratigraphy within that depth. If there are no major changes in stratigraphy, the depth beneath the anchor tip may be reduced to 1 anchor diameter. It is critical to establish whether any high permeability layers are present within or above the zone of influence for reverse end bearing under sustained loading.

The number of borings should be defined as a function of the soil variability. One boring should be performed at each anchor location when unusually large lateral variability of the soil properties is expected.

The main soil properties that are needed for the foundation design are defined in API 2SK.

9.4.5 Design issues

9.4.5.1 General principles

The following general principles are to be considered in assessing the stability of tower riser foundations.

a) Limit equilibrium methods can generally be used to evaluate the capacity of riser tower foundations. The shear strength used in the analysis should account for effects of creep and potential drainage under sustained load and cyclic degradation. The reduction in effective stresses and shear strength due to potential drainage can be studied by finite element analyses.

b) Due consideration should be incorporated into the design regarding displacement and deformation during the life of the foundation. Where displacement and deformation are critical, complex analysis methods may be warranted. The displacement analysis should include contributions from undrained shear strains due to application of the sustained load, undrained creep during the sustained load, and permanent and cyclic components from the wave loading. Displacements due to shear strain, volumetric strains and flow of water through the soil due to potential drainage during the sustained load period should also be considered.

c) The anchors should be installed within specified tolerances of tilt and mis-orientation. The design analyses should account for the effect of the tolerance limits.

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d) Installation should be planned so as to ensure the foundation can be properly seated at the intended site, without excessive disturbance to the supporting soil. Where excessive disturbance does occur, this should be considered in the assessment of foundation capacity.

e) Measures should be taken to avoid erosion and scour of the soil beneath or near the foundation base.

Where removal is anticipated, an analysis should be made of the forces generated during removal to ensure that removal can be accomplished with the means available.

9.4.5.2 Geotechnical design methodology for the foundation

The design of driven piles and gravity base is covered by the recommendations in API 2T. The design of suction caissons is covered by the recommendations in API 2SK, with more detailed aspects considered in the literature [205], [206], [208], [219], [220], [221], [222], [223], [227], [228], [238]. The following aspects should be considered:

— penetration and retrieval;

— holding capacity including long-term uplift capacity;

— long-term displacement;

— soil reactions to be used for the structural design.

The capacity should be checked both for the permanent load and for the sum of permanent and cyclic loads. If the cyclic loads are small compared to the permanent loads, the permanent load condition may be critical because the strength can be smaller for this condition than for the condition where the cyclic load is included. The following should be considered:

a) The undrained shear strength for the permanent load condition should be reduced to account for creep effects (e.g. Reference [234]). The effect of pore pressure redistribution and swelling should also be considered, since this may lead to reduction in effective stresses and undrained shear strengths, and hence reduce the capacity under transient wave loading (Reference [208]). In cases where the permanent load acts for months, it is also necessary to consider whether drained conditions may develop, and the extent to which full base suction can be maintained. The possibility of drainage channels between inside skirt stiffeners, above inside stiffeners and extended skirt wall thicknesses, and along open cracks outside the anchor at the active side of the anchor should be considered. References [223] and [226] show that the capacity of a typical suction caissons under sustained loading may be only 70 % of the capacity for rapid loading.

b) Drainage and pore pressure redistribution can also influence the undrained shear strength under cyclic loading, but the shear strength for the sum of permanent and cyclic loads will be higher than for the permanent load due to rate effects (e.g. Reference [209]).

The potential of gapping along the wall above stiffeners or sections with increased wall thickness should be evaluated. In cases with ring stiffeners it is necessary to evaluate the potential of trapped water between ring stiffeners (e.g. Reference [206]). If gapping or trapped water is possible, the effect that such gaps will have on the drainage path and on resistance due to lack of contact along the wall should be considered.

The capacity of suction caissons may depend on passive underpressure inside the caisson. If passive underpressure is relied upon, proper sealing may be critical, especially for the part of the underpressure generated by long period environmental loads such as loop currents. The anchor top may be sealed, but if the valve seals cannot be guaranteed, consideration should be given to either a back-up cap behind the valves or a monitoring program to assure the desired integrity over the lifetime of the suction caissons. If proper sealing is not ensured, the suction caissons should be designed to resist sustained uplift load without taking passive underpressure into account.

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9.4.6 Inspection and monitoring

An inspection program should be considered as integral part of the foundation design. The inspection program should include the use of instrumentation to monitor critical aspects of the foundation performance, during both installation and operation.

If at any time during the service life of the structure, the inspection program reveals conditions, or behavior, which are detrimental to the integrity of the foundation or structure, then maintenance or remedial action should be carried out as necessary.

9.5 Flowlines and pipelines

9.5.1 Introduction

Pipeline design should consider ultimate and fatigue limit states related to the stresses in the pipeline and the movements of the associated end connections including sections which transition into a catenary riser. The response is influenced by the geotechnical interaction forces between the pipeline and the seabed as well as other external loads on the pipeline and internal loads within the pipeline.

Geotechnical advice should be sought to predict the as-laid pipe embedment and the resulting force-displacement response in the axial and lateral directions. The most basic forms of axial and lateral pipe-soil model are linear elastic-perfectly plastic. More sophisticated designs require more complex pipeline-soil models. Either upper or lower bound values of the pipe-soil interaction forces can be critical for a limit state. Each bound should be assessed as necessary.

9.5.2 Loads on seabed pipelines

The loads and motions imposed during laying govern the pipeline embedment and any residual tension at the start of operation and initial in-service performance. Hydrodynamic loading and subsequent scour and seabed liquefaction processes can lead to changes in embedment during the operational life of the pipeline.

After installation, the loads imposed on an individual element of pipeline are balanced by reaction forces from the soil. The loads on the pipeline element arise from adjacent or bounding elements of the pipeline, a connected SCR, internal and external pressure, hydrodynamic or thermal loading and internal and external pressure. A pipeline may also be susceptible to external loading from debris flows and turbidity currents that arise from submarine slides and snag or impact loading from foreign objects. The compressive axial force created by operating cycles of internal pressure and temperature can lead to lateral buckling of the pipeline, or the accumulation of axial movements (pipeline ‘walking’).

9.5.3 Soil reaction forces

9.5.3.1 Pipe-soil interaction models

The interaction between the pipe and the seabed is incorporated into the structural analysis of a pipeline by attaching pipe-soil model elements at intervals along the pipe. This approach is analogous to the ‘*t-z*’ and ‘*p-y*’ load transfer methods of analyzing pile response.

For some simple pipeline design functions, pipe-soil response is represented by limiting values of axial or lateral pipe-soil resistance or bi-linear-elastic – perfectly-plastic behavior in the axial and lateral directions. The pipe-soil resistance is typically expressed as an equivalent friction factor, linking the limiting resistance to the effective pipeline weight. However, the axial and lateral resistances can depend on factors other than the pipe weight, in particular the embedment. Therefore, a friction factor is not an intrinsic soil property.

To capture the more complex effects of interaction – particularly the large displacement behavior – it is necessary to model other aspects of the response, including brittle breakout behavior and cyclic berm growth during lateral movement (Bruton et al. 2007).

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9.5.3.2 Drained and undrained soil behavior

In fine-grained sediments, pipe laying is usually an undrained process. Dissipation of the lay-induced excess pore pressure typically takes days or weeks. Lateral pipeline movements generally involve undrained deformation although consolidation between events may cause disturbed soil to regain strength. Axial pipe movements may be drained or undrained, since the relevant drainage distances are shorter than for lateral movement.

In coarse-grained sediments, pipeline installation and operation will generally occur under fully drained conditions. In a design analysis, the anticipated rates of axial and lateral pipeline movement should be compared with the relevant rates of soil drainage and consolidation to establish whether drained or undrained conditions will prevail.

9.5.4 Analysis of pipeline-soil interaction

9.5.4.1 Vertical penetration

9.5.4.1.1 Lay effects

Observations show that the as-laid pipeline embedment is typically much greater than would be expected from the static weight alone, due to over-stressing and dynamic motion in the touchdown zone during laying [245]. The contact bearing stresses (or vertical force per unit length) between the pipe and the soil in the vicinity of the touchdown point exceeds the as-laid self-weight of the pipe due to the catenary shape. The degree of over-stress is dominated by the bending rigidity of the pipeline, the apparent stiffness of the seabed, and the tension in the pipe in the vicinity of the touchdown point [239]. The applicable apparent stiffness of the seabed is a secant stiffness, for the anticipated degree of plastic penetration, and can be much lower than that customarily used for fatigue assessment within the touchdown zone of an SCR. In deep water, the overstress may be negligible because of soft seabed conditions. If the pipeline is laid empty then the maximum static loading may occur during hydrotesting, when the pipeline is heavier [249]. The as-laid embedment may therefore be governed either by the as-laid weight (with a touchdown overstress and dynamic lay effects) or by the hydrotest condition.

Vessel motion, changes in pipeline tension, and hydrodynamic loading of the hanging pipe, will induce a combination of vertical and horizontal motion of the pipeline at the seabed during laying [233] [218] [245]. Even small lateral or vertical movements can cause disturbance, local softening and erosion of the seabed in the touchdown zone, increasing the pipe embedment.

9.5.4.1.2 Static vertical penetration response

For seabed sediments where drained conditions will prevail, conventional bearing capacity approaches may be used to estimate the static pipeline penetration. The pipeline may be treated as a surface strip foundation of width equal to the (nominal) chord length of pipe-soil contact at the assumed embedment. In most cases, however, the penetration resistance will be such that minimal embedment of the pipeline will be predicted based on static loading. Other processes such as cyclic motion of the pipeline, and scour and partial liquefaction of the seabed will determine the as-laid embedment depth.

For fine-grained sediments, where undrained conditions prevail during embedment, theoretical solutions for estimating the pipe penetration resistance have been provided in References [236], [210], and [240]. These solutions use the conventional bearing capacity equation, modified for the curved shape of a pipeline. In soft soils, the enhanced soil buoyancy created by heave may be significant [239].

9.5.4.2 Axial soil resistance

Axial pipeline movement involves shear failure at or close to the pipeline-soil interface. The vertical effective contact force can be used to calculate the effective stresses and forces at the pipeline-soil interface. The integrated normal contact stresses around the pipe periphery exceed the vertical contact force, V, due to the curved shape of the pipe surface (Reference [246]). Using the enhancement factor, ζ, to account for this effect, the drained axial resistance per unit length of pipeline, T, is given by Equation 34:

$$T = \mu N = \mu \zeta V \tag{34}$$

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where μ is the pipeline-soil friction coefficient, which can be alternatively expressed in terms of an interface friction angle, δ, where μ = tan δ, and N is the integrated normal contact force.

Based on an elastic solution, Reference [246] provides the following expression for ζ:

$$\zeta = \frac{N}{V} = \frac{2\sin\theta_{D'}}{\theta_{D'} + \left(\sin\theta_{D'}\cos\theta_{D'}\right)}$$

where $\theta_{D'}$ is the half-angle of the pipe-soil contact perimeter, which varies with normalized embedment, z/D, with D the pipe diameter, according to:

$$\cos\theta_{D'} = 1 - \frac{2z}{D} \tag{35}$$

Due to the stress-dependency of soil friction angle, this parameter should be assessed by tests conducted at the correct stress level [237] [246]. During undrained axial pipe movement, the apparent friction coefficient may increase or decrease depending on whether negative or positive excess pore pressure is generated by shearing at the interface.

Laboratory tests can be used to assess an appropriate friction coefficient, μ, for drained shearing including separate values for peak and residual resistance. Tests can be conducted in a low stress shear box or on a tilt table, using a sample of the pipe surface coating. Alternatively, a model pipe section can be tested in a larger test chamber. These tests should replicate the relevant lay-induced consolidation history, and the speeds and pause periods relevant to the design situation. It should be established whether undrained or partially drained conditions may apply during pipe motion. These conditions can lead to a significant reduction in the apparent friction coefficient, μ, compared to the drained case.

9.5.4.3 Lateral soil resistance

Lateral pipe-soil resistance during breakout and large-amplitude cyclic movement is influenced by the initial pipe embedment and weight, the development of soil berms ahead of the laterally sweeping SCR or pipeline segment, and the soil properties. Two characteristic types of large-amplitude lateral response are typically observed (Figure 10) depending on the ratio of the pipeline weight to the seabed strength, V/s_uD. For values of V/s_uD below approximately 2 ('light' pipes), the pipeline tends to rise after breaking out from the as-laid position (Figure 10a). As the pipe rises, the lateral resistance reduces from the break-out value to a residual resistance. The pipeline sweeps horizontally with a berm of soil being pushed ahead of the pipe. This mechanism governs the residual resistance, H_{res}. Subsequent cycles of lateral movement lead to a steady increase in the restraint provided by the soil berms (Figure 10a). The 'light' pipe form of response is also typically observed in drained conditions.

For values of V/s_uD greater than approximately 2 ('heavy' pipes), the pipeline typically moves downwards after the initial break-out resistance is mobilized. This downward movement, coupled with the growth of a soil berm ahead of the pipe, leads to a steady increase in the lateral resistance (Figure 10b).

Empirical expressions exist for predicting lateral pipe-soil resistance, which have evolved primarily through calibration against model tests [243] [244] [215] [246]. These expressions are subject to significant uncertainty, and their relevance should be established for a particular design situation. In undrained conditions, the break-out resistance, H_{brk}, is generally divided into two contributions:

— a component proportional to the current vertical load, V, (which is essentially the pipe weight);

— a passive resistance component, linked to the embedment depth of the pipe, z,

An alternative approach is to use yield envelopes (or interaction diagrams) in vertical and horizontal load space that bound the allowable load combinations for a given pipeline embedment [235] [240].

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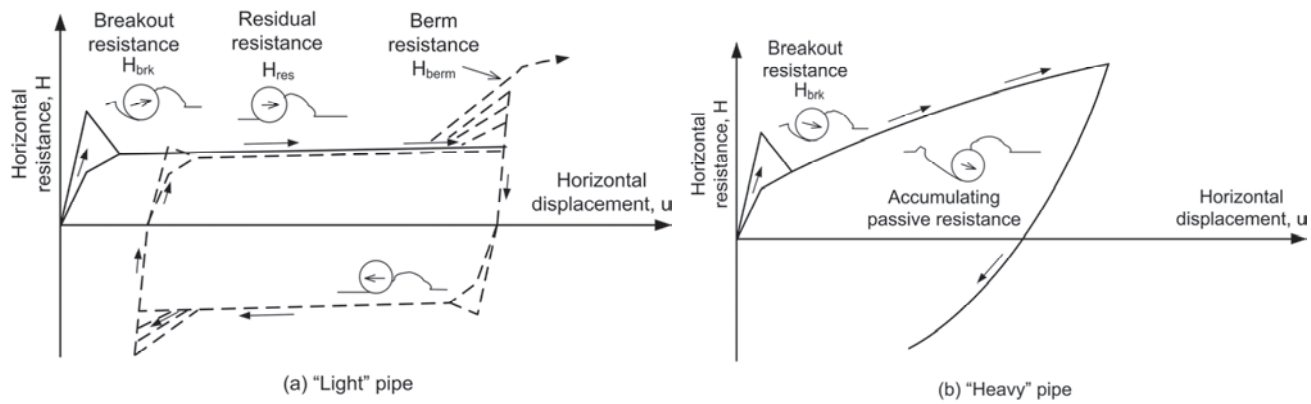

Figure 10 – Types of large-amplitude cyclic lateral response in undrained conditions

For assessment of fatigue performance of a pipeline during large amplitude cyclic movement, the size and strength of the growing berm formed ahead of the sweeping pipe segments and at the extremities of the pipe movement are significant [216].

Expressions for assessing lateral pipe-soil resistance at breakout and during cyclic motion can be calibrated and validated by laboratory models, centrifuge tests [232] [247], seabed test sections, or analytical methods.

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Annex A
(informative)

Additional information and guidance

NOTE The clauses in this annex provide additional information and guidance on clauses in the main text of this document. The title of each subclause identifies the subclause in the body of this document.

A.1 Scope

There is a large body of technical literature on offshore geoscience studies and foundation design. There are also regular conferences on these topics. The most up-to-date publications can generally be found in:

— the proceedings of the Offshore Technology Conference (OTC), Houston, Texas, which is held annually;

— the proceedings of the International Conference on the Behaviour of Offshore Structures (BOSS);

— the proceedings of various conferences of the Society of Underwater Technology, London, UK, held at less regular intervals;

— the proceedings of the ISFOG Perth 2005 Conference [63].

A review of the practice of the subjects covered by this annex may be found in Reference [1] of the Bibliography. General guidance on the application of soil mechanics theory to foundation design may be found in undergraduate and post-graduate textbooks, such as References [2] and [3], respectively.

A.2 Normative references

No guidance is offered.

A.3 Terms and definitions

No guidance is offered.

A.4 Symbols

No guidance is offered.

A.5 General requirements

No guidance is offered.

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A.6 Geotechnical data acquisition and integrated geoscience studies

A.6.1 Geotechnical assessment

Further information on using the results of integrated geoscience studies to perform a geotechnical assessment may be found in References [4] and [5].

A.6.2 Shallow geophysical investigation

No guidance is offered.

A.6.3 Geological modelling and identification of hazards

A.6.3.1 General

No guidance is offered.

A.6.3.2 Earthquakes

No guidance is offered.

A.6.3.3 Fault planes

This guidance addresses special considerations associated with the presence of non-seismogenic faults at the site of a platform, and how to account for their presence in foundation pile design.

A.6.3.3.1 Fault Identification and Mapping

Soil borings, soil sampling, and in-situ testing (e.g., cone penetrometer) typically are not effective in detecting the presence of faults in predominantly clay strata, such as are encountered in the Gulf of Mexico and in the deepwaters worldwide.

The most effective way to identify and define the extent of faults is through the use of a high-resolution geophysical survey designed to define the details of the foundation-zone stratigraphy. Examples of such surveys include subbottom profiler surveys (for example, an AUV—autonomous underwater vehicle—survey in deepwater) or other single- or multi-channel, ultra high-resolution 2-D or 3-D seismic surveys. Whenever feasible, a pair of survey cross lines should be run directly over each proposed foundation location.

If faults are identified within the foundation zone, their actual or projected intersection with the seafloor should be accurately mapped in relation to the proposed foundation locations on a seafloor topographic map or suitable seafloor image.

A.6.3.3.2 Relative Locations of Faults and Foundations

When the seafloor trace of a fault (or projected trace of any fault in the foundation zone) comes within 150 m (500 ft) of a proposed foundation location, the fault should be defined in 3-D space based on its dip angle and direction and trace azimuth. At a minimum, the fault should be indicated on the seismic cross-section(s) closest to the foundation, and the projection of the foundation should also be indicated on the seismic cross-section. The foundation should be positioned on the cross-section to clearly demonstrate the relative locations of the fault and the foundation, taking into consideration foundation offset from the seismic cross-section, geometric relationships between the fault plane and the seismic cross-section, and seismic velocity, as examples.

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A.6.3.3.3 Design considerations

A.6.3.3.3.1 Exclusion zones

It is always good engineering practice to avoid geohazards, whenever possible. If the foundation location can be moved easily so as to avoid the proximity of a fault, this should be the preferred solution. Reference [80] describes an approach to define an “exclusion zone” around each fault, in which a foundation should not be located. This design approach is that of geohazard avoidance, as further described in Reference [80] and consists of several steps, as follows.

— Map the fault trace(s) or upward projection of the fault plane(s) on a seafloor map.

— Quantify the uncertainty in the fault plane location due to survey positioning uncertainties, geological interpretation uncertainty, and fault-dip-angle uncertainty, and then define the zone in 3-D space (both in plan view and vertical cross-section below the seafloor) in which the fault plane could be located (the “fault-plane exclusion zone”). When defining the exclusion zone, the potential for possible future lateral extension (growth) of the fault plane along trend for faults that currently terminate in the area of interest should also be considered.

— Locate the foundation a “safe” distance outside the exclusion zone.

A.6.3.3.3.2 Fault-foundation intersections

In cases when the foundation location cannot be changed easily, it becomes necessary to investigate and quantify the effect(s) the presence of a nearby fault may have on the performance of the foundation, if any.

The foundation systems for several floating platforms for which some of the anchors were located near faults, or where the anchor/foundation piles were expected to intersect a fault plane, have been successfully designed and installed in the Gulf of Mexico, as documented in Reference [79], [80], and [81].

A possible approach consists of several steps, as follows.

a) Map the fault trace(s) or upward projection of the fault plane(s) on a seafloor map and quantify the uncertainty in the fault plane location as per A.6.3.3.3.1.

b) Determine the minimum offset range between the “fault-plane zone” and the foundation.

c) Determine if the fault is still moving (or “active”, but not seismically) and if it is, the expected rate of movement.

d) If the foundation intersects the fault plane, perform the necessary soil-foundation interaction analyses (e.g., Beam-Column, finite element) to evaluate the effects (if any) the maximum amount of fault movement expected to occur during the life of the platform could have on the foundation’s integrity and performance.

A.6.3.3.3.3 Effects on Soil Properties

The design of foundations in the immediate vicinity of faults should also consider the potential effects a fault may have on the soil characteristics along the fault plane and in its direct vicinity.

There is general evidence that some (but not all) faults or some points along some faults are clearly preferential and, sometimes major, fluid migration pathways. In general, the larger (deeper) the fault, the higher the potential that fluid migration is occurring or has occurred along the fault. The potential for fluid migration along fault planes, and the potential effects thereof, need to be considered on a case-by-case basis, with heavy reliance on geophysical and geological evidence.

Designers should consider the possible zone of disturbance around fault planes and the effect, if any, it might have on soil properties. However, unless proven otherwise by in situ tests or the testing of samples retrieved at or adjacent to the fault plane, there is no strong evidence that an inactive or a very slow moving compaction (or

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growth) fault would have a deleterious effect on the soil properties of the surrounding soils. Therefore, in general, no reduction in shear strength is recommended to evaluate the foundation capacity and behavior under load.

A.6.3.3.4 References

Additional guidance can be found in References [79] through [81].

A.6.3.4 Seafloor instability

No guidance is offered.

A.6.3.5 Scour and sediment mobility

No guidance is offered.

A.6.3.6 Shallow gas

No guidance is offered.

A.6.3.7 Seabed subsidence

Guidance on how to assess the magnitude of possible seabed subsidence may be found in Reference [6].

A.6.4 Geotechnical investigation

A.6.4.1 General

Shallow seismic survey data can allow geotechnical data to be extrapolated from soil borings. However, the degree of correlation between the geophysical and geotechnical data or the distance over which the stratigraphy from a borehole using geophysical results can be extrapolated depends on the nature and quality of the data, as well as on the geology of the site and the soil characteristics themselves.

A.6.4.2 Soil investigation and testing

Laboratory and *in situ* tests should be performed in accordance with procedures described in applicable national or International Standards, if available, which should be referenced in all relevant documentation.

The vertical soil profile can be assessed using continuous profile piezocone penetration test (PCPT) data and empirical correlation charts that relate PCPT results to soil classification[7]. Site-specific correlation charts, similar to the ones presented in Reference [8], can also be developed for this purpose. PCPT data can also be used to estimate trends in soil strength (for clays) and relative density (for sands) at the site. Borehole geophysical logging techniques can prove useful in identifying changes in strata and soil properties[9]. Consideration can be given to performing vertical seismic profiling in geotechnical boreholes or in seabed PCPTs for calibration of seismic survey data[10] [11] [12].

A general overview of offshore soil investigation techniques is given in References [13] and [14]. Recent advances, particularly related to deepwater investigations are described in References [15], [16] and [17]. Numerous *in situ* testing techniques other than PCPT and vane shear testing are available for determining *in situ* strength and other soil parameters. A comprehensive overview of such tests is given in Reference [18].

Significant scatter in the results of conventional laboratory shear tests can occur when performed on samples of silty clays and clayey silts. More consistent strength profiles can be achieved using the “Stress history and normalized soil engineering properties” (SHANSEP) soil strength testing described in References [19] and [20]. This methodology can be used to assess shear strength profiles in cohesive soils, as this technique addresses the critical variables that influence soil behavior, including

a) anisotropy,

b) sample disturbance,

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c) stress history changes,

d) coefficient of earth pressure at rest, and

e) rate effects.

A.6.4.3 Identification and classification of soils and rocks

Soils and rocks should be completely identified and classified with reference to a consistent classification system[21] [22] [23]. ISO 14688-1, ISO 14688-2 and ISO 14689-1 address identification and classification of soils[24] [25] and rocks[26].

Three groups of parameters are required to completely identify a soil as follows:

a) parameters necessary to define the nature of the soil, e.g. grain size, Atterberg limits, carbonate content, organic content, specific density of grains;

b) parameters related to the *in situ* state of the soil, e.g. water content, submerged unit weight, consistency of clays, relative density of sands;

c) mechanical parameters, e.g. parameters giving the strength and deformation properties of the material.

These parameters, with the exception of relative density of sands, can be obtained from laboratory tests. The soil should then be categorized according to predefined soil groups, assigned a group symbol, and thereby classified.

No universally recognized classification system is presently available for carbonate materials. A classification chart for the carbonate soils and rocks encountered in the Middle East area has been tentatively developed[27]. It is based on grain size, carbonate content, and unconfined compressive strength of materials. It is recognized today that parameters such as grain crushability or skeleton compressibility play an important role in assessing the engineering properties of carbonate materials. However, in the absence of a more definite classification scheme, the proposed chart can provide useful guidance.

The assessment of the *in situ* relative density of sands can be of particular importance for foundation design. Common practice in the offshore industry is to use empirical correlations between the relative density of siliceous sands and cone resistance as obtained from cone penetration tests. Because of the significant effects of the *in situ* stress conditions, correlations for a representative range of stress conditions should be used[28] [29] [30]. These correlations involve the use of an element of subjective judgment. In situations where this will have a significant impact on the reliability of the foundations, an alternative design analysis should be made using design procedures based directly on the cone resistance. If such procedures are not available for the particular situation being addressed, the alternative of a direct measurement of density *in situ* may be considered.

In assessing the quality and properties of rocks and rock masses, a distinction should be made between the behavior of rock material as measured on core samples and the behavior of a much larger rock mass that is affected by structural discontinuities, such as bedding plates, joints, shear zones, or solution cavities.

Useful guidance for classifying rocks may be found in the recommendations of the International Society for Rock Mechanics[31]. The following characteristics should be identified:

a) origin (e.g. sedimentary, volcanic),

b) strength of rock matrix,

c) degree of alteration of the rock mass, and

d) state of fracturation of the rock mass.

The rock quality may be quantified using the rock quality designation (RQD), which is an indication of the quality of the rock mass for engineering purposes.

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A.6.4.4 Carbonate soils

Information on carbonate soils is given in Annex B.

A.7 Stability of shallow foundations

A.7.1 General

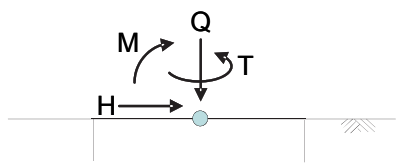
The equations provided herein are limited and are not necessarily appropriate for design in a number of situations, particularly if the soil exhibits contractancy. A common situation for which they cannot be applied arises when there is a strong layer overlying a weak layer within the zone of influence of the foundation. Irregularly shaped foundations can be difficult to analyze using the equations.

In circumstances such as these, general guidance cannot be provided and reliance should be placed on experience, published case histories, testing and numerical modelling.

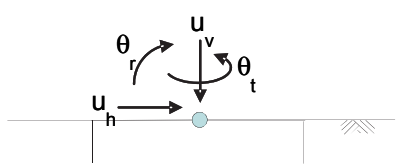
The bearing capacity factors used herein are considered the most commonly used, but alternative factors, particularly for circular foundations, are available and may be applied at the discretion of the designer.

A.7.1.1 Sign conventions

Sign conventions used in this document are illustrated in Figure A.1.



(A) Forces, Moments



(B) Displacements

Figure A.1—Shallow foundations sign conventions

A.7.2 Principles

A.7.2.1 Load transfer

For an embedded foundation, the loads applied to the top of the foundation are transferred to the foundation base level (tip of skirt for a skirted foundation). This is done by modifying the loads on the top of the foundation to account for

a) soil resistance on the sides of the embedded foundation,

b) passive and active soil resistance,

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c) foundation weight, and

d) soil weight within skirts (if applicable).

As noted in 7.2, embedded foundations or surface foundations equipped with skirts may have increased capacity under pure sliding conditions resulting from soil resistance above skirt tip level. If appropriate, this capacity may be used to offset horizontal loads applied to the foundation base, such as when calculating inclination factors. All contributions to horizontal capacity from foundation members above foundation base level should be reduced using the safety factor given in 7.3 for pure sliding conditions.

Primary contributions to resistance come from the combined effect of active and passive soil resistance, as well as potential side shear on members located parallel to the direction of lateral loading. The amount of side shear that may be adopted in design is a function of the shearing on the interface between the embedded member and the soil, and may be influenced by soil disturbance during installation and scour. Specific guidance is not provided in relation to calculating side shear.

For undrained conditions, the ultimate horizontal resistance (ΔH) that can be derived from active and passive soil resistance between seafloor and base level can be approximately calculated as follows:

$$\Delta H = K_{\mathrm{ru}} (s_{\mathrm{u\ ave}}) A_{\mathrm{h}} \tag{A.1}$$

where

$s_{\mathrm{u\ ave}}$ is the average soil strength from seafloor to base level;

A_{h} is the embedded vertical cross-sectional area of foundation;

K_{ru} is the undrained horizontal soil reaction coefficient.

K_{ru} depends on several factors, such as roughness, foundation shape, side shear, depth of embedment, and possible side gap between foundation and soil due to installation, cracking or scour. Where both active and passive resistance can be relied upon and significant scour is not expected, a value of K_{ru} = 4 is recommended. Where active soil resistance cannot be relied upon (such as due to cracking or installation disturbance) and where significant scour is not expected on the passive side of the foundation, a value of K_{ru} = 2 is recommended. In this case it may be appropriate to also account for the weight of soil within the passive soil wedge, although in such cases it should be verified that the total lateral resistance calculated does not exceed that which would be calculated using K_{ru} = 4. In strain softening soils it may not be appropriate to include the full soil resistance above skirt tip level in assessing overall sliding stability, due to strain compatibility issues, and specialist advice should be sought in these cases.

For the drained case, the ultimate horizontal resistance (ΔH) that can be derived from active and passive soil resistance between seafloor and base level can be approximately calculated as follows:

$$\Delta H = 0.5\, K_{\mathrm{rd}}\, \gamma'\, D_{\mathrm{b}}\, A_{\mathrm{h}} \tag{A.2}$$

where

γ' is the effective unit weight of soil;

D_{b} is the depth below seafloor to base level;

K_{rd} is the drained horizontal soil reaction factor.

K_{rd} depends on several factors, such as mobilized soil friction angle, roughness, foundation shape, side shear, depth of embedment, and possible side gap between foundation and soil from installation or from scour. Provided

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that the installation procedure and/or other foundation aspects do not require a more accurate assessment of the drained horizontal soil reaction factor, the following equation is recommended:

$$K_{rd} = K_p - (1/K_p) \quad \text{(A.3)}$$

where K_p is the coefficient of passive earth pressure, calculated as follows:

$$K_p = \tan^2 (45° + 0.5\,\phi') \quad \text{(A.4)}$$

A.7.2.2 Idealization of foundation area and the effective area concept

A.7.2.2.1 Idealization of foundation area

If the foundation stability problem is solved by using three-dimensional finite element analysis, the foundation area can be modelled with no or with only minor simplifications of the geometry. However, limiting equilibrium methods are generally based on a two-dimensional model (vertical slice) where the three-dimensional effects are included by defining the resistance of the vertical side areas. For irregular foundations shapes, this requires a rectangular idealization of the foundation area. This idealized area can be defined by a rectangle of width B and length L having the same area, A, and the same areal moments of inertia, I_x and I_y, as for the real area:

$$A_{\text{idealized}} = BL = A_{\text{real}} \quad \text{(A.5)}$$

$$I_{x,\text{idealized}} = I_{x,\text{real}} \text{ for action effects in the } y\text{-direction} \quad \text{(A.6)}$$

$$I_{y,\text{idealized}} = I_{y,\text{real}} \text{ for action effects in the } x\text{-direction} \quad \text{(A.7)}$$

The width B and length L of the idealized foundation area are determined by solving Equations (A.5) to (A.7).

A.7.2.2.2 Effective Area Method

Load eccentricity decreases the ultimate vertical load that a footing can withstand. This is accounted for in bearing capacity analysis by reducing the effective area of the footing according to empirical guidelines.

Figure A.2 shows footings with eccentric loads. The eccentricity, e, is the distance from the center of a footing to the point of action of the resultant, measured parallel to the plane of the soil-footing contact. The point of action of the resultant is the centroid of the reduced area. The distance e is M/Q, where M is the overturning moment and Q is the vertical load.

Where a skirted foundation incorporates a sealed base plate and the skirt compartment encapsulates soil of sufficiently low permeability, the vertical load used to calculate the effective area may include a contribution due to soil trapped within the skirted area. The following is noted.

a) Drained foundation analysis based on the effective area method should exclude the weight of the soil plug.

b) Where it is considered appropriate to use the weight of the soil plug, the effective soil weight should be used. Specialist geotechnical advice should be sought to ensure inclusion of the soil plug does not lead to less conservative foundation design. The latter comment relates specifically to soft soil sites, were the adopted soil strength is insufficient to support the effective soil weight at the depth of the skirts – sufficient support is an implicit assumption in the use of an effective area approach. In some cases it may be necessary to adopt alternative design approaches, such as those based on limit equilibrium methods.

The effective unit weight used in the analysis should be based on site investigation and laboratory data, and should incorporate an appropriate level of conservatism to account for any uncertainty. Generally, it will be conservative to adopt a lower bound profile of effective soil weight.

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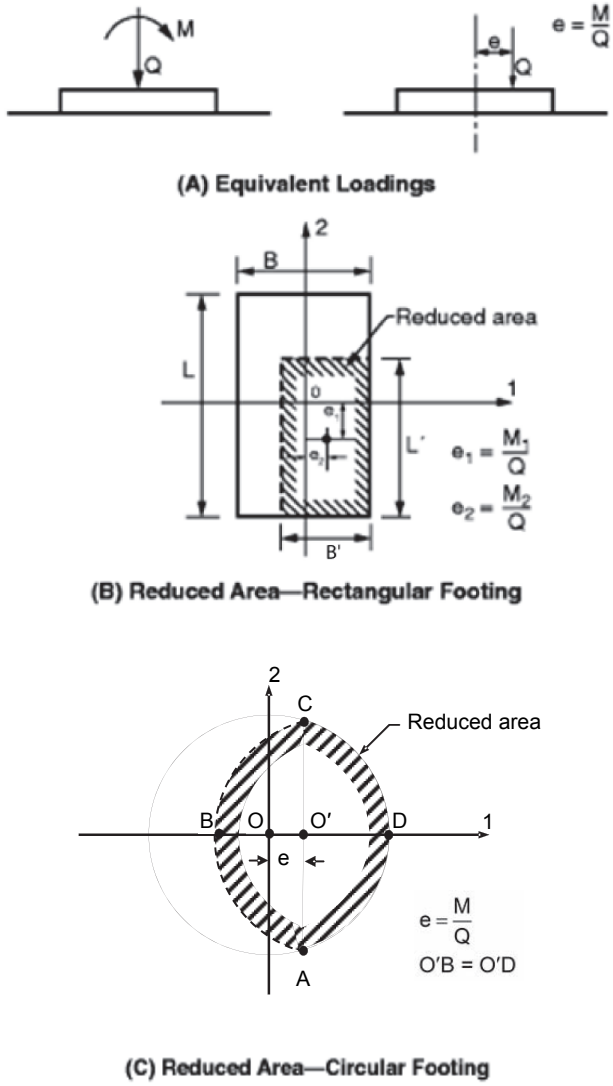

Figure A.2—Eccentrically-loaded footings

For a rectangular base area, Figure A.2b), eccentricity can occur with respect to either axis of the footing. Ideally, in conditions where eccentricity occurs in two directions specialist geotechnical advice should be sought. A simplified means of addressing this is to reduce the dimensions of the footing in both directions as follows:

$$L' = L - 2e_1$$

$$B' = B - 2e_2 \tag{A.8}$$

$$B' \leq L'$$

where L and B are the foundation length and width, respectively, the prime denotes effective dimensions, and e_1 and e_2 are eccentricities along the length and width.

Circular foundations subject to eccentric loads may be idealized as rectangular foundations by solving Equations (A.5) through (A.7). Alternatively, for a circular base with radius, R, the effective area may be assumed as shown in Figure A.2c). The centroid of the effective area is displaced a distance e from the center of the base. The effective area is then considered to be two times the area of the circular segment ADC.

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In addition, the effective area is considered to be rectangular with a length-to-width ratio equal to the ratio of line lengths AC to BD. The effective dimensions are therefore:

$$\left.\begin{aligned} A' &= 2s = B'L' \\ L' &= \left(2s\sqrt{\frac{R+e}{R-e}}\right)^{1/2} \\ B' &= L'\sqrt{\frac{R-e}{R+e}} \end{aligned}\right\} \tag{A.9}$$

where

$$s = \frac{\pi R^2}{2} - \left[e \times \left(\sqrt{R^2 - e^2}\right) + R^2 \arcsin\left(\frac{e^2}{R}\right)\right]$$

Examples of effective areas as a function of eccentricity area are shown in Figure A.3 in a dimensionless form.

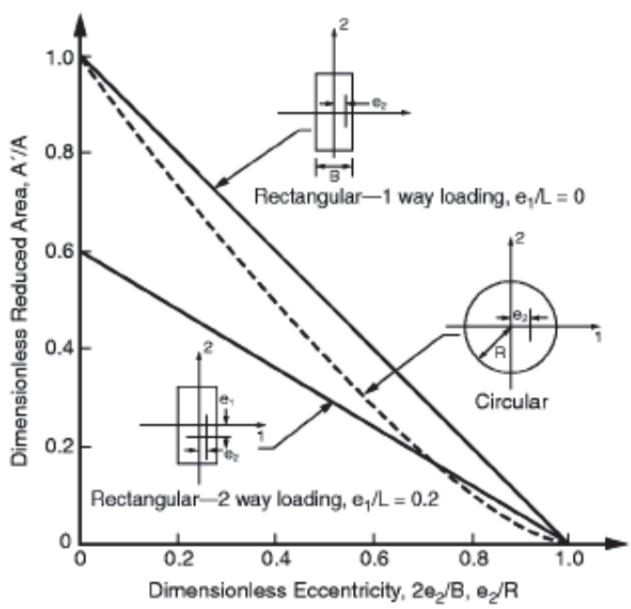

Figure A.3—Area reduction factors eccentrically loaded footings

No data are available on other foundation shapes. If appropriate, intuitive approximations may be made to find an equivalent rectangular or circular foundation when nonstandard shapes are encountered. For example, guidance for triangular shaped foundations is given in Reference [82]. Alternatively, an idealized rectangular foundation may be determined by solving Equation (A.5) through Equation (A.7).

NOTE Alternative methods may exist for assessing the effect of eccentricity in multiple direction[83] and these may be more applicable in more complex conditions.

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A.7.3 Acceptance criteria

A.7.3.1 General

Discussion on statistical analysis of soils is provided in Reference [84] and [85].

In many instances use of multiple footings can significantly increase overall foundation capacity, as illustrated in References [86] and [87].

Numerous references are available that provide additional information related to shallow foundation design in and/or general response of carbonate soils. Key references are as given in References [37], [38], and [64] (conference proceedings), and References [27], [41], [44], [47], [49], [50], [58], [60], [88], [89], [90], [91] and [92].

The methods outlined are strictly applicable to conditions of uniformly varying soil strength, although reasonable assessment of equivalent uniform properties can frequently be made. For example, the potential of a deep bearing failure depends on soil strengths at considerably greater depths than that of a sliding failure. Hence careful attention should be given to defining the soil parameters throughout the expected zone of influence.

Where foundation conditions are highly non-homogeneous or anisotropic; where load conditions deviate considerably from the simple conditions assumed in the stability formulae; where loading rates are such that the conditions are not clearly drained or undrained; or where foundation geometries are highly irregular, the use of these stability formulae is not straightforward and alternate procedures such as one or combinations of the following may be selected.

a) Use of conservative equivalent parameters along with the above mentioned formulae.

b) Use of limit analysis to determine bounds on collapse loads and to determine relative sensitivity of collapse loads to parameters of interest.

c) Use of numerical analysis to solve the governing equations directly.

d) Use of properly scaled model tests to check and verify calculation models and procedures, as described in Reference [94]. Examples of where this has been undertaken in practice are described in References [95] through [97].

A.7.3.2 Variations in safety factors

No guidance is offered.

A.7.3.3 Use in design

No guidance is offered.

A.7.3.4 Alternative method of design based on yield surfaces

A.7.3.4.1 General

An alternative method of design to assess foundation stability under combined vertical, horizontal, and overturning moment loading makes use of yield surfaces, as described below. Specialist geotechnical advice should be sought where this method is to be used for offshore design.

A.7.3.4.2 Background

Offshore foundations may experience a wide range of loading, encompassing combinations of vertical load (Q), lateral load (H) and overturning moment (M). The traditional design approach for shallow foundations (outlined in Clause 7) involves transforming the combined load into an equivalent vertical and lateral load acting on a reduced (effective) foundation area.

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Restrictions apply in the use of this design approach for offshore structures (such as in regards to allowing net tensile stress changes in the soil) which limit its general applicability. Recent literature also suggests that predicting bearing capacity of shallow foundations using the effective area method can lead to considerable under-prediction of capacity for some loading situations[97].

An alternative approach is to derive a fully encompassing yield surface in Q, H and M space. This may be used to provide an indication of the load combination required to cause the foundation to fail, as well as an explicit indication of the effect of a change in individual load components on the foundation proximity to failure. The yield surface method can also be extended to define the load-displacement response of a foundation if used in conjunction with a flow rule. In undrained conditions, when normality can be assumed, the flow rule can be directly derived from the yield surface.

A.7.3.4.3 Application to design

The general procedure for developing a yield surface for use in design involves the following.

a) Defining the 'uniaxial' ultimate limit states Q_{ult} (H = M = 0), H_{ult} (M = 0; and Q = 0 where tensile stresses are allowed beneath the foundation, or Q = Q_{ult}/2 if no tensile stresses permitted) and M_{ult} (H = 0; and Q = 0 where tensile stresses are allowed beneath the foundation, or Q = Q_{ult}/2 if no tensile stresses permitted) to define the apex points of the yield surface.

b) Defining the shape of the interaction diagram through use of closed form expressions as f(Q/Q_{ult}, H/H_{ult}, M/M_{ult}).

The magnitude of the uniaxial capacity and the shape of the yield surface depend on the soil response to loading (undrained or drained), the soil strength profile (uniform or heterogeneous), foundation shape, foundation embedment, structural connection between adjacent foundations, and inclusion of tension between the footing and the soil.

A.7.3.4.4 Prediction of uniaxial capacity

Uniaxial capacity Q_{ult}, H_{ult} and M_{ult} for undrained conditions can be derived from existing recommendations, although more accurate and rigorous solutions have become available in recent years. Key references include

— for Q_{ult}: References [98] through [103], and

— for H_{ult} and M_{ult}: References [104] through [108].

Closed-form expressions are available to describe the shape of the yield surface for a selection of cases. A review of the current literature for these cases is discussed in the following clause.

A.7.3.4.5 Yield surfaces for select cases

Historically, yield surfaces for undrained conditions have been based on analytical and numerical studies while those for drained conditions have been derived from experimental studies. The latter is due to the relative complexity of an analytical approach for drained soil conditions.

A.7.3.4.6 Undrained conditions (ϕ = 0)

A.7.3.4.6.1 General

Uplift capacity can in many instances be mobilized beneath surface or skirted foundations due to suction and can potentially be relied on for the duration over which undrained conditions prevail. Engineering judgment is required to determine whether suctions will be generated and the duration over which they can be maintained.

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A.7.3.4.6.2 Surface foundations with zero-uplift capacity along the foundation-soil interface

The general form of the three-dimensional yield surface in vertical, horizontal and moment load space for undrained failure of a surface foundation with zero-uplift along the foundation-soil interface is shown in Figure A.4 in terms of normalized loads, Q/Q_{ult}, H/H_{ult} and M/M_{ult} (vertical load Q is denoted by V in the figure). The surface is symmetrical in the H–M plane and exhibits diminishing moment capacity as vertical load (Q) falls below 0.5 Q_{ult} as the foundation begins to lift-off from the seabed.

Reference [113] presents yield surfaces for rectangular surface foundations and shows the normalized shape of the failure is unique for aspect ratios over the range 0 < B/L < 1, and presents closed-form expressions for the shape of the failure surface and the uniaxial capacities defining its apex points. Reference [111] presents a similar "scallop-shaped" yield surface for circular foundation geometry but do not present a closed-form expression to enable direct comparison of the shape of the yield surface. Existing studies have addressed homogeneous soil strength, and further work is required to extend the cases to linearly increasing shear strength profiles.

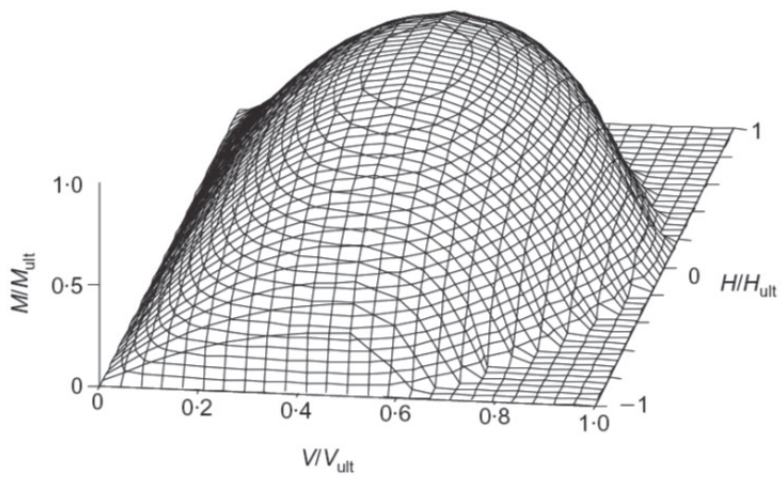

Figure A.4— Yield surface for undrained conditions for a surface foundation with zero-uplift capacity along the foundation-soil interface[113]

A.7.3.4.6.3 Surface foundations with unlimited-uplift capacity along the foundation-soil interface

Uplift capacity provided by foundation skirts can be conceptually represented by modelling a surface foundation with an unlimited tension interface. The general form of the yield surface for undrained failure of a surface foundation with an unlimited tension interface is shown in Figure A.5. The "walnut-shaped" surface is asymmetric in the H–M plane, with maximum moment capacity mobilized in conjunction with a horizontal load acting in the same direction (i.e. clockwise and left-to-right or vice versa). Moment capacity continues to increase with diminishing vertical load, contrary to the zero-tension interface case, as a foundation with unlimited uplift capacity will not lift-off from the seabed.

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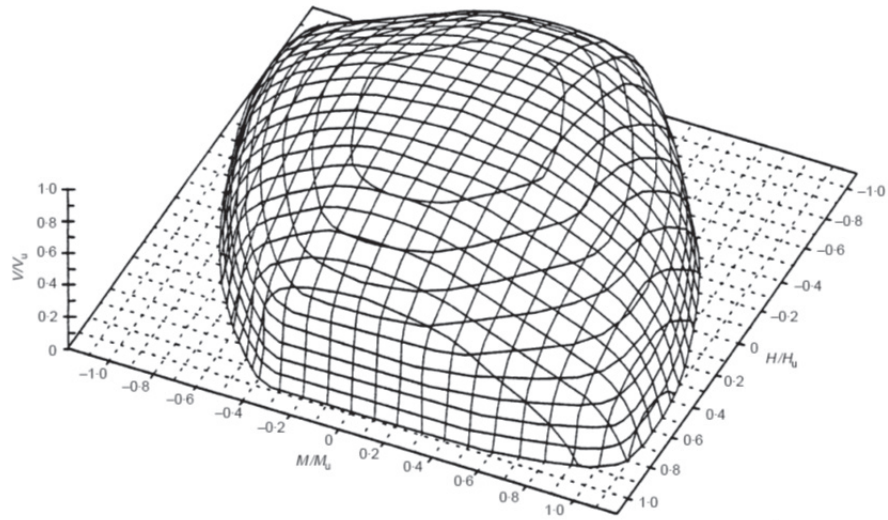

Figure A.5—Yield surface for undrained conditions for a surface foundation with unlimited-uplift capacity along the foundation-soil interface[110]

Yield surfaces have been derived for strip, rectangular and circular foundation geometry, and homogeneous and heterogeneous soil strength, although not comprehensively for all combinations (References [104], [109], [110], [106], [113], and [97]). Closed-form expressions have been derived for the selected cases of a strip foundation and linearly increasing shear strength[104] and for a circular foundation and uniform shear strength[110]).

The shape of the yield surface for foundations with unlimited-uplift capacity depends on foundation geometry and soil strength profile. In some cases the normalized size of the yield surface decreases with increasing degree of heterogeneity; therefore scaling an interaction diagram derived for homogeneous soil strength by uniaxial ultimate limit states appropriate to a heterogeneous soil strength profile would be unconservative[106] [97].

A.7.3.4.6.4 Embedded foundations

The coupling of the horizontal and moment degrees of freedom when a foundation is physically embedded leads to an asymmetric and oblique failure surface in the HM plane and the asymmetry and obliqueness become more pronounced with increasing embedment ratio. The general form of the yield surface for undrained failure of embedded foundations is shown in Figure A.6 (vertical load, Q, is denoted "V" in the figure).

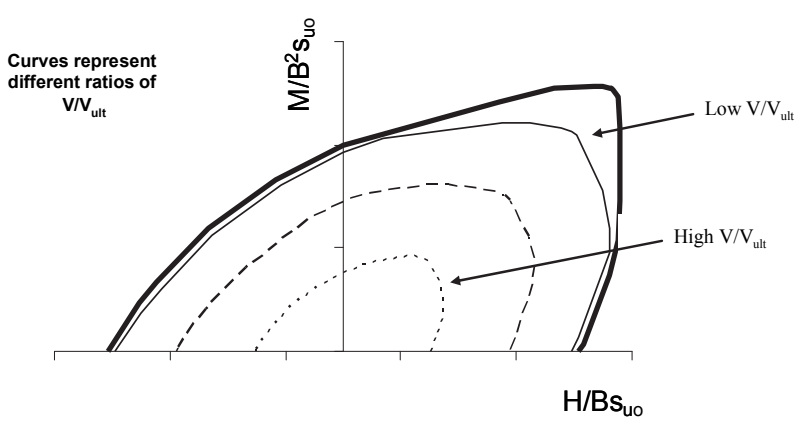

Figure A.6— Yield surface for undrained conditions for an embedded foundation[97]

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Reference [108] presents yield surfaces in general load space (Q, H, M) for shallow foundations with embedment ratios between zero and 1 for uniform soil strength. Reference [112] considers the HM interaction (Q = 0) of shallow strip foundations with embedment ratios between 0 and 1 and for both uniform and 'normally consolidated' soil strength profiles, and proposes a transformation of the moment reference point to allow the shape of the yield failure surface to be approximated by a closed-form expression.

Existing studies have considered full bonding (i.e. an unlimited tension interface) along the sides and base of the foundation, and studies are required to address the effect of gapping and the foundation-soil interface roughness.

Existing studies have considered embedment in terms of a solid plug, although it is possible that the capacity of a skirted foundation can be reduced because of the intrusion of the failure mechanism into the soil plug. Studies investigating failure of skirted foundations are also required.

Key references relating to yield surfaces in undrained conditions include Reference [97], [104], [106], and [108] through [113].

A.7.3.4.7 Drained conditions

Yield surfaces for drained conditions incorporate isotropic strain-hardening to accommodate increasing shear strength with increasing stress level. The shape of the yield surface is assumed to be unique and the isotropic expansion and contraction of the surface is defined by a hardening rule (vertical load-displacement relationship). Under drained conditions tension cannot be sustained beneath a foundation and therefore the foundation will lift-off from the seabed under moment loading in conjunction with vertical loads $Q < 0.5\ Q_{ult}$.

A.7.3.4.7.1 Surface foundations

The general form of the yield surface for drained failure of a surface foundation is shown in Figure A.7. The "rugby ball-shaped" surface is parabolic in planes of QH and QM and a rotated ellipse in planes of HM (vertical load, Q, is denoted "V" in the figure). Maximum horizontal load and moment capacity are mobilized in conjunction with a vertical load $Q = 0.5Q_{ult}$ and maximum moment capacity is mobilized in conjunction with horizontal load acting in opposition (i.e. clockwise and right-to-left or vice-versa).

Reference [119] proposed the yield surface shown in Figure A.7 along with a closed-form expression to describe its shape. The yield surface was based on results from various experimental studies on rough, rigid, plane strain and rectangular footings on dense silica sand[114], [115], [117]. A subsequent study considering circular foundations on loose carbonate sand showed a similar form of yield surface and for which a closed-form expression was proposed[121].

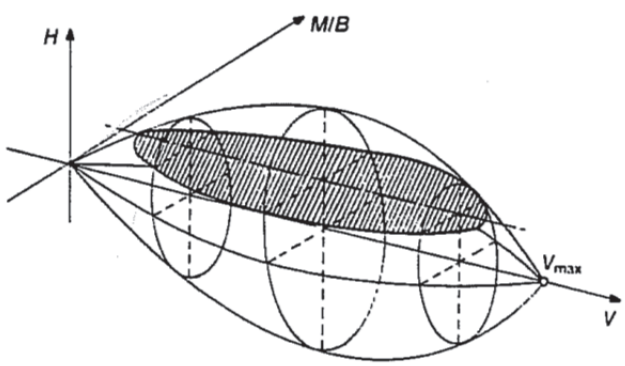

Figure A.7— Yield surface for drained conditions for a surface foundation[119]

A.7.3.4.7.2 Embedded foundations

The additional capacity available from foundation embedment is accounted for by scaling the envelope for a surface foundation (as shown in Figure A.7) by its apex points H_{max}/Q and M_{max}/Q [116], [118], [120].

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Key references relating to yield surfaces in drained conditions include References [114] through [126].

A.7.4 Undrained bearing capacity — constant shear strength with depth

A.7.4.1 Bearing capacity formulae

In general, the equations and factors outlined below should be used with care and their applicability should be checked in each case. Where appropriate, alternative methods of analysis and/or design approaches should be considered to verify the results obtained.

A.7.4.2 Bearing capacity factors

A.7.4.3 No additional guidance is offered. Bearing capacity correction factors

A.7.4.3.1 Correction Factor

The correction factor K_c is written as follows:

$$K_c = 1 + s_c + d_c - i_c - b_c - g_c \tag{A.10}$$

where s_c, d_c, i_c, b_c, and g_c are individual correction factors related to foundation shape, embedment depth, load inclination, base inclination and seafloor surface inclination respectively.

The recommended correction factors s_c, i_c, b_c, and g_c are adopted from Reference [32]. The recommended d_c is taken from Reference [36] and is slightly more cautious than specified by Brinch Hansen[32].

The relevancy of using the depth factor d_c should be evaluated in each case. If the installation procedure and/or other foundation aspects, such as scour, do not allow for the required mobilization of shear stresses in the soil above foundation base level, it is recommended that $d_c = 0$. In particular, it is recommended that $d_c = 0$ if the horizontal load leads to mobilization of significant passive earth pressure between seafloor and foundation base level.

A.7.4.3.2 Shape factor

$$s_c = 0.18\,(1 - 2\,i_c)\,(B'/L') \tag{A.11}$$

A.7.4.3.3 Depth factor

$$d_c = 0.3 \arctan (D/B') \tag{A.12}$$

A.7.4.3.4 Inclination factor

$$i_c = 0.5 - 0.5[1 - H'/(A's_u)]^{0.5} \tag{A.13}$$

$$b_c = 2\nu / (\pi + 2) \approx 0.4\nu \tag{A.14}$$

$$g_c = 2\beta / (\pi + 2) \approx 0.4\beta \tag{A.15}$$

where ν and β are base and ground inclination angles in radians. Figure A.8 defines these angles for a general foundation problem.

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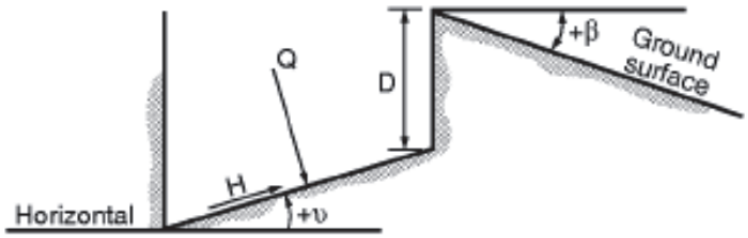

Figure A.8—Definitions for inclined base and seafloor surface[32]

NOTE H' in Equation A.13 refers to the load applied to the effective area component of the base only. This corresponds to the total lateral load applied to the foundation minus any soil resistance acting on the foundation above skirt tip level as outlined in A.7.2.1, and minus any lateral resistance that may be carried by shearing at skirt tip level outside the effective area. The safety factor for pure sliding conditions should be applied to these two resistance components prior to them being subtracted from the total lateral load.

A.7.5 Undrained bearing capacity — linearly increasing shear strength

A.7.5.1 Bearing capacity formulae

In general, the equations and factors outlined below should be used with care and their applicability should be checked in each case. Where appropriate, alternative methods of analysis and/or design approaches should be considered to verify the results obtained.

A.7.5.2 Bearing capacity correction factors

A.7.5.2.1 Correction factors

The correction factors F and K_c are defined as:

F is an empirical value taken from Figure A.9

$$K_c = 1 + s_c + d_c - i_c - b_c - g_c \quad \text{(A.16)}$$

where

s_c is a foundation shape correction factor;

d_c is a embedment depth correction factor;

i_c is a load inclination correction factor;

b_c is a base inclination correction factor;

g_c is a seafloor surface inclination factor.

NOTE In selection of F, rough conditions may generally be adopted for unpainted skirted foundations. Values of F may be approximated using the relationship:

$$F \approx a + bx - ((c+bx)^2 + d^2)^{0.5} \quad \text{(A.17)}$$

where

$x = \kappa B'/s_{uo}$ and is valid for $0 \leq \times \leq 25$;

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a, b, c, and d are constants that vary with roughness and are outlined in Table A1.

Table A.1—Correction factor for circular or square foundations

Constant	Fully rough interface (F_r)	Fully smooth interface (F_s)
a	2.56	1.372
b	0.457	0.07
c	0.713	-0.128
d	1.38	0.342

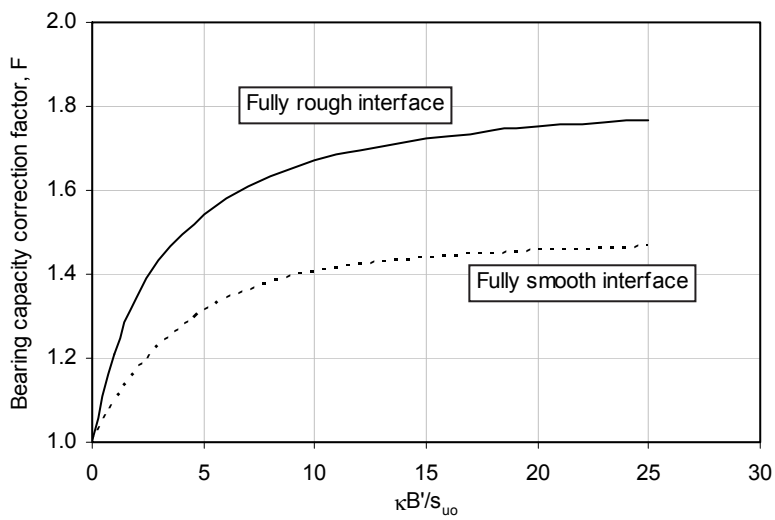

Figure A.9—Bearing capacity correction factor F for linearly increasing undrained shear strength with depth

The recommended correction factors are taken from the References [32] through [36]. The same guidance provided in A.7.4.3 for depth correction factor also applies for this case.

A.7.5.2.2 Shape factor

$$s_c = s_{cv}\,(1-2i_c)\,(B'/L') \tag{A.18}$$

where s_{cv} is taken from Table A.2. Values of s_{cv} in Table A.2 may be approximated using the relationship:

$$s_{cv} \approx 0.18 - 0.155x^{0.5} + 0.021x \tag{A.19}$$

where

x = $\kappa B'/s_{uo}$ and is valid for $0 \le x \le 10$.

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Table A.2—Shape factor for circular or square foundations for pure vertical loads

$\kappa B'/s_{uo}$	s_{cv}
0	0.18
2	0.00
4	-0.05
6	-0.07
8	-0.09
10	-0.10

A.7.5.2.3 Depth factor

$$d_c = 0.3\,(s_{u\,ave} / s_{u2}) \arctan(D/B') \tag{A.20}$$

where

$s_{u\,ave}$ is the average shear strength above base level;

s_{u2} is the equivalent shear strength below base level, and $s_{u2} = F(N_c\, s_{uo} + \kappa B'/4) / N_c$.

A.7.5.2.4 Inclination factors

$$i_c = 0.5 - 0.5[1 - H'/(A' s_{uo})]^{0.5} \tag{A.21}$$

$$b_c = 2\nu / (\pi + 2) \approx 0.4\nu \tag{A.22}$$

$$g_c = 2\beta / (\pi + 2) \approx 0.4\beta \tag{A.23}$$

NOTE H' in Equation A.21 refers to the load applied to the effective area component of the base only. This corresponds to the total lateral load applied to the foundation minus any soil resistance acting on the foundation above skirt tip level as outlined in A.7.2.1, and minus any lateral resistance that may be carried by shearing at skirt tip level outside the effective area. The safety factor for pure sliding conditions should be applied to these two resistance components prior to them being subtracted from the total lateral load.

A.7.6 Drained bearing capacity

A.7.6.1 Bearing capacity formulae

A.7.6.1.1 General

The development of the bearing capacity equation in 7.6 is predicated on the assumption that the soil is a rigid, perfectly plastic material that obeys the Mohr-Coulomb yield criterion.

The procedures recommended here are adopted from Reference [32].

A.7.6.1.2 Exclusion of effective cohesion from bearing capacity formulae

As stated in the main text, the equation for drained bearing capacity has deliberately omitted any component due to an effective cohesion, c', and accompanying bearing capacity factor, N_c. The main reason for this is that the occasions when it might be appropriate to include a component for bearing capacity due to a presumed effective cohesion are extremely rare, and specialist geotechnical advice should be sought before any such inclusion.

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The effective strength envelope for a given soil is often quoted in terms of a "cohesion intercept", c', and effective friction angle, ϕ', with the envelope fitted to results of laboratory tests conducted at different levels of effective confining stress. There has been much debate over whether the deduced c' reflects a true cohesion (or cementation) or is merely an artefact resulting from fitting a tangent to what is, in reality, a curved strength envelope. It has been well established that the friction angle for soils increases as the mean effective stress level decreases, due to increasing dilation. As such, in many cases the effective cementation is an artificial quantity arising from the interpretation of the laboratory tests rather than a true physical quantity. However, there is considerable debate over this among geotechnical specialists as documented in Reference [127].

Examples of where inclusion of an effective cohesion in estimating bearing capacity might be argued as appropriate include:

a) naturally cemented soils (particularly sands);

b) medium to heavily overconsolidated clays.

— In the first of these, care is needed because of potentially different mobilization rates for the cemented and frictional components of soil strength, and the possibility that progressive failure might occur, with the cementation reducing to zero before the full frictional strength was mobilized.

— In the second example, ignoring any effective cohesion (or dilation-induced high friction angles at low mean effective stresses) might prove over-conservative. However, inclusion of an effective cohesion in estimating bearing capacity could also prove unduly optimistic, partly because the mean effective stress levels associated with the (drained) bearing capacity might be too high to justify any non-zero effective cohesion, and partly because the level of displacement allowed in design might be too low to fully mobilize the effective cohesion (or dilation-induced high friction angles at low mean effective stresses).

A.7.6.2 Bearing capacity factors

Equations for factors N_q and N_γ are given in the main text under the discussion of Equation 6. Figure A.10 provides a plot and tabulation of these factors for varying friction angles, ϕ'. The plot is provided for ϕ' between 30 degrees and 42 degrees, which are considered reasonable limits for general use with these equations. Friction angles that fall outside of these limits may indicate non-standard soils and should be considered by specialist geotechnical engineers only. **As indicated in 7.7, $\phi'$is based on the friction angle relevant for triaxial conditions.**

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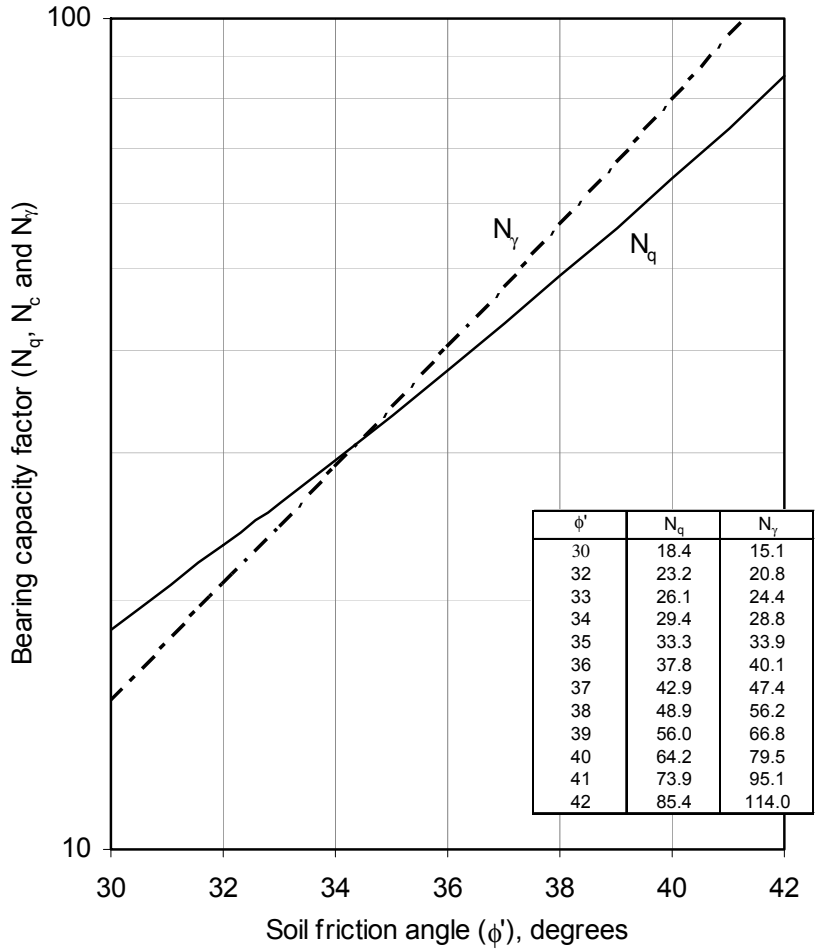

ϕ'	N_q	N_γ
30	18.4	15.1
32	23.2	20.8
33	26.1	24.4
34	29.4	28.8
35	33.3	33.9
36	37.8	40.1
37	42.9	47.4
38	48.9	56.2
39	56.0	66.8
40	64.2	79.5
41	73.9	95.1
42	85.4	114.0

Figure A.10—Recommended bearing capacity factors

A.7.6.3 Bearing capacity correction factors

A.7.6.3.1 Correction Factors

The correction factors K_q, and K_γ are written

$$K_q = i_q \times s_q \times d_q \times b_q \times g_q \tag{A.24}$$

$$K_\gamma = i_\gamma \times s_\gamma \times d_\gamma \times b_\gamma \times g_\gamma$$

where i, s, d, b, and g are individual correction factors related to load inclination, foundation shape, embedment depth, base inclination and seafloor surface inclination respectively. The subscripts q and γ identify the bearing capacity factor (N_q or N_γ) with which the correction term is associated.

Note that the factors given below for seafloor inclination may not be conservative in cases of very soft soil in sloping ground, and specialist advice should be sought in these cases.

Recommended expressions for the correction factors are described in A.7.6.3.2 through A.7.6.3.5.

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A.7.6.3.2 Load inclination factors

$$\left.\begin{aligned} i_q &= [1 - 0.5(H/Q)]^5 \\ i_\gamma &= [1 - 0.7(H/Q)]^5 \end{aligned}\right\} \quad \text{(A.25)}$$

NOTE H in A.7.6-2 refers to the load applied to the effective area component of the base only. This corresponds to the total lateral load applied to the foundation minus any soil resistance acting on the foundation above skirt tip level as outlined in A.7.2.1, and minus any lateral resistance that may be carried by shearing at skirt tip level outside the effective area. The safety factor for pure sliding conditions should be applied to these two resistance components prior to them being subtracted from the total lateral load.

A.7.6.3.3 Shape factors

$$\left.\begin{aligned} s_q &= 1 + i_q\,(B/L')\sin\phi' \\ s_\gamma &= 1 - 0.4\,i_\gamma\,(B/L') \end{aligned}\right\} \quad \text{(A.26)}$$

A.7.6.3.4 Depth factors

$$\left.\begin{aligned} d_q &= 1 + 1.2\,(D/B')\tan\phi'\,(1 - \sin\phi')^2 \\ d_\gamma &= 1 \end{aligned}\right\} \quad \text{(A.27)}$$

It should be emphasized that the effect of foundation embedment is very sensitive to soil disturbance at the soil/structure interface along the sides of the embedded base. Where significant disturbance is expected, including scour, it may be prudent to reduce or discount entirely the beneficial effect of overburden shear strength.

In addition, it is recommended that $d_c = 0$ if the horizontal load leads to mobilization of significant passive pressure between seafloor and foundation base level.

A.7.6.3.5 Base and seafloor surface inclination factors

Base inclination:

$$\left.\begin{aligned} b_q &= e^{-2\nu\tan\phi'} \\ b_\gamma &= e^{-2.7\nu\tan\phi'} \end{aligned}\right\} \quad \text{(A.28)}$$

Seafloor slope:

$$g_q = g_\gamma = (1 - 0.5\tan\beta)^5 \quad \text{(A.29)}$$

A.7.7 Shear strength used in bearing capacity calculations

As indicated in 7.7, the value of ϕ' used in calculating the drained bearing capacity equations in A.7.6 should be relevant to triaxial conditions (ϕ'_{TX}). It is noted that Reference [32], used as the basis of the calculation approach for drained bearing capacity, recommends use of plane strain ϕ'_{PS}. Further, it is commonly assumed that ϕ'_{PS} is 10 % higher than ϕ'_{TX}. **Use of triaxial conditions in this case is recommended so as to obtain comparable safety margins when using this document versus the partial factor design method in ISO 19901-4 (where ϕ'_{PS} is used).**

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One approach for deriving appropriate cyclic soil strengths for use in pseudo-static stability analysis, taking account of load history and average shear stress, is outlined in Reference [128].

Non-cohesive soil may be assessed using cyclic undrained soil strength derived in a manner similar to cohesive soils. A procedure for how this can be done is presented in Reference [96].

A.7.8 Response of shallow foundations to static and pseudo-static loading

A.7.8.1 General

Static deformations are generally considered to be of two types. Short term deformation is the more or less instantaneous response of a foundation to loading and primarily results from shear deformation (shear straining) of the soil. Long term deformation occurs over a period of time and is primarily associated with a gradual dissipation of excess pore pressure and attendant volume changes of the soil. In addition, secondary displacement due to creep may occur.

A.7.8.2 Short term displacement (undrained loading)

Because soils exhibit non-linear, path dependent behavior under load the short term deformation problem is quite complex. For monotonic, low level loads (with respect to failure loads) estimates of deformation can be made assuming the soil to be a homogeneous linearly elastic material.

Solutions for conditions other than those given by Equation 9 through Equation 12, including point displacements within the soil mass itself can be found in Reference [129].

Considerable care must be exercised in determining the elastic constants of the soil since the elastic moduli of soils depend strongly on the level of effective mean stress. This is particularly significant for highly permeable granular soils where equivalent moduli must be selected from some weighted average mean stress taken over the volume of soil subjected to significant stresses. For relatively impermeable soils such as clays a correlation of modulus with strength and overconsolidation ratio may lead to satisfactory results. Further discussion of these points is presented in Reference [130]

Where the foundation base is flexible or the loading is sufficiently severe to create high stresses throughout a significant volume of soil, Equations 9 through 12 are inappropriate and numerical analyses may be required. Finite element and finite difference techniques have the capability of including complex geometries and loadings and nonlinear, variable soil profiles. Special consideration should be given to the potential effects of softening of the soil (reduction in modulus) under cyclic loading.

A.7.8.3 Long-term displacement (primary settlement)

The long-term settlement of a foundation on clay is a three-dimensional problem in which stress distributions and pore pressures are coupled. Complex numerical schemes are therefore necessary to determine acceptable solutions. Such schemes may be necessary to determine such things as creep, load redistributions, and differential settlements; and to account for important initial conditions such as excess pore pressures. Equation 13 is a widely used simplified estimate of long term or consolidation settlement obtained by assuming a one-dimensional compression of soil layers under an imposed vertical stress.

Because of the finite extent of the foundation, the vertical stress imposed by the structure should be attenuated with depth. An estimate of such attenuation can be determined from elastic solutions such as those referenced above by Reference [129]. This approximate method is particularly appropriate where settlement is governed by thin, near-surface layers.

The rate at which settlement will occur can be estimated according to methods described in many soil mechanics texts, for example, Reference [131] and Reference [132].

A.7.8.4 Long-term displacement (secondary settlement)

No guidance is offered.

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A.7.8.5 Long-term displacement (regional)

No guidance is offered.

A.7.9 Response of shallow foundations to environmental loading

A.7.9.1 General

In many cases the foundation can be treated as an elastic half space subject to the restrictions outlined in A.7.8, and the loading is considered pseudo-static. Consequently, the stiffness of the soil can be calculated in a manner similar to that suggested by Equations 9 through 12.

Note that half space solutions can be considerably in error where non-uniform soil profiles exist. In addition, for large amplitude environmental loading non-linear soil behavior may be significant. In such cases a numerical analysis may be required or at least a study of a range of soil stiffness properties should be considered.

In some cases it will not be appropriate to treat the foundation as an elastic half space, such as where it becomes necessary to model the energy loss in the soil. Specialist advice should be sought in these cases.

A.7.9.2 Cyclic loading

A special case of environmental loading is the response of offshore foundations to cyclic loading arising from waves. The various displacement components and how they may be evaluated are discussed in References [133] and [134].

A.7.10 Hydraulic stability

No guidance is offered.

A.7.11 Installation and removal

No guidance is offered.

A.7.12 Shallow foundations equipped with seabed penetrating skirts

A.7.12.1 Penetration of skirts

Skirts can provide a significant resistance to penetration. This resistance, Q_r can be estimated as a function of *depth by the following:*

$$Q_r = Q_f + Q_p + = f A_s + q A_p \tag{A.30}$$

where

Q_f is the shaft friction resistance;

Q_p is the total end bearing resistance;

f is the unit shaft friction;

A_s is the side surface area of skirt embedded at a particular penetration depth (including both sides);

q is the unit end bearing pressure on the skirt tip;

A_p is the end area of skirt tip.

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The end bearing components can be estimated by bearing capacity formulae or alternatively by the direct use of cone penetrometer resistance q_c corrected for shape difference. The latter may not be directly applicable for wide concrete skirts. The shaft friction resistance can be determined by laboratory testing or other suitable experience. In most cases it is highly desirable to achieve full skirt penetration. This should be considered in selecting soil strength properties (or CPT q_c values, see C.8.1.4.2) for use in analysis as low estimates of strength are non-conservative in this case.

General guidance on assessing skirt penetration in North Sea soils based on cone penetrometer resistance can be found in Reference [36].

For non-standard soils, such as deep water clays found offshore West Africa and carbonate soil, specialist advice may be required.

The foundation surface should be prepared in such a way to minimize high localized contact pressures. If this is not possible grout can be used between the structure foundation and soil to ensure intimate contact. In this case the grout must be designed so that its stiffness properties are similar to that of the soil.

In general, water will be trapped within the shear skirt compartments. The penetration rate should be such that removal of the water can be accomplished without forcing it under the skirts and damaging the foundation. In some cases a pressure drawdown (i.e. negative excess pore pressures) can be used to increase the penetration force, although analysis should be carried out to ensure that damage to the foundation will not result.

In assessing the penetration of skirts careful attention should be given to site conditions. An uneven seafloor, lateral soil strength variability, existence of boulders etc., can give rise to uneven penetration or structural damage of skirts. In some cases site improvements may be required such as levelling the area by dredging or fill emplacement.

A.7.12.2 Removal

During removal suction forces will tend to develop at the foundation base and the tips of skirts. These forces can be substantial but can usually be overcome by sustained uplift forces or by introducing water into the base compartments to relieve the suction.

NOTE Set-up effects may result in higher extraction resistance than installation resistance.

A.7.13 Shallow foundations without seabed penetrating skirts

No guidance is offered.

A.7.14 Installation effects

The use of relatively small installation vessels can result in foundation failure; vessel heave motions greater than crane pay-out speed can result in the structure having excessive velocity at impact with the seafloor and multiple set-downs. The result is often that the foundation is pulled up after first set-down, thus generating a pullout failure in the soil (reverse bearing capacity failure). The soil condition under the structure after such an event may be close to a remolded state and normal safety factors can be too small to prevent foundation failure during final set-down.

If there is a risk of significant heave motions or impact of touchdown, higher safety factors than outlined in 7.3 should be applied.

A.7.15 Sliding stability

No guidance is offered.

A.7.16 Torsional stability

Methods that may be used for assessing torsional stability are provided in References [135] and [136].

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Annex B
(informative)

Carbonate soils

B.1 General

Carbonate soils cover over 35 % of the ocean floor. For the most part, these soils are biogenic. That is, carbonate soils are composed of large accumulations of the skeletal remains of plant and animal life, such as coralline algae, coccoliths, foraminifera, and echinoderms. To a lesser extent, carbonate soils also exist as non-skeletal material in the form of oolites, pellets, grape-stone, etc. These carbonate deposits are abundant in the warm, shallow water of the tropics, particularly between the 30° north and south latitudes. Deep-sea carbonate oozes have been reported at locations considerably outside these mid-latitudes. Since temperature and water conditions (water depth, salinity, etc.) have varied throughout geological history, ancient deposits of carbonate material can be found buried under more recent terrestrial material outside the present zone of probable active deposition. Major carbonate deposits are known to exist in the Gulf of Mexico along the Florida coastline and in the Bay of Campeche.

The comments below are focused primarily on carbonate silts and sands. Clay soils with varying proportions of carbonate content are common offshore, but there is little guidance as to how conventional design approaches for clay soils should be modified for different carbonate content. Local experience is important in making such assessments.

B.2 Characteristic features

Carbonate soils differ in many ways from silica-rich soils. An important distinction is that the major constituent of carbonate soils is calcium carbonate, which has a low hardness value compared to quartz (the predominant constituent of the silica-rich sediments). Susceptibility of carbonate soils to disintegration (crushing) into smaller fractions at relatively low stress levels is partly attributed to this condition. Typically, carbonate soils have large interparticle and intraparticle porosity, resulting in high void ratio and low density and, hence, are more compressible than soils from a terrigenous silica deposit. Furthermore, carbonate soils are prone to post deposition alterations by biological and physiochemical processes under normal pressure and temperature conditions. This results in the formation of irregular and discontinuous layers and lenses of cemented material. These alterations, in turn, profoundly affect mechanical behavior.

The fabric of carbonate soils is an important characteristic feature. Generally, particles of skeletal material will be angular to subrounded in shape, with rough surfaces, and have intraparticle voids. Particles of non-skeletal material, on the other hand, are solid with smooth surfaces and without intraparticle voids. It is generally understood that uncemented carbonate soils consisting of rounded non-skeletal grains that are resistant to crushing are stronger foundation materials than carbonate soils, that show partial cementation but allow a moderate degree of crushing. There is information that indicates the importance of carbonate content as it relates to the behavior of carbonate sediments. A soil matrix that is predominantly carbonate is more likely to undergo degradation due to crushing and compressibility of the material than soil that has low carbonate fraction in the matrix. Other important characteristic features that influence the behavior of the material are grain angularity, initial void ratio, compressibility, and grain crushing. These characteristic features are interrelated parameters in the sense that carbonate soils with highly angular particles often have a high *in situ* void ratio due to particle orientation. These soils are more susceptible to grain crushing due to angularity of the particles and thus will be more apt to be compressed.

This clause gives a general overview of the mechanical behavior of carbonate soils. For a more detailed understanding of material characteristics, information can be found in References [37] to [64].

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B.3 Properties

Globally, it is increasingly evident that there is no unique combination of laboratory and *in situ* testing program that is likely to provide all the appropriate parameters for design of foundations in carbonate soils. Some laboratory and *in situ* tests have been found useful. As a minimum, a laboratory testing program for carbonate soils should determine the following:

a) material composition, particularly carbonate content;

b) material origin to differentiate between skeletal and non-skeletal sediments;

c) grain characteristics, such as particle angularity, porosity, and initial void ratio;

d) compressibility of the material;

e) soil strength parameters and volume change characteristics on shearing, including effects of cyclic actions;

f) formation cementation, at least in a qualitative sense.

For site characterizations, maximum use of local experience is important, particularly in the selection of an appropriate soil investigation and testing program. In new unexplored territories, where the presence of carbonate soils is suspected, selection of an *in situ* test program should draw upon any experience with carbonate soils where geographical and environmental conditions are similar.

B.4 Foundations

B.4.1 Driven piles

Several case histories have been reported that describe some of the unusual characteristics of foundations on carbonate soils and their often poor performance. It has been shown from numerous pile load tests that piles driven into weakly cemented and compressible carbonate sands and silts mobilize only a fraction of the capacity (as low as 15 %) predicted by conventional design and/or prediction methods for siliceous material. On the other hand, dense, strongly cemented carbonate deposits can be very competent foundation material. Unfortunately, the difficulty in obtaining high-quality samples and the lack of generalized design methods sometimes make it difficult to predict where problems can occur. With clays, care should be taken when the carbonate content exceeds 50 % and where no pile test data or local experience exist.

B.4.2 Other deep foundation alternatives

The current trend for deep foundations in carbonate sands and silts is a move away from driven piles. However, because of lower installation costs, driven piles still receive consideration for support of lightly loaded structures or where extensive local pile load test data and experience exists to substantiate the design premise. Furthermore, driven piles can be appropriate in moderately competent carbonate soils. At present, the preferred alternative to the driven pile is the drilled and grouted pile. Drilled and grouted piles mobilize significantly higher unit skin friction. The result is a substantial reduction in the required pile penetration compared with driven piles. Because of the high construction cost of drilled and grouted piles, an alternative driven and grouted pile system has received some attention in the recent past. This system has the potential to reduce installation costs while achieving comparable capacity. For any type of grouted pile, consideration should be given to the potential for reduction in shaft capacity due to cyclic actions, especially once slip has been initiated between pile and soil.

B.4.3 Shallow foundations

Shallow foundations are suitable for use on carbonate sediments, although any evaluation of such foundations should account for the important difference that exists with such material compared with silica sands or normal clays. Carbonate sands and silts generally have higher friction angles than silica sands and silts, but are more compressible, and these two factors influence bearing capacity in opposite ways. Carbonate sands and silts are also generally less permeable than equivalent silica material, leading to longer drainage times for a given size of foundation. The tendency for volume reduction on shearing, particularly under cyclic actions, combined with

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longer drainage times, leads to potential for bearing failure induced by soil liquefaction. It should also be noted that undrained cyclic strength of carbonate sands is generally lower than for most silica sands. The high compressibility of most carbonate sediments results in relatively large consolidation settlements, and can give rise to large settlements induced by cyclic actions. Shallow foundations are attractive for carbonate sediments that exhibit a significant degree of cementation, since they give high bearing capacities, good resistance to cyclic actions and low potential for settlements. However, layered profiles of variably cemented and uncemented sediments should be treated cautiously, with due account taken of the risks and consequences of a punch-through type of failure. Novel foundation systems such as those using suction assistance for penetration of skirts should be evaluated carefully on a case-by-case basis.

Some evidence indicates that the bearing capacity of shallow foundations in weakly cemented and compressible granular carbonate deposits can be significantly lower than the capacity in the siliceous material generally encountered in the Gulf of Mexico. On the other hand, higher bearing capacities have been reported where the soil is dense, strongly cemented, competent material.

B.5 Assessment

To date, general design procedures for foundations in carbonate soils are not available. Acceptable design methods have evolved but remain highly site-specific and dependent on local experience. Stemming from some recent publications describing poor foundation performance in carbonate soils and the financial consequences of the remedial measures, there is a growing tendency to take a conservative approach to design in carbonate soils, even if the carbonate content in the sediment fraction is relatively low. This is not always warranted. As with other designs, the judgment of knowledgeable engineering remains a critical link in economic design of offshore foundations in carbonate soil environments.

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Annex C
(informative)

Pile foundation design commentary

NOTE The clauses in this annex provide additional information and guidance on Clause 8 in the main text of this document. The title of each subclause identifies the subclause in Clause 8 of this document.

C.8 Foundation design

C.8.1 Pile capacity for axial compression

C.8.1.1 General

No guidance is offered

C.8.1.2 Ultimate bearing capacity

It is not always correct to add the ultimate end bearing capacity to the ultimate shaft friction to obtain the ultimate axial bearing capacity of a pile. This subject is addressed in References [137], [138], and [139]. For the particular case of a belled pile, this matter is discussed in Reference [139].

C.8.1.3 Shaft friction and end bearing in cohesive soils

C.8.1.3.1 General

Estimating pile capacity in clay soils requires considerable judgment in selecting design parameters and in interpreting calculated capacities. Some of the items that should receive design consideration are detailed in C.8.1.3.2 and C.8.1.3.3.

C.8.1.3.2 Axial pile capacity in clay

C.8.1.3.2.1 Load test database for piles in clay

A number of studies, References [71] and [140] through [144], have been carried out and aimed at collecting and comparing axial capacities from relevant pile load tests to those predicted by traditional offshore pile design procedures. Studies such as these can be very useful in calibrating one's judgment in the design process. It is clear, for example, that there is considerable scatter in the various plots of measured vs. predicted capacities. The designer should be aware of the many limitations of such comparisons when making use of these results. Limitations of particular importance include the following.

a) There is considerable uncertainty in the determination of both predicted capacities and measured capacities. For example, determination of the predicted capacities is very sensitive to the selection of the undrained shear strength profile, which itself is subject to considerable uncertainty. The measured capacities are also subject to interpretation as well as possible measurement errors.

b) The conditions under which the pile load tests are conducted generally vary significantly from the design loads and field conditions. One clear limitation is the limited number of tests on deeply embedded, large diameter, high capacity piles. Generally, pile load tests have capacities that are 10 % or less of the prototype capacities. Another factor is that the rate of change and the cyclic history of the loads are usually not well represented in the load tests (see C.8.3.2). For practical reasons, the pile load tests are often conducted before full set-up occurs (see C.8.1.3.3). Furthermore, the pile tip conditions (closed vs. open-ended) can differ from offshore piles.

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c) In most of the studies an attempt has been made to eliminate those tests that are thought to be significantly affected by extraneous conditions in the load test, such as protrusions on the exterior of the pile shaft (weld beads, cover plates, etc.), installation effects (jetting, drilled out plugs, etc.), and artesian conditions. Nevertheless, it is not possible to eliminate all extraneous factors in all cases.

The database includes a number of tests that were specially designed for offshore applications as well as a number of published tests that are fortuitously relevant to offshore conditions (appropriate pile type, installation method, soil conditions, etc.). The former are generally higher quality and larger scale, and hence are particularly important in calibrating the design method. The tests most relevant to offshore applications have all been conducted in the United States or in Europe. As regional geology and particularly operating experience are considered very important in foundation design, care should be exercised in applying these results to other regions of the world. In addition, the designer should note that certain important tests in silty clays of low plasticity, such as at the Pentre site[145], indicate overprediction of frictional resistance by the Equations 16 through 18. The reason for this overprediction is not well understood and has been an area of active research. The designer is thus cautioned that pile design for soils of this type should be given special consideration.

Additional considerations that apply to drilled and grouted piles are discussed in References [146] and [147].

C.8.1.3.2.2 Alternative methods of determining pile capacity

Alternative methods of determining pile capacity in clays exist, based on sound engineering principles and consistent with industry experience, and may be used in practice. One such method is described below.

For piles driven through clay, $f(z)$ may be less than or equal to, but should not exceed, the undrained shear strength of the clay, s_u, as determined by unconsolidated-undrained (UU) triaxial tests and miniature vane shear tests.

Unless test data indicate otherwise, $f(z)$ should not exceed s_u or the following limits.

a) For highly plastic clays, $f(z)$ may be equal to s_u for underconsolidated and normally consolidated clays. For overconsolidated clays, $f(z)$ should not exceed 48 kPa (1 kips/ft^2) for shallow penetrations or the equivalent value of s_u for a normally consolidated clay for deeper penetrations, whichever is greater.

b) For other types of clay:

$$f(z) = s_u \text{ for } s_u < 24 \text{ kPa } (0{,}5 \text{ kips/ft}^2) \quad \text{(C.1)}$$

$$f(z) = s_u/2 \text{ for } s_u > 72 \text{ kPa } (1{,}5 \text{ kips/ft}^2) \quad \text{(C.2)}$$

$f(z)$ varies linearly for values of s_u between the above limits.

For other methods, see References [72], [140], [141], and [143].

It has been shown in Reference [144] that, on the average, the above cited methods predict the available but limited pile load test database results with comparable accuracy. However, capacities for specific situations computed by different methods can differ by a significant amount. In such cases, pile capacity determination should be based on engineering judgment, which takes into account site-specific soils information, available pile load test data, and industry experience in similar soils.

C.8.1.3.2.3 Establishing design strength and effective overburden stress profiles

The axial pile capacity in clay determined by these procedures is directly influenced by the undrained shear strength and effective overburden stress profiles selected for use in analyses. The wide variety of sampling techniques and the potentially large scatter in the strength data from the various types of laboratory tests complicate appropriate selection.

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UU triaxial compression tests on high quality samples, preferably taken by pushing a thin-walled sampler with a diameter of 75 mm (3 in) or more into the soil, are recommended for establishing strength profile variations because of their consistency and repeatability. In selecting the specific shear strength values for design, however, consideration should be given to the sampling and testing techniques used to correlate the shear strength to any available relevant pile load test data. The experience with pile performance is another consideration that can play an important role in assessing the appropriate shear strength interpretation.

Miniature vane tests on the pushed samples should correlate well with the UU test results and will be particularly beneficial in weak clays. In situ testing with a vane or cone penetrometer will help in assessing sampling disturbance effects in gassy or highly structured soils. Approaches such as the SHANSEP technique (stress history and normalized soil engineering properties; see References [19] and [20]) can help provide a more consistent interpretation of standard laboratory tests and will provide history information used to determine the effective overburden stress in normally or underconsolidated clays.

C.8.1.3.2.4 Pile length effect

Long piles driven in clay soils can experience capacity degradation due to

a) continued shearing of a particular soil horizon during pile installation,

b) lateral movement of soil away from the pile due to “pile whip” during driving, and/or

c) progressive failure in the soil due to strength reduction with continued displacement (softening).

The occurrence of degradation due to these effects depends on many factors related to both installation conditions and soil behavior. Methods of estimating the possible magnitude of reduction in capacity of long piles can be found in References [137] through [141] and [143].

C.8.1.3.3 Changes in axial capacity in clay with time

Existing axial pile capacity calculation procedures for piles in clay are based on experience assisted by the results of axial pile load tests. In these tests, few of the piles were instrumented and in most cases little or no consideration was given to the effects of time after driving on the development of pile-soil shear resistance. Axial capacity of a driven pipe pile in clay computed according to the guidelines set forth in 8.1.1 and 8.1.2 is intended to represent the long-term static capacity of piles in undrained conditions when subjected to axial loads until failure, after dissipation of excess pore water pressure caused by the installation process. Immediately after pile driving, pile capacity in a cohesive deposit can be significantly lower than the ultimate static capacity. Field measurements[145] [148] [139] have shown that the time required for driven piles to reach ultimate capacity in a cohesive deposit can be relatively long — as much as two to three years. However, it should be noted that the rate of strength gain is highest immediately after driving, and this rate decreases during the dissipation process. Thus a significant strength increase can occur in a relatively short time.

During pile driving in normally to lightly overconsolidated clays, the soil surrounding a pile is significantly disturbed, the stress state is altered, and large excess pore pressures can be generated. After installation, these excess pore pressures begin to dissipate, i.e. the surrounding soil mass begins to consolidate and the pile capacity increases with time. This process is usually referred to as set-up. The rate of excess pore pressure dissipation is a function of the coefficient of radial (horizontal) consolidation, pile radius, plug characteristics (plugged versus unplugged pile), and soil layering.

In the case of driven pipe piles supporting a structure where the design loads can be applied to the piles shortly after installation, the time-consolidation characteristics should be considered in pile design. In such cases, the capacity of piles immediately after driving and the expected increase in capacity with time are important design variables that can impact the safety of the foundation system during early stages of the consolidation process.

A number of investigators [150] [151] have proposed analytical models of pore pressure generation and the subsequent dissipation process for piles in normally consolidated to lightly overconsolidated clays. Since excess pore pressures are generated by pile driving operations, any dissipation of the excess pore pressures after installation should correspond to an increase in the shear strength of the surrounding soil mass and hence an increase in the capacity of the pile. After dissipation of excess pore pressures, the capacity of a pile approaches

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long-term capacity, although some strength gain may continue due to secondary processes. In some overconsolidated clays, pile capacity can decrease as pore pressures dissipate, provided the rate of change of radial total stress decreases faster than the rate of change of pore pressure. The analytical models account for the degree of plugging by assuming various degrees of plug formation, ranging from closed- to open-ended pile penetration modes. Input necessary for the analysis includes the soil characteristics (compressibility, stress history, strength, etc.) and the initial site conditions.

In Reference [148] the behavior of piles subjected to significant axial loads in highly plastic, normally consolidated clays was studied using a large number of model pile tests and some full scale pile load tests. From the study of pore pressure dissipation and load test data at different times after pile driving, empirical correlations were obtained between the degree of consolidation, degree of plugging, and pile shaft shear transfer capacity. The analysis is dependent on the shear strength of the surrounding soil mass. The method is presently limited to use in highly plastic, normally consolidated clays of the type encountered in the Gulf of Mexico, since validation data have been published only for those soils.

In Reference [149], in highly overconsolidated glacial till, capacity was shown to undergo significant short-term reduction associated with pore pressure redistribution and reduction in radial effective stresses during the early stages of the equalization process. The capacity at the end of installation was never fully recovered. Test results for closed-ended steel piles in heavily overconsolidated London clay indicate that there is no significant change in capacity with time[153]. This is contrary to tests on 0.273 m (10.75 in.) diameter closed-ended steel piles in overconsolidated Beaumont clay, where considerable and rapid set-up (in four days) was found[153].

Caution should be exercised in using the above-mentioned procedures to evaluate set-up, particularly for soils with different plasticity characteristics and under different states of consolidation (especially overconsolidated clays) and piles with D/WT (pile outer diameter / wall thickness) ratios greater than 40.

C.8.1.4 Shaft friction and end bearing in cohesionless soils

C.8.1.4.1 General

Estimating axial pile capacity in cohesionless soils requires considerable engineering judgment in selecting an appropriate method and associated parameter values. Some of the items that should be considered by geotechnical engineers are detailed in the following subclauses.

Clause C.8.1.4.2 discusses four CPT-based methods for axial pile capacity that incorporate length effects and friction fatigue. Offshore experience with some of these methods is given in References [200] through [203]. Some of these methods have only recently been made available in the literature; they have not yet been frequently compared for routine offshore pile projects. Hence, geotechnical engineering judgment is needed to select the most appropriate method for the design case under consideration. Additional care is required in cases of clay layers at or near pile tip level.

The piles are assumed to be open-ended steel piles of uniform outer diameter. Installation is by impact driving into significant depths of clean siliceous sand. In general, such piles drive unplugged (i.e. they core). However, when they are statically loaded in compression, sufficient inner friction is generally mobilized to cause the pile to act as fully plugged (i.e. the soil plug does not undergo overall slip relative to the pile wall during compression pile loading).

The term sand is used hereafter for all cohesionless siliceous soils. Exceptions (e.g. carbonate sands and gravels) are addressed in C.8.1.4.4.

The appropriate safety factors to be used with the methods discussed in C.8.1.4.2 are not provided in C.8.1.4. The designer should carefully evaluate, for each design case, whether the safety factors provided in the applicable structural design standard are appropriate or not.

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C.8.1.4.2 CPT-based methods for pile capacity

C.8.1.4.2.1 General

In 8.1.4 a simple method for assessing pile capacity in cohesionless soils is presented, which is a modification of methods recommended in API 2A-WSD. Changes were made to remove potential unconservatism. C.8.1.4.2.1 through C.8.1.4.2.5 present recent and more reliable CPT-based methods for predicting pile capacity. These methods are all based on direct correlations of pile unit friction and end bearing data with cone tip resistance values from cone penetration tests (CPT). These CPT-based methods cover a wider range of cohesionless soils, are considered fundamentally better and have shown statistically closer predictions of pile load test results.

Friction and end bearing contributions to pile capacity are assumed to be uncoupled. Hence, for all methods, the ultimate axial pile capacity in compression, Q_c, and in tension, Q_t , of plugged open-ended piles is determined by Equations C.3 and C.4:

$$Q_c = Q_{f,c} + Q_p = \pi D \int f_c(z)\,dz + qA_p \tag{C.3}$$

$$Q_t = Q_{f,t} = \pi D \int f_t(z)\,dz \tag{C.4}$$

where

Q_c is the axial pile ultimate capacity in compression, in force units;

Q_t is the axial pile ultimate capacity in tension, in force units;

$Q_{f,c}$ is the shaft friction capacity in compression, in force units;

$Q_{f,t}$ is the shaft friction capacity in tension, in force units;

Q_p is the end bearing capacity, in force units;

$f_c(z)$ is the unit shaft friction in compression, in stress units, which is a function of depth, geometry and soil conditions;

$f_t(z)$ is the unit shaft friction in tension, in stress units, which is a function of depth, geometry and soil conditions;

z is the depth below the original seafloor;

q is the unit end bearing at the pile tip, in stress units;

D is the pile outside diameter;

A_p is the gross end area of the pile, $A_p = \frac{\pi D^2}{4}$.

Since the friction component, $Q_{f,c}$ or $Q_{f,t}$ involves numerical integration, results are sensitive to the depth increment used, particularly for CPT-based methods. As guidance, depth increments for CPT-based methods should be in the order of 1/100th of the pile length (or smaller). In any case, the depth increment should not exceed 0.2 m (0.5 ft).

The four recommended CPT-based methods discussed herein are

— Method 1, Simplified ICP-05 (as described in this document),

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— Method 2, Offshore UWA-05, References [154] and [155],

— Method 3, Fugro-05, References [154] and [156], and

— Method 4, NGI-05, References [154] and [157].

Method 1 is a simplified version of the design method recommended by Jardine et al. [158]), whereas Method 2 is a simplified version of the UWA-05 method applicable to offshore pipe piles. Methods 2, 3, and 4 are summarized in Reference [154]. Friction and end-bearing contributions should not be taken from different methods. A general description of methods 1, 2, and 3 is given below, after which details of the various methods are presented separately.

The unit shaft friction formulae for open-ended steel pipe piles for CPT-based methods 1, 2, and 3 can all be considered as being special cases of the general formula:

$$f(z) = u \times q_c(z) \times \left(\frac{p'_o(z)}{p_a}\right)^a \times A_r^b \times \left[\max\left(\frac{L-z}{D}, v\right)\right]^{-c} \times \left[\tan\delta_{cv}\right]^d \times \left[\min\left(\frac{L-z}{D} \times \frac{1}{v}, 1\right)\right]^e \quad \text{(C.5)}$$

where, in addition to prior definitions,

$f(z)$ is the unit shaft friction, in stress units, which is a function of depth, geometry and soil conditions;

$q_c(z)$ is the CPT cone-tip resistance at depth, z, in stress units;

$p'_o(z)$ is the effective vertical stress at depth z;

p_a is the atmospheric pressure, in stress units, (e.g. p_a = 100 kPa);

A_r is the pile displacement ratio, $A_r = \frac{A_w}{A_p} = 1 - \left(\frac{D_i}{D}\right)^2$;

A_w is the area of the rim of the steel pile, $A_w = \frac{\pi}{4}\left(D^2 - D_i^2\right)$;

D_i is the pile inside diameter, $D_i = D - 2WT$;

WT is the pile wall thickness;

L is the embedded length of the pile below the original seafloor;

δ_{cv} is the sand constant volume friction angle at the interface between the sand and the pile wall.

Recommended values for the parameters, a, b, c, d, e, u, and v, for compression and tension are given in Table C.1.

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Table C.1—Unit shaft friction parameter values for driven open-ended steel piles for Methods 1, 2, and 3

Method	Parameter						
	a	*b*	*c*	*d*	*e*	*u*	*v*
Method 1:							
compression	0.1	0.2	0.4	1	0	0.023	$4\sqrt{A_r}$
tension	0.1	0.2	0.4	1	0	0.016	$4\sqrt{A_r}$
Method 2:							
compression	0	0.3	0.5	1	0	0.030	2
tension	0	0.3	0.5	1	0	0.022	2
Method 3:							
compression	0.05	0.45	0.90	0	1	0.043	$2\sqrt{A_r}$
tension	0.15	0.42	0.85	0	0	0.025	$2\sqrt{A_r}$

Additional recommendations for computing unit friction and end bearing of all four CPT-based methods are presented in C.8.1.4.2.2 to C.8.1.4.2.5.

C.8.1.4.2.2 Method 1

C.8.1.4.2.2.1 Friction

Reference [158] presents a comprehensive overview of research work performed at Imperial College on axial pile design criteria for open- and closed-ended piles in clay and sand. The design equations for unit friction in sand in this reference include a soil dilatancy term, implying that unit friction is favorably influenced by soil dilatancy. This influence diminishes with increasing pile diameter. The unit shaft friction, $f(z)$, of open-ended pipe piles in Method 1, given by Equation C.5 and the parameter values in Table C.1, are a conservative approximation of the full calculation in Method 1, since dilatancy is ignored and some parameter values were conservatively rounded up/down.

The original "full" design equations in Reference [158] may be used, particularly for small diameter piles [D < 0.76 m (30 in.)], provided that larger resistance factors are considered. See Reference [158] for a discussion on reliability based design using the "full" Method 1.

C.8.1.4.2.2.2 End bearing

The unit end bearing, q, for open-ended pipe piles follows the recommendations of Reference [158]. These specify a unit end bearing for plugged piles given by:

$$q = q_{\text{c,av,1.5D}}\left[0.5 - 0.25\ \log_{10}\ \left(\frac{D}{D_{\text{CPT}}}\right)\right] \geq 0.15\, q_{\text{c,av,1.5D}} \tag{C.6}$$

where, in addition to the general definitions given in C.8.1.4.2.1,

$q_{\text{c,av,1,5D}}$ is the average value of $q_{\text{c}}(z)$ between 1.5D above the pile tip and 1.5D below the pile tip;

$$q_{\text{c,av,1.5D}} = \tfrac{1}{3D} \times \left[\int_{\text{L-1.5D}}^{\text{L+1.5D}} q_{\text{c}}(z)\, dz\right];$$

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D_{CPT} is the diameter of the CPT tool; D_{CPT} = 36 mm for a standard cone with a base area of 10 cm^2.

Reference [158] specifies that plugged pile end bearing capacity applies, which means that the unit end bearing, q, acts across the entire pile tip cross-section, provided both of the following conditions are met:

$$D_i < 2\,(D_r - 0.3) \tag{C.7}$$

$$D_i / D_{CPT} < 0.083\,\frac{q_c(z)}{p_a} \tag{C.8}$$

where

D_r is the relative density of the sand ($0 \le D_r \le 1.0$).

NOTE Equation C.7 is the equation given in Reference [158]. It implies that the maximum pile inside diameter for plugged behavior in Method 1 is 1.4 m. However, experience indicates that offshore piles with appreciably larger inside diameters can still behave as plugged.

If either of the above conditions is not met, then the pile will behave unplugged and Equation C.9 should be used for computing the end bearing capacity:

$$Q_p = \pi\,(D - WT)\,WT\,q_{c,\,tip} \tag{C.9}$$

where

$q_{c,\,tip}$ is the cone tip resistance at the pile tip.

The full pile end bearing computed using Equation C.6 for a plugged pile should not be less than the end bearing capacity of an unplugged pile computed according to Equation C.9.

C.8.1.4.2.3 Method 2

C.8.1.4.2.3.1 Friction

Reference [154] summarizes the results of recent research work at the University of Western Australia on development of axial pile design criteria for open- and closed-ended piles driven into silica sands. The full design method (described in Reference [154] and [155]) for unit friction on pipe piles includes a term allowing for favorable effects of soil dilatancy (similar to Method 1) and an empirical term allowing for partial soil plugging during pile driving. The authors of Reference [155] recommend for offshore pile design to ignore these two favorable effects (dilatancy and partial plugging), resulting in the recommended Equation C.5 and associated Table C.1 parameter values.

The original "full" design equations in Reference [154] may be used, particularly for small diameter piles [D < 0.76 m (30 in.)], provided that larger resistance factors are considered. See Reference [154] for a discussion on reliability-based design using Method 2.

C.8.1.4.2.3.2 End bearing

References [154] and [155] present design criteria for ultimate unit end bearing of plugged open-ended pipe piles. Their "full" design method for pipe piles includes an empirical term allowing for the favorable effect of partial plugging during pile driving. For offshore pile design, Reference [154] and [155] recommend to ignore this effect, resulting in the recommended design equation for plugged piles in Method 2:

$$q = q_{c,\,av1,5D}\,(0.15 + 0.45A_r) \tag{C.10}$$

where, again in addition to the general definitions given in C.8.1.3.2.1,

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$q_{c,\ av1,5D}$ is the average value of $q_c(z)$ between 1.5D above the pile tip and 1.5D below the pile tip;

$$q_{c,av,1.5D} = \int_{L-1.5D}^{L+1.5D} q_z(z)\,dz / (3D)\ .$$

Since Method 2 considers non-plugging under static loading to be exceptional for typical offshore piles, the method does not provide criteria for unplugged piles. The unit end bearing, q, calculated in Equation C.10 is therefore acting across the entire tip cross-section. The use of $q_{c,\ av1,5D}$ in Equation C.10 is not recommended in sand profiles where the CPT $q_c(z)$ values show significant variations in the vicinity of the pile tip or when penetration into a founding stratum is less than five pile diameters. For these situations, Reference [155] provides guidance on the selection of an appropriate average $q_c(z)$ value for use in place of $q_{c,\ av1,5D}$.

C.8.1.4.2.4 Method 3

C.8.1.4.2.4.1 Friction

Method 3 is a modification of Method 1 and was developed as part of a research project for API. The unit friction equations were unfortunately misprinted in Reference [156] and [159] and these references are not to be used in design. However, the correct equations are presented both by Reference [154] and by Equation C.5 and the parameter values in Table C.1. Like the “full” Method 1 and the “full” Method 2, it is recommended that larger safety factors be considered when using Method 3. See Reference [160] for a discussion on reliability-based design using Method 3.

C.8.1.4.2.4.2 End bearing

The basis for the ultimate unit end bearing on pipe piles according to Method 3 is presented in a research report to API[159] and summarized in Reference [156]. The recommended value of the unit end bearing for plugged piles is given by:

$$q = 8.5\,p_a \left(\frac{q_{c,av,1.5D}}{p_a} \right)^{0.5} A_r^{0.25} \qquad \text{(C.11)}$$

Both Method 2 and Method 3 do not specify unplugged end bearing capacity because typical offshore piles behave in a plugged mode during static loading[160]. It can be shown that plugged behavior applies if either:

— the cumulative thickness of sand layers within a soil plug is in excess of 8D, or

— the total end bearing Q_p is limited as follows:

$$Q_p \le Q_{f,i,clay}\ e^{L_s/D} \qquad \text{(C.12)}$$

where

$Q_{f,\ i,\ clay}$ is the cumulative shaft friction capacity of the clay layers within the soil plug, in force units;

L_s is the length of the soil plug in the sand layers.

The cumulative frictional capacity of the clay layers within the soil plug, $Q_{f,\ i,\ clay}$, can be estimated using similar procedures as for computing estimated pile friction in clay (see 8.1.3).

Equation C.12 applies for fully drained behavior of sand within the pile plug. Criteria for undrained/partially drained sand plug behavior are presented in Reference [161].

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For the exceptional case of unplugged end bearing behavior in fully drained conditions, reference is made to Reference [160] and [162] for estimating end bearing capacity.

C.8.1.4.2.5 Method 4

C.8.1.4.2.5.1 Friction

Ultimate unit shaft friction values for tension, $f_t(z)$, and compression, $f_c(z)$, for driven open-ended steel pipe piles in Method 4 are given in Reference [157]:

$$f_t(z) = (z/L)\, p_a\, F_{sig}\, F_{Dr} > 0.1\, p'_o\,(z) \tag{C.13}$$

$$f_c(z) = 1.3\,(z/L)\, p_a\, F_{sig}\, F_{Dr} > 0.1\, p'_o\,(z) \tag{C.14}$$

where, in addition to the general definitions given in C.8.1.3.1,

$$F_{sig} = (p'_o\,(z)/p_a)^{0.25} \tag{C.15}$$

$$F_{Dr} = 2.1\,(D_r - 0.1)^{1.7} \tag{C.16}$$

$$D_r = 0.4 \times \ln\left(\frac{q_c(z)}{22 \cdot \left(p'_o(z)\, p_a\right)^{0.5}}\right) \geq 0.1 \tag{C.17}$$

NOTE $D_r > 1$ should be accepted and used

As with the "full" Methods 1, 2 and 3, higher safety factors should be considered when using Method 4.

C.8.1.4.2.5.2 End bearing

The recommended equation for the ultimate unit end bearing of plugged open-ended steel pipe piles in Method 4[158] is:

$$q = \frac{0.7\, q_{c,av,1.5D}}{1 + 3D_r^2} \tag{C.18}$$

where,

$q_{c,av,1.5D}$ is the average value of $q_c(z)$ between 1.5D above the pile tip and 1.5D below the pile tip;

$$q_{c,av,1.5D} = \int_{L-1.5D}^{L+1.5D} q_z(z)\, dz \,/\, (3D)\,. \tag{C.19}$$

D_r is derived as described in Equation C.17.

NOTE Values of $D_r > 1$ should be accepted and used.

The resistance of non-plugging piles should be computed using a unit end bearing value for the steel pile rim, $q_w(z)$, shown in Equation C.20.

$$q_w(z) = q_c(z) \tag{C.20}$$

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Likewise, the unit friction, $f_p(z)$, between the soil plug and the inner pile wall is shown in Equation C.21.

$$f_p(z) = 3 f_c(z) \tag{C.21}$$

The lower of the plugged resistance, q, of Equation C.18, the unplugged resistance determined by Equation C.20, and Equation C.21 should be used in design.

C.8.1.4.3 Parameter value assessment

The geotechnical site investigation should provide information that is adequate to capture the spatial variability, horizontally and vertically, of the boundaries and parameter values of all layers.

For any CPT method, the computed pile capacity in sand is most sensitive to cone penetration resistance, $q_c(z)$, followed by tan δ_{cv} and $p'_o(z)$. Since an accurate capacity assessment is a function of the accuracy of both the model and the parameters, guidance regarding selecting appropriate parameter values is given below.

C.8.1.4.3.1 Parameter $q_c(z)$

The CPT should measure $q_c(z)$ with apparatus and procedures that are in general accordance with those published by Reference [164]. In particular, Reference [164] prescribes cones with a base area in the range of 500 mm^2 to 2 000 mm^2 and a penetration rate of 20 ± 5 mm/s.

It should be noted that the CPT-based design methods were established for cone resistance values up to 100 MPa. Caution should be exercised when applying the methods herein to sands with higher resistances.

A measured, continuous profile of $q_c(z)$ is preferable to an assumed/interpolated discontinuous profile, but is generally not achievable offshore at large depths below the seafloor with a downhole CPT apparatus. This is generally due to factors such as limited stroke and/or maximum resistance being achieved. When (near) continuous $q_c(z)$ profiles are needed, one can consider overlapping CPT push strokes.

With discontinuous CPT data, a “blocked” $q_c(z)$ profile can be used: the soil profile is divided into layers, in each of which $q_c(z)$ is assumed to vary linearly with depth. “Blocked” profiles should be carefully assessed, particularly when they contain maximum $q_c(z)$ values at the ends of CPT push strokes. When the push strokes contain no maximum $q_c(z)$ data, a moving window may be used to determine the average profile (and its standard deviation), through which a straight line can be fitted. If present, thin layers of weaker material (e.g. silt or clay) need to be modelled conservatively.

For geotechnical investigations where several vertical CPT profiles have been made (e.g. one per platform leg), it is suggested that at least two approaches be employed: capacity should first be based on the combined averaged $q_c(z)$ profile and then based on individual $q_c(z)$ profiles. Judgment is required to select the most appropriate $q_c(z)$ profile and to determine the associated final axial capacity.

C.8.1.4.3.2 Parameter $p'_o(z)$

Usually, pore water pressures in sands are hydrostatic and in this case $p'_o(z)$ equals $(\gamma' z)$, where γ' is the submerged soil unit weight. Offshore sands are generally very dense and often silty. In general, design γ' values in sands should be based on measured laboratory values (corrected for sampling disturbance effects), which should be compatible with relative density, D_r , estimated from $q_c(z)$ and maximum and minimum dry unit weight values determined in the laboratory.

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C.8.1.4.3.3 Parameter D_r

Common practice is to use the relationship for Ticino-Toyura-Hokksund sand between $q_c(z)$ and D_r as proposed by Reference [164]:

$$D_r = \frac{1}{2.96} \ln \left(\frac{q_c(z) / p_a}{24.94 \left(\frac{p'_m(z)}{p_a} \right)^{0.46}} \right) \tag{C.22}$$

where

$p'_m(z)$ is the effective mean stress at depth z, $p'_m = (1/3)\,(p'_o(z) + 2p'_{oh}(z))$;

$p'_{oh}(z)$ is the effective horizontal stress at depth z;

p_a is the atmospheric pressure, in same units as $p'_m(z)$ and $q_c(z)$, (e.g. p_a = 100 kPa).

Ticino sand is a medium grained silica sand with no fines. A reasonably comprehensive database is available for this sand[165]. However, D_r assessment for Method 4 should be according to Equation C.17 and Equation C.19. Most $q_c(z) - D_r$ relationships are not valid for silty sands. However, $q_c(z)$ may be adjusted for such materials to derive a "clean sand equivalent normalized cone resistance" (see, for example, Reference [166]).

C.8.1.4.3.4 Parameter tan δ_{cv}

The constant volume interface friction angle, δ_{cv}, should be measured directly in laboratory interface shear tests. The recommended test method is by ring shear apparatus, but the direct shear box may also be used. Guidance on test procedures is provided in Reference [158].

If site-specific tests cannot be performed, the constant volume interface friction angle may be estimated as a function of mean effective particle diameter, D_{50}, using Reference [159]. An upper limit of tan δ_{cv} = 0.55 (δ_{cv} = 28.8°) applies to all methods as shown in Figure C.1. However, for materials with unusually weak grains or compressible structures this method is not always appropriate. Of particular importance are sands containing calcium carbonate, for which specific advice is given in C.8.1.4.4.

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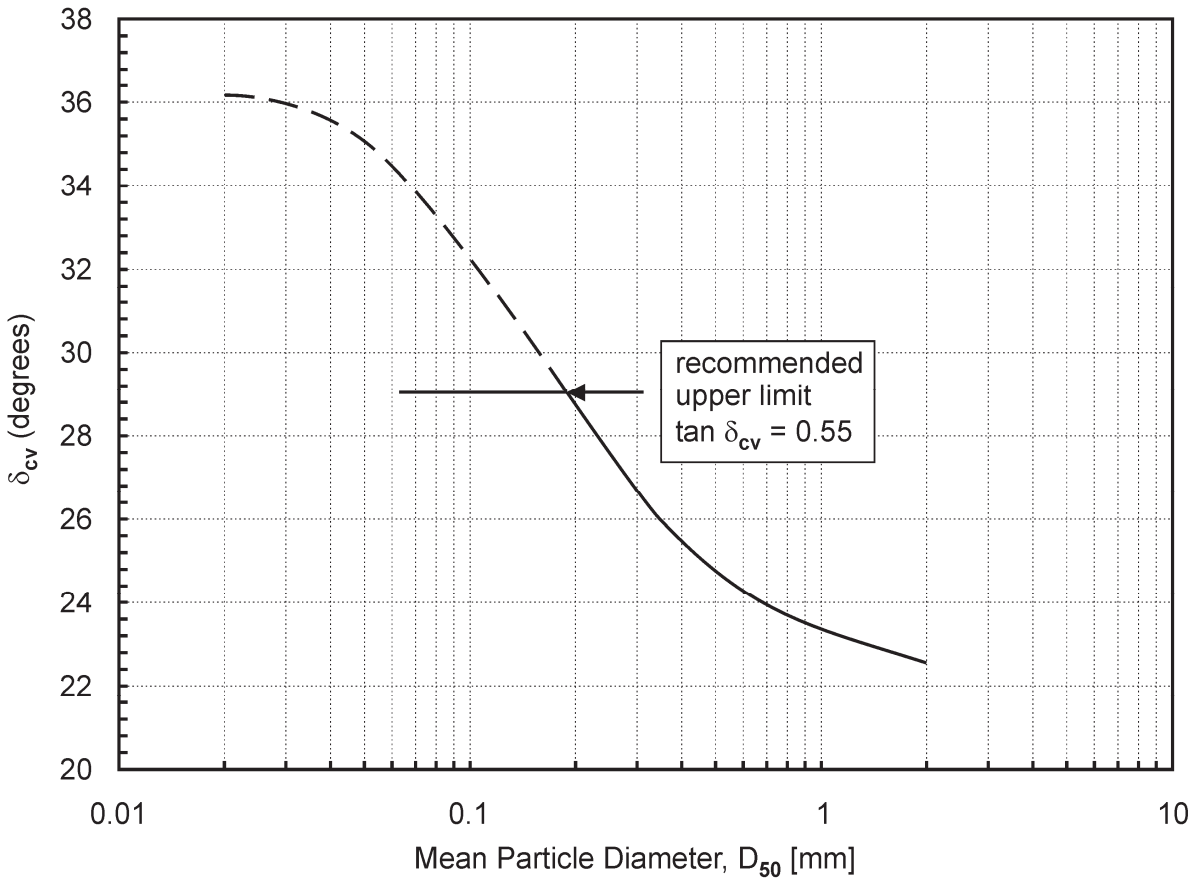

Key

δ_{cv} is the sand constant volume friction angle at the interface between the sand and the pile wall;

D_{50} is the mean particle diameter.

Figure C.1—Interface friction angle in sand, δ_{cv}, from direct shear interface tests

C.8.1.4.4 Application of CPT-based methods

C.8.1.4.4.1 General

Clauses C.8.1.4.2 and C.8.1.4.3 provide four methods for computing pile capacity in silica sands using CPT data. Clauses C.8.1.4.4.1.1 through C.8.1.4.4.1.7 give guidance on the following aspects of pile design using CPT-based methods:

— axial load-displacement behavior;

— application to soils other than silica sands;

— application to piles with different geometries than typical offshore piles;

— effects of scour on pile capacity.

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C.8.1.4.4.1.1 t–z data for axial shear transfer-displacement response

No strain softening is applicable. However, unlike for the simplified method given in Equation C.5, the peak unit shaft frictions in compression and tension at given depths, $f_c(z)$ and $f_t(z)$, are not unique and both depend on pile geometry. They depend not only on the pile diameter and wall thickness, but also on the total pile penetration. An increased pile penetration will decrease these values at a given depth.

C.8.1.4.4.1.2 q–z data for end bearing-displacement response

Unit end bearing, q, is assumed to be fully mobilized at a pile tip displacement value of 0.1D. This displacement is consistent with the manner in which pile load test data were interpreted.

C.8.1.4.4.1.3 Application to sands other than siliceous sands

Other sands include carbonate sands, micaceous sands, glauconitic sands, volcanic sands, silts, and clayey sands.

Some cohesionless soils have unusually weak structures/compressible grains. These usually require special in situ and/or laboratory tests for selection of an appropriate design method with associated design parameters. See Reference [167] and [168] and Annex B for pile design in carbonate sand and Reference [168] for guidelines on pile design in other sands and silts. Consideration should be given to using a design method for clays for cases of low permeability sands and silts. All former methods should be applied cautiously since limited data are available to support their reliability in these sediments.

C.8.1.4.4.1.4 Cone resistance, $q_c(z)$, in gravel

The measured $q_c(z)$ data should not be taken at face value in this cohesionless soil type and appropriate adjustments should be made. For example, CPTs made in (coarse) gravels, especially when particle sizes are in excess of 10 % of the CPT cone diameter, are misleading and one possible approach could be to use the lower bound $q_c(z)$ profile. Alternatively, one may estimate an appropriate design $q_c(z)$ profile from adjacent sand layers.

C.8.1.4.4.1.5 End bearing, Q_p, in presence of weaker clay layers near pile tip

The $q_c(z)$ data used can have a substantial impact on the fully plugged unit end bearing, q. The use of $q_c(z)$ data averaged between 1.5D above the pile tip to 1.5D below the pile tip level should generally be satisfactory, providing $q_c(z)$ does not vary significantly. This is not necessarily the case when clay layers occur. If significant $q_c(z)$ variations occur, then Figure 2.2 of Reference [154] should be used to compute a suitable average $q_{c,av}$ value.

Thin clay layers (less than around 0.1D thick) are problematic, particularly when CPT data are discontinuous vertically and/or not all pile locations have been investigated. Factors to be considered include the variance of layer thickness and of strength and compression parameters. If no direct data are available, a cautious interpretation should be made based on the engineering geology of the surrounding sand soil unit. Offshore piles usually develop only a small percentage of end bearing under extreme conditions. Hence, capacity and settlement calculations, using a finite element model of a pile tip on sand containing weaker layers, may be considered to adequately assess axial pile response under such conditions.

For thick clay layers, shallow geophysical data can be useful to assess layer thickness and elevation. The recommendation in 8.1.4 is to reduce the end bearing component if the pile tip is within a zone up to ± 3D from such layers. If averaging of $q_c(z)$ data is applied to this ± 3D zone, the combined effects can be unduly cautious and such results should be critically reviewed. Similarly, for large diameter piles (say D > 2 m), the reduction method in 8.1.4 should be carefully reviewed.

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C.8.1.4.4.1.6 Near-shore and onshore piles

In general, for near-shore and onshore piles, the assumptions in C.8.1.4.1 and C.8.1.4.2 are not necessarily valid and should be checked.

Near-shore and onshore pipe piles can respond unplugged when loaded due to insufficient mobilization of inner friction. Similarly, dilatancy effects (which are neglected for offshore piles) may be considered for smaller diameter piles. Scour (especially general scour) can be significant for near-shore pile foundations. In addition, driven closed-ended instead of open-ended steel piles are sometimes used.

The original publications of References [154], [157], [158], and [160] should be consulted for assumptions made and for further guidance; most of these references include methods to provide the capacity of unplugged pipe piles and of closed-ended piles.

C.8.1.4.4.1.7 Scour

Scour (seabed erosion due to wave and current action) can occur around offshore piles. Common types of scour are general scour (overall seabed erosion) and local scour (steep-sided scour pits around single piles or pile groups). There is no generally accepted method to account for scour in axial capacity for offshore piles. Publications such as Reference [169] give techniques for scour depth assessment. In addition, general scour data can be obtained from national authorities.

In lieu of project specific data, C.8.5 gives advice on local scour depth.

Scour decreases axial pile capacity in sand. Both friction and end bearing components are usually affected. This is because scour reduces both $q_c(z)$ and $p'_o(z)$ (vertical effective stress). For excavations (which may be considered equivalent to general scour), Reference [170] recommends that $q_c(z)$ is simply proportional to $p'_o(z)$, i.e.,

$$q_{c,f}(z) = \chi\, q_{c,0}(z) \tag{C.23}$$

where

$q_{c,f}(z)$ is the reduced CPT cone-tip resistance at depth, z, in stress units, to account for general scour;

$q_{c,0}(z)$ is the CPT cone-tip resistance at depth, z, in stress units, before general scour;

χ is the dimensionless scour reduction factor ($\chi = p'_{of}/p'_o$),

where

p'_{of} is the final vertical effective stress value, after scour;

p'_o is the original vertical effective stress value, before scour.

For large general scour depths and normally consolidated sands, an alternative and conservative approach[172] can be to determine χ from Equation C.24.

$$\chi = \frac{1}{1+2K_o}\sqrt{\frac{z'+2K_o\sqrt{\Delta z_{GS}\times z'+z'^2}}{\Delta z_{GS}+z'}} \tag{C.24}$$

where

Δz_{GS} is the general scour depth;

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z' is the final depth below seafloor, after general scour, ($z' = z - \Delta z_{GS}$);

K_o is the coefficient of lateral earth pressure at rest (the ratio of the effective horizontal to vertical *in situ* soil stresses, $K_o = p'_{oh}(z)/p'_o(z)$.

Clause C.8.5 gives a method to reduce the effective stress for both general and local scour.

C.8.1.5 Shaft friction and end bearing of grouted piles in rock

No guidance is offered.

C.8.2 Pile capacity for axial pullout loads

No guidance is offered.

C.8.3 Axial pile performance

C.8.3.1 Static axial behavior of piles

An analytical method for determining axial pile performance is provided in Reference [172]. This method makes use of t–z curves of local transfer of axial pile shear, t, against local pile displacement, z, to model the axial support provided by the soil along the side of the pile. An additional Q–z curve is used to model the tip end bearing, Q, against tip displacement, z. Methods for constructing t–z and Q–z curves are given in Clause 8.4.

In some circumstances, i.e. for soils that exhibit strain-softening behavior and/or where the piles are excessively axially flexible, the actual capacity of the pile can be less than the ultimate capacity given by Equation 15. In these cases, an explicit consideration of these effects on axial capacity is warranted.

C.8.3.2 Cyclic axial behavior of piles

C.8.3.2.1 Qualification

Modeling cyclic effects explicitly can improve the designer's insight into the relative importance of the cyclic characteristics of the loads. On the other hand, extreme care should be exercised in applying this approach; historically, cyclic effects have been taken into account implicitly rather than explicitly. Design methods developed and calibrated on an implicit basis generally need extensive modification where explicit algorithms are employed.

C.8.3.2.2 Loads

Axial loads on piles are developed from a wide variety of operating, structural, and environmental sources. Permanent and variable loads are generally long duration loads and are often referred to as static loads. Environmental loads are developed by winds, waves and currents, earthquakes and ice floes. These loads can have both low and high frequency cyclic components in which the rates of change of loads and load durations are measured in seconds. Storm and ice can cause several thousand cycles of (relatively speaking) low frequency loads, while earthquakes can induce several tens of cycles of high frequency loads[173].

C.8.3.2.3 Static capacity

For most fixed offshore structures supported on piles, experience has proven the adequacy of determining pile penetration based on static capacity evaluations, with static design loads and commonly accepted working stress design (WSD) factors of safety that, in part, account for the cyclic effects[172].

Detailed consideration of cyclic effects can be warranted when there are unusual limitations on pile penetrations or when certain soils, conditions related to loads or novel structures (e.g. compliant towers) are involved.

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C.8.3.2.4 Cyclic effects

Compared with long-term static loads, cyclic loads can have the following important influences on pile axial capacity and stiffness. They can

— decrease capacity and stiffness due to repeated loads[174], or

— increase capacity and stiffness due to high rates of change of loads[175].

The resultant effect on capacity is primarily influenced by the pile properties (stiffness, length, diameter, material), the soil characteristics (type, stress history, strain rate and cyclic degradation) and the load characteristics (numbers and magnitudes of repeated loads). Cyclic loads can also cause accumulation of pile displacements and either stiffening and strengthening or softening and weakening of the soils around the pile. Hysteretic and radiation damping dissipate the energy provided by the loads in the soil[173]. For earthquakes, the free-field ground motions (independent of the presence of the piles and structure) can develop important cyclic straining effects in the soils; these effects can influence pile capacity and stiffness[176].

C.8.3.2.5 Analytical models

C.8.3.2.5.1 General

A variety of analytical models have been developed and applied to determine the cyclic axial behavior of piles. These models can be grouped into two general categories, as below.

C.8.3.2.5.2 Discrete element models

The soil around the pile is idealized as a series of uncoupled “springs” or elements attached between the pile and the far field soil (usually assumed rigid). The material behavior of these elements can vary from linearly elastic to non-linear, hysteretic and rate dependent. The soil elements are commonly referred to as t–z (shaft resistance-displacement) and Q–z (tip resistance-displacement) elements, see Reference [177] to [179]. Linear or non-linear dashpots (velocity dependent resistances) can be placed in parallel and in series with the discrete elements to model radiation damping and rate of change of load effects[180]. The pile can also be modelled as a series of discrete elements, e.g. rigid masses interconnected by springs, or modelled as a continuous rod, either linear or non-linear. In these models, material properties (soil and pile) can vary along the pile.

C.8.3.2.5.3 Continuum models

The soil around the pile is idealized as a continuum attached continuously to the pile. The material behavior can incorporate virtually any reasonable stress-strain rules the analyst can devise. Depending on the degree of non-linearity and heterogeneity, this model can be quite complicated. Again, the pile is typically modeled as a continuous rod, either linear or non-linear. In these models material properties can vary in any direction, see Reference [181] and [182].

There is a wide range of assumptions that can be used regarding boundary conditions, solution characteristics, etc. which lead to an unlimited number of variations for either of the two approaches.

Once the idealized model is established and the relevant equations are developed, then a solution technique should be selected. For simple models, a closed form analytical approach is sometimes possible. Otherwise, a numerical procedure should be used. In some cases, a combination of numerical and analytical approaches is helpful. The most frequently used numerical solution techniques are the finite difference method and the finite element method. Either approach can be applied to both the discrete element and continuum element models. Discrete element and continuum element models are occasionally combined[173][180]. Classical finite element models have been used for specialized analyses of piles subjected to monotonic axial loads (Reference [181]).

For practical reasons, discrete element models, solved numerically, have seen the most use in evaluation of piles subjected to high intensity cyclic load. Results from these models are used to develop information on accumulated pile displacements and on pile capacity following high intensity cyclic loads[178] [179].

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Elastic continuum models solved analytically (similar to those used in machine vibration analyses) have proven to be useful for evaluations of piles subjected to low intensity, high frequency cyclic loads at or below design working levels[181] [182]). At higher intensity loads, where material behavior is likely to be non-linear, the continuum model solved analytically can still be used by employing equivalent linear properties that approximate the non-linear, hysteretic effects[183].

C.8.3.2.6 Soil characteristics

A key part of developing realistic analytical models to evaluate cyclic effects on piles is the characterization of soil-pile interaction behavior. High quality *in situ*, laboratory, and model-prototype pile load tests are essential in such characterizations. In developing (soil) characterizations relevant for soil-pile interaction, it is important that pile installation and relevant conditions of the loads on a pile be integrated into the testing programs[173] [179].

In situ tests (e.g. vane shear, cone penetrometer, and pressuremeter) can provide important insights into in-place soil behavior and stress-strain properties[184]. Both low and high amplitude stress-strain properties can be developed. Long-term (static, creep), short-term (dynamic, impulsive) and cyclic (repeated) loads sometimes can be simulated with in situ testing equipment.

Laboratory tests on representative soil samples permit a wide variety of stress-strain conditions to be simulated and evaluated[185]. Soil samples can be modified to simulate pile installation effects (e.g. remolding and reconsolidating to estimated in situ stresses). The samples can be subjected to different boundary conditions (triaxial, simple-shear, interface shear) and to different levels of sustained and cyclic shear time histories to simulate in-place conditions of applied loads.

Tests on model and prototype piles are another important source of data for developing soil characterizations for cyclic analyses. Model piles can be highly instrumented and repeated tests can be performed in soils and for a variety of loads[179] [186]. Geometrical scale, time scale and other modelling effects should be carefully considered in applying results from model tests to analyses of prototype behavior.

Data from load tests on prototype piles are useful for calibrating analytical models[187] [190]. Such tests, even if not highly instrumented, can provide data to guide development of analytical models. These tests can also provide data for verifying results of soil characterizations and analytical models, as shown in References [173], [179], [180], [191], and [192]. Prototype pile load testing coupled with in situ and laboratory soil testing and realistic analytical models can provide an essential framework for making realistic evaluations of the responses of piles to cyclic axial loads.

C.8.3.2.7 Analysis procedure

C.8.3.2.7.1 General

The primary considerations in performing an analysis of cyclic axial effects on a pile using discrete element models are as below.

C.8.3.2.7.2 Loads

The loads on the pile head should be characterized in terms of their magnitudes, durations, and numbers of cycles. This includes both long-term loads and short-term cyclic loads. Typically, the design static and cyclic loads expected during a design event are chosen.

C.8.3.2.7.3 Pile properties

The properties of the pile including its diameter, wall thickness, stiffness, weight, and length should be defined. This will require an initial estimate of the pile penetration that is appropriate for the design loads. Empirical, pseudo-static methods based on pile load tests or soil tests can be used to make such estimates.

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C.8.3.2.7.4 Soil properties

Different analytical approaches will require different soil parameters. For the continuum model, the elastic and damping properties of the soil are required. In the discrete element model, soil resistance–displacement relationships along the pile shaft t–z and at its tip Q–z should be determined. In situ and laboratory soil tests and model and prototype pile load tests can provide a basis for such determinations. These tests should at least implicitly include the effects of pile installation, types of loads and time scales. In addition, the test should be performed so as to provide insight regarding the effects of the characteristics of the loads on the pile. Most importantly, the soil behavior characteristics should be appropriate for the analytical model(s) used, duly recognizing the empirical bases of these models.

C.8.3.2.7.5 Cyclic analyses

Analyses should be performed to determine the response (resistance and displacement) characteristics of the pile subjected to its design static and cyclic loads. Recognizing the inherent uncertainties in evaluations of pile loads and soil-pile behavior, parametric analyses should be performed to evaluate the sensitivities of the pile response to these uncertainties. The analytical results should develop realistic predictions of pile resistance and accumulated displacements for design loads. In addition, following the simulation of static and cyclic design loads, the pile should be further analysed so as to estimate its reserve capacity.

C.8.3.2.8 Performance requirements

A primary objective of these analyses is to ensure that the pile and its penetration are adequate to meet the structure's requirements.

The pile performance for explicit cyclic analyses should be evaluated separately. The pile should have a capacity that provides an adequate margin of reserve above the design loads. In addition, the pile should not settle or pull-out, nor accumulate displacements to the extent that could constitute failure of the structure foundation system.

C.8.3.3 Overall axial behavior of piles

No guidance is offered.

C.8.4 Soil reaction for piles under axial compression

C.8.4.1 General

No guidance is offered.

C.8.4.2 Axial shear transfer t–z curves

Theoretical curves may be constructed in accordance with Reference [71]. Empirical t–z curves based on the results of model and full-scale pile load tests may follow the procedures for clay soils described in Reference [72].

C.8.4.3 End bearing resistance–displacement Q–z curves

No guidance is offered.

C.8.5 Soil reaction for piles under lateral loads

Generally, under lateral loads, clay soils behave as a plastic material which makes it necessary to relate pile-soil deformation to soil resistance. To facilitate this procedure, lateral soil resistance-displacement p–y curves should be constructed using stress-strain data from laboratory soil samples. The ordinate for these curves is soil resistance, p, and the abscissa is pile wall displacement, y. By iterative procedures, a compatible set of lateral resistance-displacement values for the pile-soil system can be developed.

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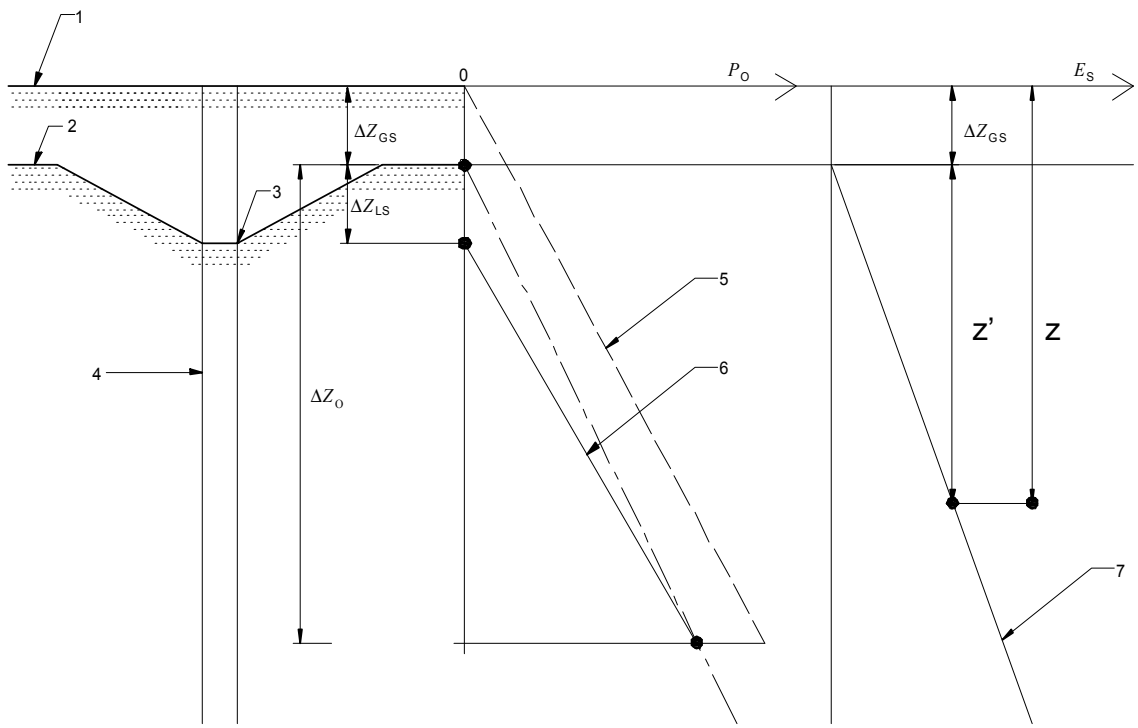
For a more detailed study of the construction of p–y curves, see Reference [193] for soft clay, and References [194] and [199] for structured and relatively unstructured stiff clays, Reference [195] for sand and Reference [196] for layered soils.

Scour (seabed sediment erosion due to wave and current load) can occur around offshore piles. Scour reduces lateral soil support, leading to an increase in pile maximum bending stress. Scour is generally not a problem for cohesive soils, but should be considered for cohesionless soils. Common types of scour are

— general scour (overall seabed erosion), and

— local scour (steep sided scour pits around single piles).

Publications such as Reference [169] give techniques for scour depth assessment. In addition, general scour data may be obtained from national authorities. In the absence of project specific data, for an isolated pile a local scour depth equal to 1.5D and an overburden reduction depth equal to 6D may be adopted, D being the pile outside diameter; see Figure C.2.



Key

1	Original seafloor level	Δz_{GS}	General scour depth
2	Level after general scour	Δz_{LS}	Local scour depth (1.5 × D typical)
3	Level of local scour	Δz_O	Overburden reduction depth (6.0 × D typical)
4	Pile	P'_O	Vertical effective stress
5	No scour case	E_S	Initial modulus of subgrade reaction
6	Local scour case	z	Depth below original seafloor
7	$E_S = k\,z'$ (see Table 5 for k)	z'	Final depth below seafloor, after general scour

Figure C.2—*P-Y* lateral support – scour model

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Reduction in lateral soil support is due to two effects:

— a lower ultimate lateral pressure caused by decreased vertical effective stress p'_o (z);

— a decreased initial modulus of subgrade reaction modulus, E_S.

There is no general accepted method to allow for scour in the p–y curves for offshore piles. Figure C.2 suggests one of the methods for evaluating $p'_o(z)$ and E_S as a function of scour depths. In this method, general scour reduces the $p'_o(z)$ profile uniformly with depth, whereas local scour reduces $p'_o(z)$ linearly with depth to a certain depth below the base of the scour pit. Subgrade modulus reaction values E_S may be computed assuming the general scour condition only.

Other methods, based upon local practice and/or experience, may be used instead.

C.8.6 Pile group behavior

C.8.6.1 General

Routine numerical analysis of pile groups can be divided into two main categories.

The first category, which is computationally the simplest, uses algebraic expressions to define the elastic single pile resistance to general (axial, lateral, and torsional) loads[197]. The group resistance is determined by modifying the single pile expressions to account for elastic pile-soil-pile interaction.

The second analysis category, which is normally performed for offshore pile groups, is more rigorous. Methods are usually hybrid, employing a mixture of discrete p–y curves (Winkler approach) and continuum soil behavior, first described in Reference [77] for lateral analysis. Since then, numerous programs have been developed worldwide for general types of load. Typically, the non-linear single pile resistances to general loads are computed using axial t–z and lateral p–y curves and combined with elastic interaction expressions similar to the first category. The resulting equations are solved for various pile head fixity conditions and/or pile cap restraint to determine the non-linear group resistance and individual pile forces and moments, plus the so-called "z- and y-modifiers".

For more detailed discussions see Reference [197] and [198].

C.8.6.2 Axial behavior

In general, group effects depend considerably on pile group geometry and penetrations and thickness of any bearing stratum underneath the pile tips[197][198].

C.8.6.3 Lateral behavior

Experience confirms that the available tools for analysis of pile groups subjected to lateral loads provide approximate answers that sometimes deviate significantly from observed behavior, particularly with regard to displacement calculations. Also, limitations in site investigation procedures and in the ability to predict soil-pile interaction behavior for a single pile produce uncertainty regarding proper soil input to group analyses. Therefore, multiple analyses should be performed for pile groups using two or more methods of analysis and upper-bound and lower-bound values of soil properties in the analyses. By performing such analyses, the designer will obtain an appreciation for the uncertainty involved in his predictions of foundation performance and can make more informed decisions regarding the structural design of the foundation and structure elements.

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